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ACRP REPORT 150

NextGEN for Airports

AIRPORT COOPERATIVE RESEARCH PROGRAM

Volume 5

Airport Planning and Development



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ACRP REPORT 150

NextGEN for Airports Volume 5

Airport Planning and Development

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in TRB Special Report 272: Airport Research Needs: Cooperative Solutions in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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Foreword

By **Joseph D. Navarrete**
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The Next Generation Air Transportation System (NextGen) refers to the federal programs (predominately airspace, air traffic, or avionics related) that are designed to modernize the National Airspace System (NAS). ACRP's NextGen initiative aims to inform airport operators about some of these programs and how the enabling practices, data, and technologies resulting from them will affect airports and change how they operate.

This project is part of an ACRP NextGen initiative that comprises five distinct projects, which have been conducted simultaneously. The scope, ideas, and preliminary results have been shared among all five projects. The titles of the ACRP projects included in the ACRP NextGen Initiative are as follows:

- ACRP 01-27, "NextGen—A Primer";
- ACRP 01-28, "NextGen—Guidance for Engaging Airport Stakeholders";
- ACRP 03-33, "NextGen—Airport Planning and Development";
- ACRP 03-34, "NextGen—Understanding the Airport's Role in Performance-Based Navigation (PBN)"; and
- ACRP 09-12, "NextGen—Leveraging NextGen Spatial Data to Benefit Airports."

ACRP Report 150: NextGen for Airports, Volume 5: Airport Planning and Development provides information and guidance to help airport industry practitioners understand and incorporate NextGen capabilities into their planning efforts for all categories of airports. The report will be a particularly valuable resource to airport planning practitioners wishing to ensure that airport facilities are designed in a manner that not only can accommodate NextGen but also take full advantage of its potential benefits.

Airports rely on planning to ensure that safety, efficiency, and capacity needs are met in a responsible and cost-effective manner. The FAA is in the process of developing and implementing various NextGen capabilities, many of which have direct implications for airport facilities and operations. While many airports could benefit from incorporating NextGen capabilities, there has been limited information and guidance on how to incorporate them into their planning process. In addition, NextGen capabilities are complex and continue to mature, and there is uncertainty regarding industry priorities and the timing of potential initiatives. Research was needed to provide guidance for airport industry practitioners to incorporate NextGen capabilities into their planning and development process.

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The research, led by LeighFisher, Inc., featured extensive collaboration with the other research teams involved with ACRP's NextGen initiative. A key element of their research approach was an examination of case studies of airports and regions where various NextGen initiatives have been implemented. The team then developed a preliminary draft of the guidebook based on their findings and expertise. To ensure the guidebook would be practical and easy to use, the team vetted the preliminary draft with airport industry practitioners and produced the final version based on their feedback.

The *Airport Planning and Development* guidebook will help airport planning practitioners:

- Determine which NextGen capabilities are applicable for their particular planning project;
- Identify the steps and roles of stakeholders needed for accommodating and implementing NextGen-related initiatives;
- Determine the likelihood and timing of NextGen capabilities;
- Provide planning flexibility to account for the risk and uncertainty associated with NextGen capabilities;
- Identify opportunities for airport participation in NextGen implementation;
- Identify strategies for engaging with FAA and other stakeholders, including the community, involved in various NextGen capabilities; and
- Integrate airport layout plans, GIS, online mapping, FAA data resources, and other "big data" tools into the planning process.

The guidebook provides specific guidance for large, medium, and small airports. The guidebook also contains appendices that provide lessons learned from case studies as well as a summary of NextGen resources.

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1 Introduction and Background

Since the inception of the Next Generation Air Transportation System (NextGen), airports have been challenged to understand its potential benefits, costs, and impacts on their operations and on their relationships with industry stakeholders, the public, and other external interests. Nowhere has this been more evident than in the field of airport planning, where anticipation of long-term factors affecting airport development and impacts requires a more detailed understanding of NextGen's promise and processes. This guidebook is intended to provide airport planning professionals with detailed resources to assist them in understanding how NextGen will affect the traditional airport planning process and their airport's future development.

Objectives

The overall objective of this research is to develop a guidebook to help airport industry planning practitioners understand and incorporate NextGen capabilities into planning for all categories of airports. Below is a list of specific objectives that were established at the outset of this project:

- Determine which NextGen capabilities are applicable for a particular airport planning project and identifying the role the airport sponsor has for implementing a particular NextGen capability.
- Identify airport-led planning initiatives that use NextGen capabilities to address gaps that will not be addressed by FAA NextGen plans [e.g., ground-based augmentation system (GBAS), multilateration outside the movement area].
- Identify, justify, and develop strategies for funding airport infrastructure that rely on and accommodate NextGen capabilities.
- Determine the likelihood and timing (near-term and intermediate-term initiatives as well as potential, long-range future concepts) of NextGen capabilities.
- Assess local user readiness for particular NextGen capabilities.
- Provide planning flexibility to account for the risk and uncertainty associated with NextGen capabilities, in order to develop risk-adjusted strategies to guide airport development plans.
- Identify opportunities for airport participation in NextGen forums.
- Identify and engage various FAA offices and other stakeholders involved in various NextGen capabilities.
- Develop strategies for airport participation when prioritizing NextGen capabilities affecting their airport.
- Identify a mechanism for obtaining FAA's navigation and surveillance facility plans for an airport.
- Incorporate FAA-estimated capacity improvements from NextGen projects at the airport.

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- Explain how airports may stay informed of local and regional airspace efficiency improvements that could impact the airport.
- Incorporate NextGen capabilities into the environmental process, including impacts from aircraft operator-initiated and FAA-initiated projects.
- Integrate airport layout plan (ALP), GIS (geographic information system), online mapping, FAA data resources, and other "big data" tools.

Relationship to Other ACRP NextGen Initiative Projects

After considerable consultation with the FAA and industry representatives, ACRP has developed what is referred to as the "ACRP NextGen Initiative" comprising five concurrent and coordinated NextGen research projects, each covering a critical NextGen need and audience, as follows:

- ACRP Project 01-27, "NextGen—A Primer."
- ACRP Project 01-28, "NextGen—Guidance for Engaging Airport Stakeholders."
- ACRP Project 03-33, "NextGen—Airport Planning and Development."
- ACRP Project 03-34, "NextGen—Understanding the Airport's Role in Performance-Based Navigation (PBN)."
- ACRP Project 09-12, "NextGen—Leveraging NextGen Spatial Data to Benefit Airports."

ACRP Project 03-33 has been coordinated with the other four ACRP NextGen Initiative projects, primarily through a face-to-face meeting of the five panels and research teams at a 1-day on-site NextGen Initiative workshop.

There is considerable and intentional overlap among the scopes of these five projects, although the intended audiences and levels of detail of the five projects differ. Three of the ACRP NextGen Initiative projects are highly focused on specific NextGen capabilities and airport planning elements (e.g., PBN, stakeholder engagement, and spatial data) that will be of great interest to the other two projects (the resource guide/primer and airport planning), which have much broader scopes. The level of technical depth of the three closely related projects on the treatment of airport planning and development issues different significantly: (1) Project 01-27 presents a brief overview; (2) Project 03-33 is like an undergraduate survey course on "NextGen for professional planners 101"; and (3) Project 03-34 is like an undergraduate course focused only on PBN procedures.

Summarized below is a brief description of the five ACRP NextGen Initiative projects and how they relate to one another:

ACRP Project 01-27, "NextGen—A Primer," started earlier than the other four projects because it was intended to set the stage for defining NextGen at a high level and developing a NextGen information guide for airport practitioners, airport decision makers, and the public. It provides a high-level summary of NextGen capabilities, data, and related technologies that airport executives need to understand, and that are presented in greater detail in the other four projects, as appropriate. Project 01-27 provides a high-level introduction to NextGen issues that is intended to give airport senior executives, airport general staff, and the public an overview of NextGen capabilities, issues, and impacts. This airport planning guidebook will give more specific guidance to airport planners, planning consultants, and others with a need to understand the more detailed aspects of NextGen and airport

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planning. It delves deeply into the information needed to incorporate NextGen capabilities into airport master planning, airport environmental planning, state, and regional/metropolitan airport system planning, airspace redesign studies [e.g., OAPM (optimization of airspace procedures in the metroplex)] and airport financial and strategic planning.

ACRP Project 01-28, "NextGen—Guidance for Engaging Airport Stakeholders," had the objective of developing guidance for airports for engaging with the FAA and other airport stakeholders on the NextGen development and implementation lifecycle including, but not be limited to, planning, environmental review, design, and deployment. This project contributes elements not considered in detail in the other four NextGen Initiative projects in that it provides communication and collaboration tools, strategies and methods for engaging stakeholders, and example applications of these tools, strategies, and methods to particular NextGen initiatives through the use of case studies. In particular, Project 01-28 comprehensively addresses the communication issues and methods for interacting with various stakeholders, such as airlines, members of the public, and the press. This airport planning guidebook will focus more narrowly on the interactions between the process of incorporating NextGen into airport planning activities and working with various internal and external stakeholders.

ACRP Project 03-33, "NextGen—Airport Planning and Development," had the objective of incorporating NextGen capabilities into airport planning and development processes. In this research, expectations and recommendations are provided for implementing various NextGen capabilities at airports of all sizes, and case studies combined with industry outreach will inform airports of what benefits and challenges they should expect. The other four ACRP NextGen Initiative projects provide essential inputs to the following critical elements of airport planning and development:

A common high-level understanding and vocabulary for NextGen technologies and operational improvements by practitioners, decision makers, and the communities (ACRP Project 01-27).

New tools for airports to engage with the FAA and other airport stakeholders in the planning, review, design, and deployment of NextGen capabilities (ACRP Project 01-28).

An understanding of the role airports should play in the implementation of PBN flight procedures and the benefits, impacts, and tradeoffs that airports will need to address in their planning for future development (ACRP Project 03-34).

Guidance on the potential benefits of collecting, maintaining, and using spatial data to enable and support the implementation of NextGen at airports (ACRP Project 09-12).

ACRP Project 03-34, "NextGen—Understanding the Airport's Role in Performance-Based Navigation (PBN)," provides a user-friendly "Airport PBN Resource Guide" that will provide input to this ACRP 03-33 project on relevant federal policies, the potential tradeoffs between efficiency and capacity, and potential impacts on airport stakeholders. This project provides both high-level descriptions of PBN for ACRP 01-27 and more comprehensive information for ACRP 03-33 and ACRP 01-28 on the potential benefits, impacts, and priorities for PBN, as well as criteria for airports to use in evaluating proposed PBN flight procedures.

ACRP Project 09-12," NextGen—Leveraging NextGen Spatial Data to Benefit Airports," has the objectives of (1) identifying the benefits that can be derived from spatial data that are to be collected, maintained, and shared in support of the FAA's NextGen effort and (2) providing guidance on how airports can maximize the use of these data. In particular, this project provides guidance on how airports will participate in the Airports Surveying Geographic Information System (Airports GIS) program in collecting airport and aeronautical data to meet the demands of NextGen.

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NextGen Background and Why This Guidebook Is Needed

As stated in the RFP for ACRP Project 01-27, "NextGen—A Primer":

> For the ACRP NextGen Initiative, the term "NextGen" shall refer to a number of federal programs (predominately airspace, air traffic, or avionics related) that are designed to modernize the NAS (National Airspace System).

NextGen is the culmination of a number of air traffic modernization programs that the FAA has initiated over many years. The original concept was developed by a broad FAA/industry effort published in October 1995 as the "Final Report of RTCA (Radio Technical Commission for Aeronautics) Task Force 3 Free Flight Implementation." The current initiative started in December 2003 with the *Vision 100—Century of Aviation Reauthorization Act*, which authorized plans for a new, multi-year, multi-agency effort to develop an air transportation system for the year 2025 and beyond, which would be planned by a Joint Planning and Development Office (JPDO) at the FAA composed of representatives from FAA, NASA, United States Department of Transportation (U.S.DOT), Department of Defense (DOD), Department of Homeland Security (DHS), Commerce Department, and the White House Office of Science and Technology Policy. One year later, JPDO published the Integrated Plan for the Next Generation Air Transportation System, which laid out goals, objectives, and requirements for NextGen. This was redefined and expanded in September 2009 by another RTCA industry task force as the "NextGen Mid-Term Implementation Task Force 5 Report," which advocated for the focus to be on near- to mid-term benefits of NextGen.

From the beginning there has been limited but increasing opportunity for airport planning practitioners to participate in the development of NextGen. For example:

- JPDO established an Airports Working Group to represent the needs and concerns of airports.
- The airport associations, e.g., American Association of Airport Executives (AAAE) and Airports Council International–North America (ACI–NA), began sponsoring conference sessions (and whole conferences) on NextGen and established their own NextGen working groups.
- Airports secured representation on various industry NextGen committees at FAA and RTCA, among other organizations.
- The FAA established a new Surface Operation Office with the mission to enhance the efficiency of the nation's airports.

NextGen success stories [e.g., surface management at John F. Kennedy International Airport (JFK), the Greener Skies over Seattle program at SeaTac, and wake turbulence recategorization (Wake RECAT)] began to generate increased interest in the potential near-term benefits of NextGen at airports.

Despite increasing forums for participation and involvement by airports, there are still significant challenges to airports being able to incorporate NextGen into airport planning and development processes, including the following:

- The perception that NextGen implementation is driven primarily by the FAA Air Traffic Organization (ATO), the airlines, and manufacturers of airplanes, airplane avionics, and air traffic control (ATC) systems is widely held among airports. Moreover, many airport planning practitioners believe that they have little or no influence, or stake, in the implementation of NextGen.
- The substantial investment required by the airport users [i.e., the operators of commercial and general aviation (GA) aircraft] to equip their aircraft with the avionics required to realize the potential benefits of certain NextGen technologies. This concern was validated somewhat by recent expe-

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riences of certain airlines that have equipped their aircraft only to find out that very few benefits have resulted from their investment.

- The uncertainty about the timing of NextGen implementation and its potential benefits to airports has been a factor in many airports' reluctance to incorporate NextGen into their existing facility, environmental, and financial planning processes. As a result, until recently many in the airports community have chosen not to become engaged with the NextGen planning and implementation process.
- The lack of FAA advisory circulars (ACs) and policy documents containing comprehensive guidance on methods to assess NextGen improvements at airports and how they should be addressed in planning and environmental studies.
- At a 2013 ACI-NA conference session on NextGen, panel members were asked what they thought was the greatest challenge to the implementation of NextGen at their airports. The almost unanimous response was gaining community acceptance of the potential NextGen PBN flight procedures over the communities surrounding the airport. Two other challenges mentioned by the panel members were (1) convincing aircraft operators to equip their airplanes in order to realize potential benefits and (2) the uncertainty about the timing and benefits that NextGen will actually deliver.

The premise of this guidebook is that greater involvement by airports is essential to the acceptance and implementation of NextGen technologies by airport and community stakeholders. In particular, the target audience for this guidebook is the "airport planning practitioner." For purposes of this guidebook, "airport planning practitioners" are defined as aviation professionals practicing in airport planning and development whose employers could include, but are not limited to, airport sponsors, airport consulting firms, airlines, state or regional aviation organizations, airport and airline associations, FAA Airports Division and Air Traffic Organization, federal contractors, aircraft users groups, and aviation research organizations.

Airport planning practitioners have the local knowledge of their airports and communities, which could be extremely valuable in (1) organizing community support and reducing the potential for community opposition; (2) designing and setting local priorities for PBN flight procedures within the broader framework established by industry groups such as the NextGen Advisory Committee; (3) identifying NextGen operational improvements (e.g., closely spaced parallel runway) that could potentially defer investment in capacity-enhancement projects or reduce the cost of those projects; and (4) providing the information needed to counter misunderstandings by airport development opponents of the ability of certain NextGen technologies and operational improvements to reduce or eliminate the need for capital improvement projects.

Moreover, it may be possible for airports to provide economic and/or operational incentives for aircraft equipage or early adoption of NextGen-enabling infrastructure. Airports stand to benefit from increased equipage by their users because such increased equipage could (1) increase the achievable benefits of NextGen operational improvements, such as reduced aircraft delays and emissions; (2) reach a threshold of feasibility for implementing certain advanced NextGen flight procedures (e.g., such implementation may require in excess of 80% average); and (3) reduce the airline operating costs at the airport, making the airport more attractive for maintaining and increasing air service.

Unfortunately, however, at present there is no single source or guidebook that airport planning practitioners can use to understand and incorporate NextGen capabilities into their planning efforts. Therefore, the timing is right for the publication of a guidebook for airport industry planning practitioners on how to adapt their facility, operational, environmental, and financial planning processes to incorporate NextGen technologies and operational improvements despite uncertainties in terms of when they will be available, how much benefit they will provide, and how much they will cost.

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In summary, this guidebook is needed to define what NextGen means to airport planning practitioners to help them decide how to properly prepare for NextGen.

Who Should Use This Guidebook?

This guidebook is aimed at airport planning practitioners responsible for master planning, strategic and financial planning, airspace capacity, and procedure development as well as for state and regional/metropolitan airport system planners.

What Existing Guidance on NextGen at Airports Is Available?

Although FAA has several helpful publications and online resources describing NextGen for airports, they are not specific enough to provide the level of detail needed for an airport planner to use effectively. For example, in June 2014 FAA published a brochure titled "NextGen for Airports" available at https://www.faa.gov/nextgen/qanda/airports/. More recently, FAA added an update to its NextGen website, "NextGen Update 2016," which includes a page devoted to airports available at https://www.faa.gov/nextgen/update/operator_investments_and_airports/airport_enhancements/. This new webpage includes NextGen programs of interest to airports such as the Airports GIS program, PBN, Surface Surveillance and Data Sharing, and Closely Spaced Parallel Runways.

Another useful source of information is the FAA webpage on the *runway*Simulator program. In particular, many of the basic NextGen operational assumptions (e.g., 7110.308 staggered approach procedures, Wake RECAT, etc.) are built into the *runway*Simulator program for capacity evaluations. This program can be found at http://www.faa.gov/airports/planning_capacity/runwaysimulator/.

This guidebook will serve as a "desk reference" for airport planners trying to integrate NextGen into their routine planning responsibilities.

Organization of This Guidebook

The guidebook is organized into five substantive chapters and a number of appendices as described below:

Chapter 2—NextGen Technologies and Operational Improvements

This chapter provides an in-depth description of the technologies and operational improvements promised by NextGen and emerging non-NextGen technologies, beginning with a high-level review of NextGen technologies, operational improvements, and potential benefits. It continues with a more detailed planning practitioner discussion of the NextGen elements of most interest to airports and how they could affect the airport planning process.

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Chapter 3—Incorporating NextGen into Airport Planning and Development

This chapter provides a working-level discussion of the likely changes to planning practices to reflect the expected gradual introduction of NextGen capabilities to airports. As NextGen becomes more widely implemented, additional and more specific guidance will become available from FAA in the form of revised ACs, orders, and other documents. Not all of the information needed to plan for NextGen is likely to be available from the FAA Office of Airports. As is currently the case, much of the information needed to consider NextGen effects is distributed in other places, such as the Office of NextGen, the ATO, Flight Standards Service, RTCA, and elsewhere.

Chapter 4—Applicability of NextGen to Medium and Large Airport Planning and Development

This chapter describes how NextGen could affect medium and large airport planning and development, starting with a clarification of the definition of medium and large airports for NextGen purposes. Because the traditional set of airport size definitions (hubs—large, medium, small, non-hubs, and GA) does not accommodate the issues facing airports in NextGen very well, this chapter will develop an approach to defining medium and large airports that is derived from the NextGen capabilities that are most likely to be of interest to those airports.

Chapter 5—Applicability of NextGen to Small Airport Planning and Development

This chapter describes how NextGen could affect small airport planning and development, starting with a clarification of the definition of small airports for NextGen purposes. Because the traditional set of airport size definitions (hubs—large, medium, small, non-hubs, and GA) does not accommodate the issues facing airports in NextGen very well, this chapter will develop an approach to defining small airports that is derived from the NextGen capabilities that are most likely to be of interest to smaller airports, whether medium/small/non-hubs or GA airports.

Chapter 6—Role of Airports in NextGen Development, Implementation, and Community Outreach

This chapter provides insight to airport planning practitioners into how to identify and engage the key individuals or interests who may need additional outreach to assure effective communication on NextGen issues, as well as some recommended best practices to assist with this problem.

Appendices

The following appendices have been included to provide additional detail on the information and concepts introduced in the foregoing chapters.

Appendix A—ACRP NextGen Initiative Projects

Appendix B—Best Practices and Lessons Learned from Airport Case Studies

Appendix C—NextGen Elements and Applicability by Airport Sizes and Issues

Appendix D—NextGen Resources, Organizations, and Contacts

Appendix E—Relevant Airport Planning and NextGen References and Guidance Documents

Appendix F—NextGen—Airport Planning and Development List of Acronyms

The final technical report of Project 03-33 includes additional technical details on the research conducted, data collected, and contacts made with the industry during the project.

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2 NextGen Technologies and Operational Improvements

Chapter 2 describes NextGen technologies and operational capabilities of interest to airport planning practitioners and examines how they could impact and change the airport planning and development process. The NextGen capabilities identified in this chapter are the result of the ACRP 03-33 team's research efforts into how these capabilities could functionally impact airport operations and the relationship between timing of implementation of the technology, the airport planning process, and strategic airport expansion decisions. The technology discussion is organized to establish the near-term NextGen-enabling technologies, existing and emerging NextGen technologies, and emerging non-NextGen technologies likely to be implemented over the next 20 years. Each of these sections includes a breakdown of how these technologies may affect airports in the near- and mid-term (0 to 5 years), and the long-term (5 years and beyond) planning horizons as defined in the FAA FACT3: Airport Capacity Needs in the National Airspace System publication (www.faa.gov/airports/planning_capacity/media/FACT3-Airport-Capacity-Needs-in-the-NAS.pdf). These planning horizon definitions differ from those used for typical airport planning efforts. Airport planning practitioners should correlate the timing of the technologies presented in this chapter to the typical short-term (0 to 5 years), mid-term (5 to 10 years), and long-term (10 to 20 years) master planning horizons.

Near-Term Capabilities That Are Enablers of the NextGen Capabilities

This section is focused on core NASA-wide technologies, including automatic dependent surveillance-broadcast (ADS-B), Data Communications (Data Comm), System Wide Information Management, and NextGen Weather, that are enablers of the other more specific FAA NextGen operational improvements and technologies to be discussed in the next section. These core capabilities will enable the other technologies and efficiencies to be developed and implemented at airports throughout the nation. Later in this chapter, the research team will identify other enablers of specific NextGen operational improvements and technologies.

Automatic Dependent Surveillance-Broadcast (ADS-B)

ADS-B is a precise satellite-based surveillance system that uses GPS (global positioning system) technology to determine an aircraft's location, airspeed, and other data, and broadcasts that information via ADS-B Out capabilities integrated into the aircraft transponders to a network of ground stations. The ground stations relay the data via ADS-B Out to air traffic control displays and to nearby aircraft equipped to receive the data via ADS-B In. Operators of aircraft equipped with ADS-B In can also receive weather and traffic position information delivered directly to the cockpit.

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ADS-B Out will emerge as the core surveillance tool in the future. Although aircraft equipage is required, FAA has mandated a requirement for all aircraft operating in certain airspace. Under the rule, ADS-B Out performance will be required to operate in:

1. Class A, B, and C airspace.
2. Class E airspace within the 48 contiguous states and the District of Columbia at and above 10,000 feet mean sea level (MSL), excluding the airspace at and below 2,500 feet above the surface.
3. Class E airspace at and above 3,000 feet MSL over the Gulf of Mexico from the coastline of the United States out to 12 nautical miles.
4. Around those airports identified in 14 CFR part 91, Appendix D.

The rule requires all aircraft operation in the designated airspace to be equipped with ADS-B Out by January 1, 2020. As of August 2016, more than 15,000 GA aircraft and 650 commercial aircraft have been equipped with ADS-B Out avionics. These numbers represent approximately 18 percent of the IFR (instrument flight rules) GA fleet and approximately 33% of the commercial fleet in the U.S. ADS-B ground infrastructure will be added along Mexico's Yucatan Peninsula in 2016 and 2017, providing increased coverage over the Gulf of Mexico.

The additional surveillance accuracies of ADS-B Out will, in combination with PBN and Wake RECAT, enable more efficient operations in offset, parallel, simultaneous (dual and triple) configurations in the future. Implementation of ADS-B In and cockpit display of traffic information (CDTI) will, ultimately, provide pilots with the ability to maintain their own separation from other aircraft, which will enable increased efficiencies for operations in the en route, terminal, and arrival environments, especially for closely spaced parallel runways.

Data Communications (Data Comm)

The Data Comm program is a key element in the implementation of NextGen because it is the first phase of the transition from the current analog voice system to full digital communication. Completion of this transition is considered necessary to handle the projected increases in traffic over the next decade. Data Comm will initially enable increased capacity and efficiencies for ground movements at airports in the NAS, while reducing voice congestion. Data Comm will be implemented in the en route and terminal environments as the technology matures enabling aircraft re-route required navigation performance (RNP) procedures to be uploaded directly from ATC to the aircraft flight management system (FMS). This technology combined with NextGen weather predictive tools and traffic flow modernization technology, will enable controllers to manage weather events impacting en route and terminal flight paths.

System Wide Information Management (SWIM)

SWIM is an information-sharing platform that will enable increased common situational awareness throughout the NAS. It relies on a standard data format so information from unrelated computer systems may be shared efficiently, which will enable airline operators, air traffic controllers, and airports to share information in near real time. It provides the basic design for all new data-sharing systems in the NAS. This information-sharing platform offers a single point of access of aviation data from multiple sources. Data sharing from airport GIS systems combined with PBN, ADS-B, Data Comm, and NextGen Weather data will serve to reduce system error and increase efficiencies in all flight domains. SWIM will provide airports with access to real-time information from the ATC system and airlines.

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NextGen Weather: The 4D Weather Cube

According to FAA, weather accounts for about 70% of traffic delays in the NAS. The FAA is working in conjunction with the National Oceanic and Atmospheric Administration (NOAA) and the National Weather Service (NWS) on the 4D Data Cube, which is formed by merger of model data, automated gridded algorithms, climatology and observational data, and meteorologist input/data manipulation to ensure consistency and accuracy. The 4D weather cube will contain:

- Continuously updated weather observations (surface to low earth orbit, including space weather and ocean parameters).
- High resolution (space and time) analysis and forecast information (conventional weather parameters from numerical models).
- Aviation impact parameters necessary for initial operational capability (IOC), including:
  - Turbulence,
  - Icing,
  - Convection,
  - Ceiling and visibility, and
  - Winds (surface and aloft).

It provides a common weather picture for NAS participants (Airlines, DOD, FAA, and airports), and is expected to enable controllers and operators to develop more reliable flight plans, make better decisions, and improve on-time performance. These increased capabilities are expected to result in less weather delay, less frequent flight cancellations and refueling stops, and more dependable flight schedules.

En Route Automation Modernization (ERAM)

As of March 27, 2015, ERAM has replaced the 40-year-old En Route Host computer and backup system used at 20 FAA Air Route Traffic Control Centers nationwide. ERAM technology supports the transition from a ground-based system of air traffic control to a satellite-based system of air traffic management and also is the foundational platform required for FAA to enable SWIM, Data Comm, and ADS-B. Also, with ERAM, controllers benefit from increased collaboration and seamless data sharing between Centers.

ERAM enables controllers to digest data from 64 radars compared with just 24 for HOST radar systems, allowing them the ability to see beyond their specific center, therefore handling traffic more efficiently. Additionally, ERAM can track 1,900 aircraft at a time, whereas the computer HOST could only track 1,100 aircraft. ERAM has eliminated the need for the FAA HOST computer and consolidates multiple functions of previous en route computer systems.

Existing and Emerging NextGen Technologies

The descriptions of NextGen technologies included in this section were culled from several FAA and industry documents, including (1) the FAA's October 2014 Report to Congress titled *NextGen Priorities Joint Implementation Plan*, (2) the May 2015 *NextGen Implementation Plan*, (3) various pages of the FAA's NextGen website (https://www.faa.gov/nextgen/), and (4) various reports and presentations

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prepared for the NextGen Advisory Committee (NAC) established under the auspices of RTCA. In addition, many of these technologies are included in the case studies conducted by the research team, which are documented in Appendix B of this guidebook.

The five FAA programs described in the previous section provide the building blocks for multiple NextGen capabilities that may provide benefit for independent, dependent, and closely spaced parallel runway operations; terminal area navigation and operations; surface movements; landing systems; separation management; and surveillance. Table 2-1 highlights the specific applications and capabilities these NextGen technologies offer by function for airports and identifies the scheduled timing of the technology based on a near-, mid-, or long-term planning horizon.

Table 2-1. NextGen capabilities related to airports.

FUNCTION	CAPABILITY	TIME FRAME
Multiple Runway Operations (MRO)-Independent Runways	Wake Turbulence Recategorization (Wake RECAT)	NM
	Phase I (Aircraft Grouping Reclassification)	NM
	Phase II (Aircraft specific)	NM
MRO-Dependent Runways	Wake Turbulence Avoidance Procedures	NM
	Wake Turbulence for Parallel Runways (<2,500′ spacing—small/large leading)	NM
	Wake Turbulence Mitigation for Arrivals-Procedures (<2,500′ spacing—B757/heavy leading)	NM
	Wake Turbulence Mitigation for Departures (WTMD) (Upwind runways)	NM
MRO-Closely Spaced Parallel Operations	Dual Independent Parallel Operations (>3,600′ spacing)	NM
	Dual Dependent Parallel Operations (2,500′–3,600′ spacing)	NM
	Triple Dependent Parallel Operations (>3,900′ spacing)	NM
	Dual Independent Parallel Operations with Offset (>3,000′ spacing)	NM
	Dependent Parallel Operations (>4,300′ spacing)	NM
	RNP Parallel Approaches with Transition (RPAT)	L
	Paired Approaches	L
Performance-Based Navigation (Terminal Area)	Lateral Navigation (LNAV)	NM
	LNAV/Vertical Navigation (VNAV)	NM
	RNP	NM
Surface Operations & Data Sharing	Collaborative Decision Making/Terminal Flight Data Management (TFDM)	NM
	SWIM	NM

continued

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Table 2-1. Continued

FUNCTION	CAPABILITY	TIME FRAME
Improved Landing Systems	LPV (Localizer Performance with Vertical Guidance)	NM
	LP (Localizer Performance)	NM
	LNAV	NM
	LNAV/VNAV	NM
	GBAS	NM
	CAT I Approaches (multiple runway ends)*	NM
	CAT II/III Approaches (multiple runway ends)	NM
Separation Management	Advanced Technologies and Oceanic Procedures (ATOP)	NM
	Terminal Automation Modernization and Replacement (TAMR)	NM
	ERAM	NM
	Established on RNP (EoR)	NM
	Equivalent Lateral Spacing Operations (ELSO) National Standard (reduced divergence angle)	NM
	Unified Departure Operational Spacing (UDOS)	NM
	CDTI Based Separation	L
Multilateration (Surveillance)	Wide Area Multilateration (WAM)—En Route and Terminal Airspace	NM
	Airport Surveillance Detection Equipment-Model X (ASDE-X), ASDE-3/Airport Movement Area Safety System (AMASS), Airport Surface Surveillance Capability (ASSC)	NM
ADS-B (Surveillance)	Traffic Information Services Broadcast (TIS-B)	NM
	Automatic Dependent Surveillance-Rebroadcast (ADS-R)	NM
	Flight Information Services Broadcast (FIS-B)	NM
	CDTI	L

Notes: *Currently available for Category (CAT) I instrument landing systems (ILSs) as a non-federal system. Research and development ongoing for CAT II/III operations.
NM = Near- or mid-term implementation, 0–5 years.
L = Long-term implementation, 5 years and beyond.

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Near- and Mid-Term NextGen Capabilities with Benefits for Airports

The following sections highlight the near and mid-term NextGen capabilities and describe their effects on airport planning and development.

Wake Turbulence Recategorization

Through research conducted by the FAA, the U.S.DOT Volpe National Transportation Systems Center, the aviation industry, and EUROCONTROL (the 39-nation European Organization for the Safety of Air Navigation) experts found that regrouping aircraft types according to similarities in their wake turbulence characteristics, rather than by maximum gross takeoff weights, allows safe reduction in separation between aircraft, which will increase efficiency and airport capacity.

The FAA previously used five wake turbulence separation categories based primarily on weight: Super, Heavy, B757, Large, and Small. Wake RECAT applies advances in knowledge of wake vortex characteristics of the breadth of aircraft types in terms of both the strength of the wake vortices generated by each aircraft type and the ability of each aircraft type to withstand a wake encounter. Using the results of this data collection and analysis program, the FAA Wake Turbulence Office has changed wake turbulence separation minima from the current standard to a new standard, which involved placing aircraft into six categories (labeled A–F) for defining minimum separations between each lead-trail pair of the six aircraft types.

These new separation minima recognize that there are significant differences in the wake characteristics of the various aircraft previously classified as "Heavy" jets. Similarly, there are significant differences in the way characteristics of the various aircraft previously classified as "large."

Phase 2 of Wake RECAT will determine the optimal wake separation between any two given aircraft from a sample of 100 types that represent 99 percent of the global traffic. Wake RECAT Phase 2 is planned for completion in fiscal year 2017.

Effects on Airport Planning and Development

Wake RECAT could affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

These new separation minima could result in significant improvements in maximum arrival and departure throughput at airports that have significant numbers of heavy Jets and B757s in their aircraft-fleet mix.

In November 2012, the FAA implemented the new wake separation standards at Memphis International Airport. Since then FedEx has experienced an increase in airfield capacity of 20% at Memphis. The airline burns 4.2 million fewer gallons of fuel each year and emits 39,992 fewer metric tons of CO_2. FedEx also saves an average of 3.3 minutes per flight in taxi-out time and 2.6 minutes per flight in aircraft delay. Airport planners should be cautioned that the benefit of increased throughput is heavily dependent on the fleet mix and specific conditions at their individual airports. At some airports the increase in throughput due to Wake RECAT is on the order of 4%.

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FACILITY REQUIREMENTS

Wake RECAT is strictly a change in the ATC separation rules. No additional facility requirements are needed on the ground, and no additional equipment is needed in the airplane. Wake RECAT is supported by advanced decision support tools and displays for air traffic controllers, but these tools had been developed before Wake RECAT.

ALTERNATIVES

Wake RECAT would not affect the analysis of alternatives at an airport except that the FAA Wake Turbulence Office has been asking representatives of both airports and local FAA air traffic whether they would be interested in implementing Wake RECAT at their airport. In some cases, the airport and local ATC representatives have decided not to participate in the Wake RECAT program, most likely because they thought it would not make much difference because of the aircraft fleet mix at the airport, particularly if it did not include a significant number of heavy jets or B757s.

Another reason for declining to participate in the Wake RECAT program has been the existence of severe departure airspace constraints at an airport due to limited departure headings or interactions with other traffic. In such cases, these departure airspace constraints govern the achievable departure capacity, not the existing wake turbulence separations.

TIMING AND IMPLEMENTATION

- Memphis was the first airport where Phase 1 of Wake RECAT separation standards were implemented. Below is the implementation schedule for subsequent and Phase 2 Wake RECAT installations:
- January 2015—Current RECAT airports [Memphis International Airport (MEM), Louisville International Airport (SDF), Cincinnati/Northern Kentucky International Airport (CVG), and Hartsfield-Jackson Atlanta International Airport (ATL)].
- March 2015—New York Metroplex [John F. Kennedy International Airport (JFK), LaGuardia International Airport (LGA), New York/Newark Liberty International Airport (EWR), Teterboro Airport (TEB), Westchester County Airport (HPN), and Long Island MacArthur Airport (ISP)].
- March 2015—Charlotte Metroplex [Charlotte Intenational Airport (CLT)].
- June 2015—Chicago Metroplex [Chicago O'Hare International Airport (ORD) and Chicago Midway International Airport (MDW)].
- September 2015—Northern California Metroplex [San Francisco International Airport (SFO), Oakland International Airport (OAK), and San Jose International Airport (SJC)] and Ted Stevens Anchorage International (ANC).
- March 2016—Southern California Metroplex [Los Angeles International Airport (LAX)].
- Fiscal Year 2017—Minneapolis-Saint Paul International Airport, Wold-Chamberlain Field (MSP), Miami International Airport (MIA), and Washington Dulles International Airport (IAD).

Closely Spaced Parallel Runways

There are multiple airports in the NAS with closely spaced parallel runways. These runways can be constrained in low weather conditions because of current runway separation standards. Efforts are underway to apply NextGen PBN, GBAS, and ADS-B technologies to enable increased capacity during marginal visual flight rules (VFR) and IFR conditions for closely spaced parallel runways served by precision instrument approach procedures with vertical guidance. This could include instrument landing system (ILS) approaches, RNAV/RNP approaches, or RNAV (GPS) approaches such as localizer performance with vertical guidance (LPV) approaches. Below is a summary of those efforts:

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- FAA Order 7110.308 (http://www.faa.gov/documentLibrary/media/Order/JO%207110.308.pdf), 1.5-Nautical Mile Dependent Approaches to Parallel Runways Spaced Less Than 2,500 Feet Apart, allows a reduction in the required wake separations for dependent operations for runways spaced less than 2,500 feet apart when small or large category aircraft are leading in the dependent pair.
- Dual Independent Parallel Operations: This capability allows dual simultaneous operations for runways spaced greater than 3,600 feet.
- Dependent Parallel Operations Between 2,500 Feet and 3,600 Feet: This capability reduces the dependent stagger separation from 1.5 nautical miles (nm) to 1.0 nm for runways separated by more than 2,500 feet.
- Triple Independent Parallel Operations: This capability allows triple simultaneous operations for runways spaced greater than approximately 3,900 feet.
- Dual Independent Parallel Operations with Offset: This capability allows dual simultaneous operations with the use of an offset for runways spaced greater than approximately 3,000 feet.
- Dependent Parallel Operations for Runways Greater than 4,300 Feet: This capability reduces the dependent stagger separation from 2.0 nm to 1.5 nm for runways greater than 4,300 feet.
- WTMD: When wind conditions allow, WTMD permits any aircraft to depart from the "upwind" runway without waiting two or three minutes after Heavy or B757 aircraft depart the "downwind" runway. This technology is specific to San Francisco International Airport (SFO) and will remain for further testing of dependent concepts.
- Wake Turbulence Mitigation for Arrivals-Procedures: This capability allows a reduction in required wake separations for dependent operations for runways spaced less than 2,500 feet apart when Heavy or B757 aircraft are leading in the dependent pair.

The foregoing programs directly related to closely spaced parallel operations (CSPO) are illustrated in Figures 2-1 and 2-2 below:

An additional concept for RPAT has been developed by NextGen working groups and is in the process of implementation. The concept is based on the use of RNP to ensure separation in a dual simultaneous arrival scenario. Figure 2-3 highlights the RPAT concept.

Effects on Airport Planning and Development

CSPO concepts combined with Wake RECAT could affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

The foregoing rules affecting wake turbulence separations and the spacing requirements for CSPO could enable significant capacity increases at airports that (1) have significant volumes of Heavy jet aircraft, (2) already have closely spaced parallel runways that could be used more efficiently under the new rules, or (3) could implement closely spaced parallel runways on available airport property and provide either more efficient independent or dependent parallel operations that could not otherwise have been provided within the available space.

FACILITY REQUIREMENTS

CSPO are currently conducted very efficiently at airports in visual meteorological conditions (VMC) but experience a significant capacity drop in instrument meteorological conditions (IMC) because they do not meet the current spacing requirements for dependent or independent approaches. The

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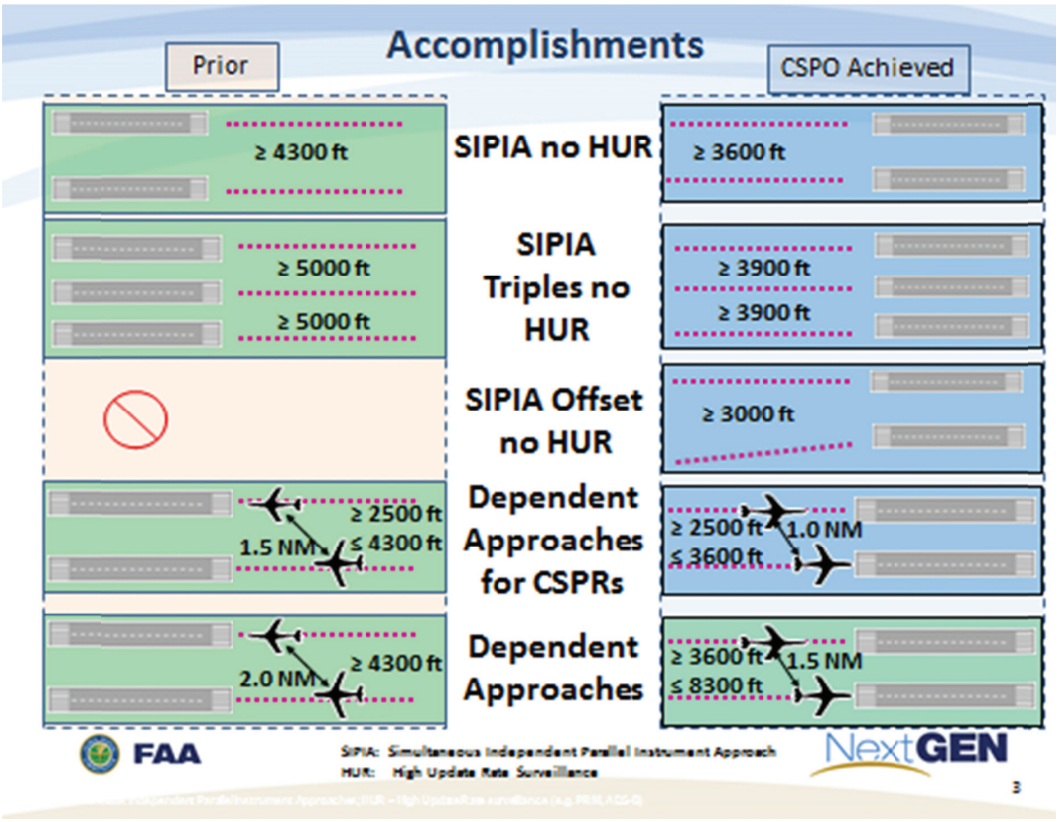


Figure 2-1. *Expected changes in CSPO independent separation standards.*

foregoing potential rule changes affecting spacing requirements for CSPO assume that the associated parallel runways would be served by precision instrument approach procedures with vertical guidance, which could include ILS approaches, RNAV/RNP approaches, or RNAV (GPS) approaches such as LPV approaches and adequate terminal radar approach control (TRACON) automation. Advanced concepts will require a high-update radar or fused ADS-B surveillance, NextGen-enabling technology, and advanced FMS capabilities.

ALTERNATIVES

The potential changes in the rules for the minimum spacing between parallel runways required for dependent or independent approaches could open up new opportunities at airports that previously would not qualify for such approaches. Therefore, new alternatives could be considered in the airport planning and development process for such airports. Such alternatives would have to be evaluated in terms of how the spacing between the parallel runways could facilitate the development of passenger terminal facilities between those runways and also how the new capabilities could affect noise exposure in the surrounding communities and operations at other nearby airports.

TIMING AND IMPLEMENTATION

CSPO changes focusing on dependent runways with separations greater than 2,500 feet and less than 3,600 feet have been implemented into FAA Order 7110.65W as of December 2015 primarily. FAA has plans to implement these capabilities at multiple airports by 2017. Further enhancement of offset, simultaneous triple, and independent operations capability will extend into the mid-term. Advanced

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concepts to include RPAT and paired approaches will likely be implemented in the long-term horizon. The most recent anticipated timing for the foregoing changes in the rules associated with CSPO is summarized in the FAA's *NextGen Priorities Joint Implementation Plan*.

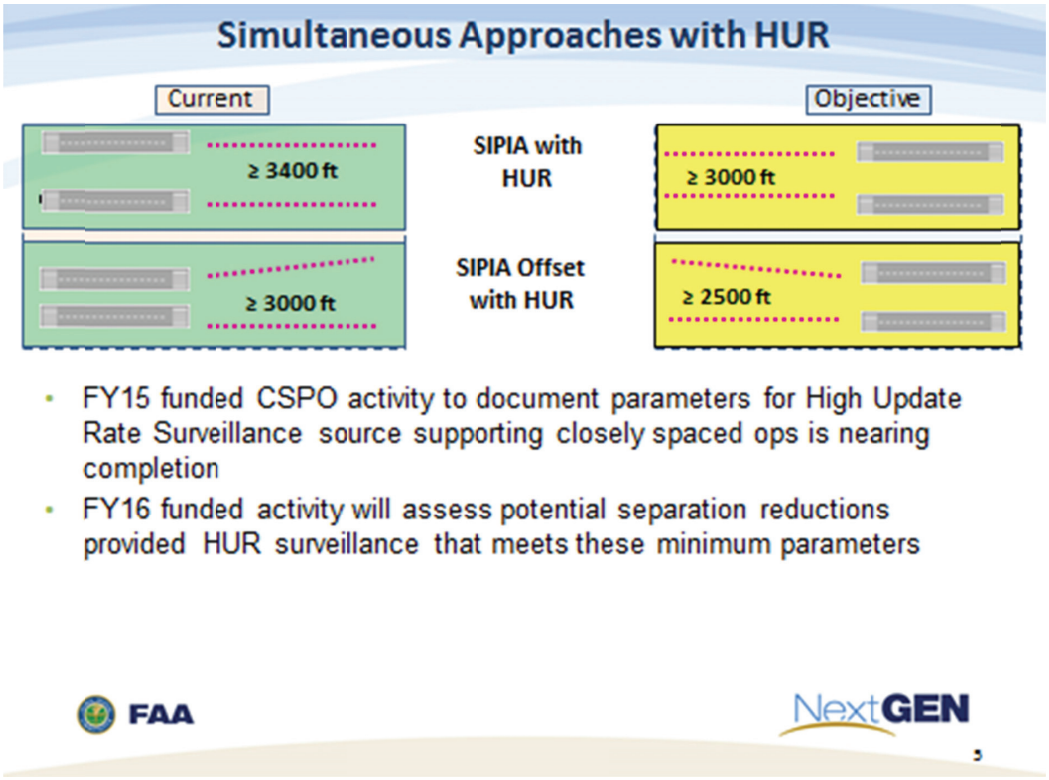


Figure 2-2. *Expected changes in CSPO dependent separation standards.*

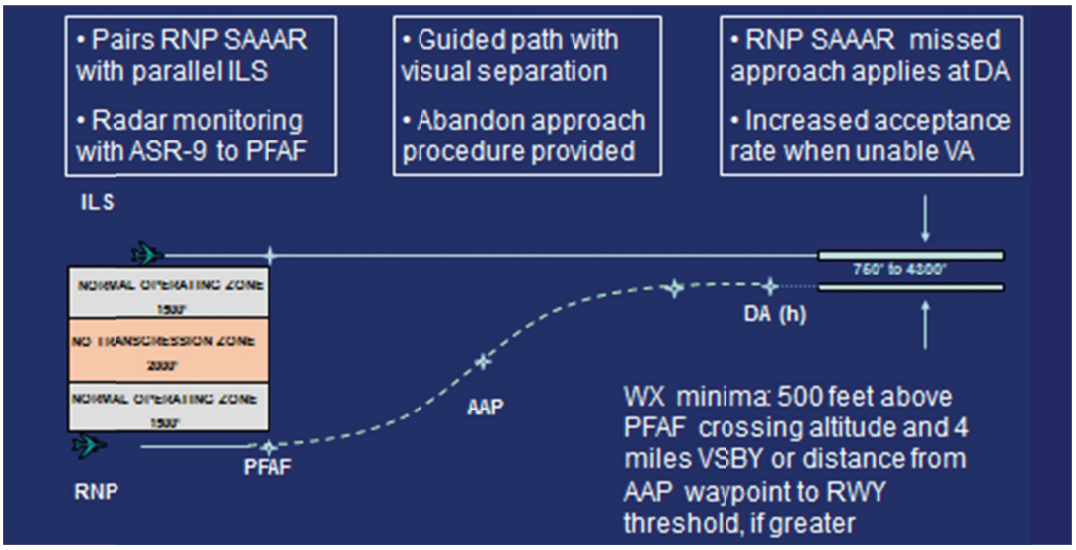


Figure 2-3. *RPAT approach.*

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PBN

PBN refers to instrument flight rules and procedures that primarily use satellite-based navigation and on board aircraft equipment to navigate with greater precision and accuracy than with electronic ground-based navigation. PBN defines the performance requirements for routes and procedures enabling aircraft to navigate directly to points in space. Performance requirements include the navigation accuracy, integrity, continuity, and functionality requirements for an aircraft to operate in the NAS. PBN may define required aircraft- and ground-based sensors and equipment. The aircraft's FMS is typically the primary tool for conducting PBN operations.

Two key components of PBN are area navigation (RNAV) and RNP. RNAV is a broad term that refers to flight paths within the limits of space-, ground-, or aircraft-based navigational aids (NAVAIDs). RNAV is split into two categories: LNAV and lateral and vertical navigation (LNAV/VNAV). LNAV has lateral guidance only based on GPS or distance measuring equipment (DME) navigational input. LNAV/VNAV systems use the same inputs for lateral navigation as well as barometric sensing for vertical guidance. RNAV enables the aircraft to follow the route of flight with a certain degree of lateral and/or vertical precision. RNP extends LNAV/VNAV capability with aircraft onboard monitoring of its navigation performance and alerting to the flight crew if the required precision cannot be met.

PBN is leveraged to design instrument flight procedures (IFP) for the departure, en route, arrival, and approach phases of flight. IFP comprise lateral routes, altitude restrictions, speed restrictions, and other specifications for aircraft guidance.

PBN procedures most relevant to airports include SID procedures from the airport, and standard terminal arrival route (STAR) procedures and standard instrument approach procedures (SIAP) to the airport. PBN procedures provide a foundation for flight paths, airspace design, route separation, and obstacle clearance. RNAV procedures provide routing flexibility, efficient flight paths, and airport access in congested airspace or bad weather. RNAV LNAV, RNAV LNAV/VNAV, and RNAV LP may achieve non-precision minimums (as low as 250 feet above the threshold). RNP procedures increase airport access in bad weather and mountainous terrain and help procedurally separate multi-airport traffic, particularly with curved paths, achieving non-precision minimums (as low as 250 feet above the threshold). PBN arrival procedures enabling optimized profile descent (OPD) reduce fuel burn, emissions, and pilot-controller communications. There can be additional efficiency benefits from new PBN procedures closer to airport.

The RNAV-enabled ELSO national standard capitalizes on the increased navigational precision of RNAV departure operations to provide a reduced divergence angle while maintaining the established minimum lateral spacing between departure paths. This capability, depicted in Figure 2-4, is currently in use at ATL and is being considered at other major airports including Seattle-Tacoma International Airport (SEA) and Denver International Airport (DEN). The capability could enable substantial increases in departure rates at airports that currently have difficulty meeting the minimum diversions requirements for efficient successive departures.

Two other RNAV-enabled departure procedures also offer the promise of increasing the departure capacity of single or multiple runway operations: UDOS and established-on-departure operation (EDO). Both UDOS and EDO (like ELSO) are intended to enable departure divergence at locations that previously could not provide the divergence necessary for efficient departure throughput. UDOS and EDO are intended to provide such divergence at points several miles after takeoff at a defined divergence point. Applications of these new systems are expected to reduce in-trail spacing in the terminal and transition airspace. Figure 2-5 highlights the UDOS capability.

The effects of PBN procedures have caused significant concerns over increased noise exposure at some airports where they have been implemented. The concentration of flight paths over a narrow geographic area, which is a characteristic of RNP, has caused an increase in noise exposure and significant

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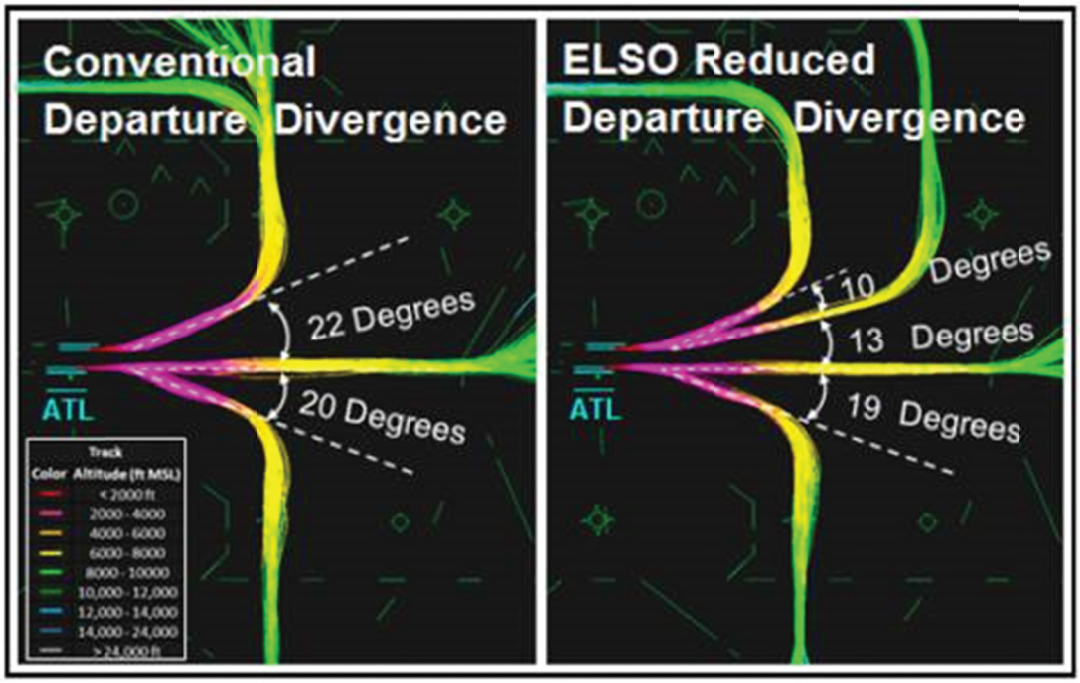


Figure 2-4. *ELSO.*

adverse community reaction and litigation in several PBN implementations. The current environmental screening methods used by the ATO do not capture noise effects below the DNL 65 level, which has been established as the federal threshold of significance for airplane noise by FAA. However, noise impacts below that DNL value can provoke significant adverse community reaction. On the plus side, PBN procedures following precise flight paths can be designed to avoid noise sensitive areas and possibly provide multiple departure headings.

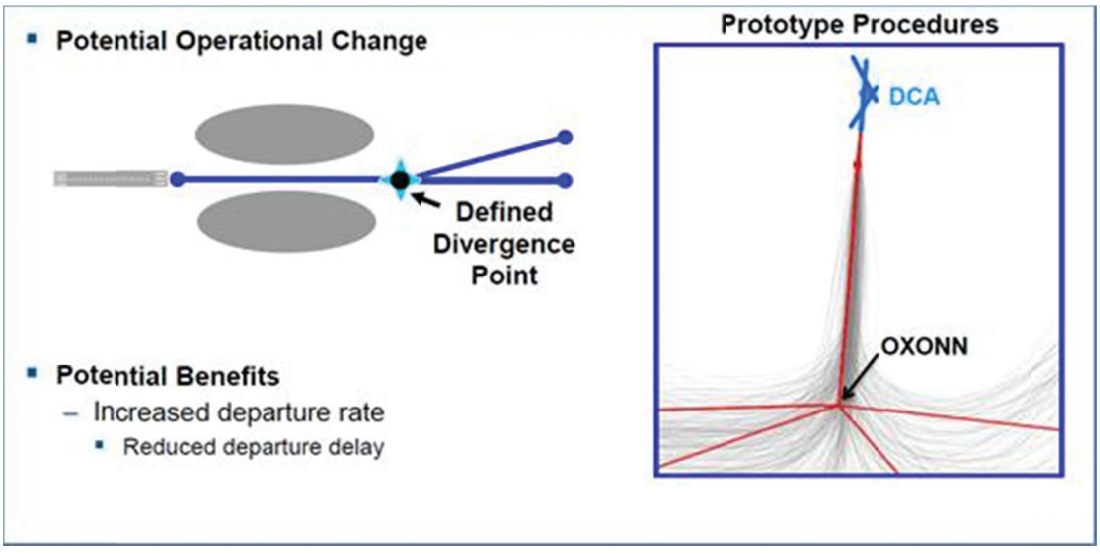


Figure 2-5. *UDOS.*

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The FAA's plans call for PBN procedures to be implemented at all 35 Operational Evolution Plan (OEP)/Core 30 airports, 35 Non-OEP/Non-Core airports, and other airports throughout the NAS. The FAA has also undertaken the implementation of PBN procedures through its Metroplex initiative, 3rd-Party Vendor Process, and other case-by-case projects. The FAA's Metroplex program calls for design and implementation of PBN procedures in the following Metroplexes throughout FY11–FY18: Washington D.C., North Texas, Houston, Charlotte, Atlanta, Northern California, Southern California, Florida, Phoenix, Chicago, Memphis, Cleveland/Detroit, Las Vegas, and Boston. Airport planners should be aware that PBN procedures implemented as part of a Metroplex project will entail a full EA (environmental assessment) and public process, while PBN procedures adopted as local initiatives may not afford that level of public process unless the airport intercedes and requests an EA by citing the potential for significant adverse public opposition on environmental grounds (CEQ regulations at 40 CFR 15056(c) 1 and 2).

Effects on Airport Planning and Development

PBN can affect airport planning and development through increased reliability of approaches and efficiency of operations (positive functional and economic benefits). Moreover, PBN combined with other NextGen technologies, such as ADS-B Out, could be an enabler of other potential operational improvements that could ultimately reduce the required separation between parallel runways and the required minimum separations between successive aircraft on approach or departure. Nevertheless, PBN has become controversial because of its potential concentration of flight tracks over noise sensitive areas near an airport.

PBN could significantly affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

PBN combined with improved TRACON and surveillance and automation system provides controllers the ability to sequence aircraft into tighter intervals and better control traffic flow. On the negative side, the concentration of flight paths over a narrow band can cause an increase in noise exposure and community reaction close into the airport. With the tighter tolerances, controllers will be able to significantly reduce delay across the entire system and at the local level. The PBN systems will allow aircraft to be continuously monitored and tracked via new flight management technologies therefore increasing situational awareness for both pilots and controllers alike. The airlines will see an added benefit in fuel cost, reserve fuel, and aircraft operating expenses as a result of increased efficiency (direct routing) across the NAS. An additional added benefit is reduced aircraft emissions and reducing individual aircraft carbon footprints.

Weather rerouting will also be improved for pilots and controllers with FAA staff having the ability to open and close departure and arrival routes in real time. The majority of aircraft delays originate on the East Coast of the United States, often due to convective weather or periods of low visibility and ceilings. NextGen allows better sequencing for aircraft and the ability to route traffic in narrower pathways alleviates stress on the system from delayed aircraft. High-precision GPS technologies may also reduce cancellations and diversions. Some airports may see extra departures per hour from the increase in available departure routes. Others may see an increase in capacity derived through better efficiencies in the terminal area or through better approach minimums, which may drive the need for additional hold and parking areas on the airport to manage terminal bottlenecks.

FACILITY REQUIREMENTS

It is FAA's policy not to add new systems to the network of ILSs in place. Airports should plan for PBN approaches. Airports will generally not have to add any facilities to take advantage of these advanced

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PBN procedures. However, they may assume responsibility for increased obstacle clearance if the procedures result in lower minimums.

ALTERNATIVES

When airport planners are looking at alternative development scenarios, the impact of PBN could have a material effect on how those alternatives stack up. For example, one of the case studies presented in Section 9 (for Friedman Memorial Airport) actually dealt with an RNP approach and its resultant benefits compared to relocating the airport entirely for purposes of increased reliability.

TIMING AND IMPLEMENTATION

The PBN program is well underway and new applications are emerging each year. Implementation has occurred through collaborative local initiatives as well as the FAA Metroplex program efforts including ATL, Ronald Reagan Washington National Airport (DCA), DEN, CLT, IAD, McCarran International Airport (LAS), John Wayne Airport (SNA), and SEA.

PBN has begun its rollout across the United States with implementation across a wide range of airport sizes and functions. New procedures are being developed and implemented for airports continuously and aircraft equipage is forecast to increase. Figure 2-6 depicts the current equipage status for the U.S. air transportation and air taxi fleet.

Significant efforts are required for the development and implementation of PBN. The FAA and airlines have dedicated personnel who continuously are developing and flight checking IFP. Airports have used consulting services associated with PBN development, modeling, and implementation as well.

PBN requires avionics capable of receiving and transmitting precise GPS signals to determine the aircraft's spatial reference. Air carriers absorb significant costs to upgrade existing systems and outfit cockpits for PBN avionics. Airports incur very little, if any, capital costs and are generally not required to locate navigational aids on-field. Some local augmentation systems will be required for ground sensing, though most of the navigation system will be guided through satellite-based systems.

The average cost of equipping aircraft with NextGen avionics systems is between $35K and $140K per aircraft for basic systems. Advanced systems may cost $500K or more. (Source: www.washingtonpost.com).

Surface Operations and Data Sharing/Collaborative Decision Making/SWIM

Improved surface operations will improve safety, efficiency, and flexibility on the airport surface by implementing new traffic management capabilities for pilots and controllers using shared surface movement and en route data. The capabilities address surface movement and the exchange of information between controllers, pilots, and air traffic managers that occur from before the aircraft pushes back from its gate up to the departure of the aircraft from the airport, and for landing traffic, from exiting the runway to arriving at the terminal gate.

The sharing of data has enabled the establishment of collaborative decision making (CDM) efforts between FAA and industry stakeholders and between stakeholders at individual airports. Federal CDM groups follow guidelines established in the Collaborative Decision Making Leadership, Strategies, Structure, and Guidelines V 4.0 document. CDM is enabled through the dissemination of traffic flow data management and other SWIM-enabled data to CDM group to enable:

- Improved real-time decision making,
- Tools and procedures to be established to enable air navigation service providers and the flight operators to more easily respond to changing conditions, and

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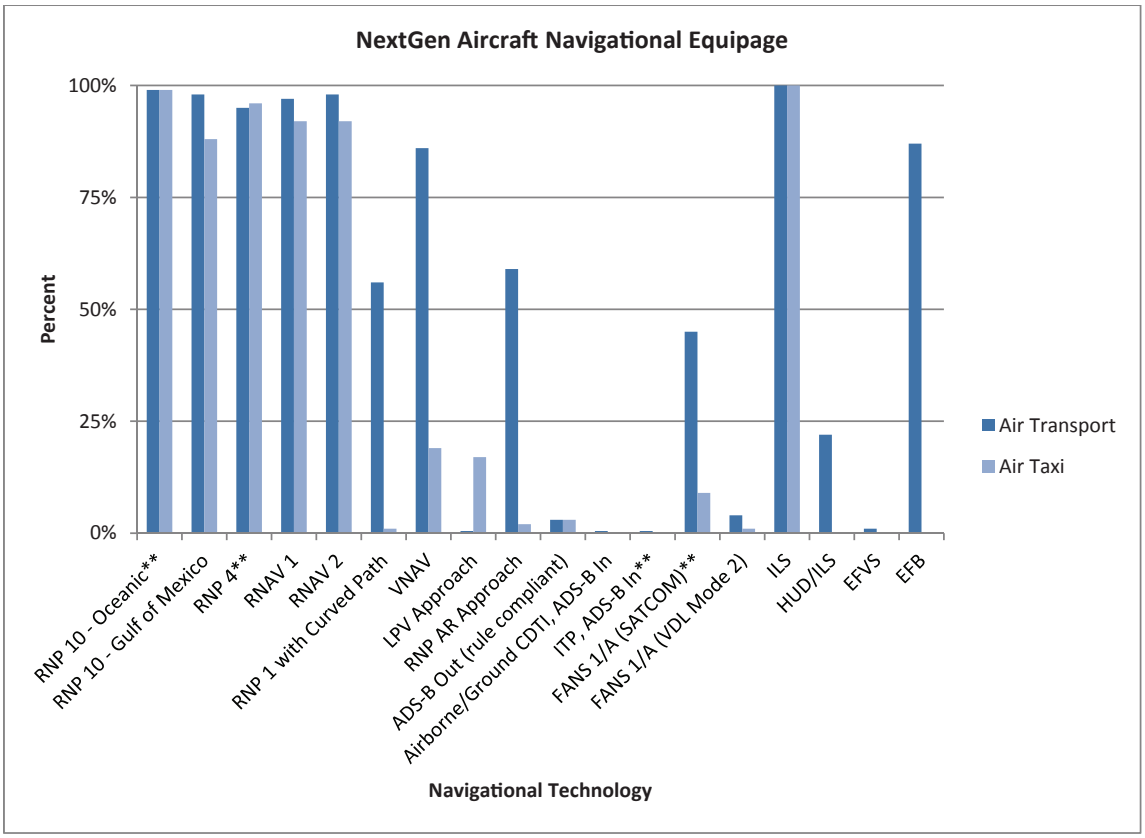


Figure 2-6. *Current NextGen aircraft equipage.*

Notes:

1. Source: FAA.gov, March 10, 2015
2. ** Indicates oceanic capable aircraft equipage
3. Air Transport includes FAR Part 121 Operators. Air Taxi includes FAR Part 91 and 135 operators.
4. Further detail included in ACRP 03-34: NextGen - The Airport's Role in Performance-Based Navigation.

RNP AR = RNP authorization required, ITP = in-trail procedures, FANS = future air navigation system, HUD = heads-up display, EFVS = enhanced flight vision systems. EFB = electronic flight bag.

- Advanced technological solutions that evolve the NAS and influence global CDM for all stakeholders.

Surface operation and CDM data sharing technology track the movement of surface vehicles and aircraft, incorporating the movement data into the airport surveillance infrastructure and sharing the information with controllers, pilots, and airline operations managers. This is being done as part of the Terminal Flight Data Modernization Program which aims to integrate electronic flight data, traffic flow management data, and CDM for the replacement and consolidation of multiple NAS systems. In spring of 2016, FAA awarded a 13-year contract for the development and implementation of TFDM into the NAS.

Effects on Airport Planning and Development

Surface operations, data sharing, and CDM could significantly affect the following major elements of airport planning and development programs:

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DEMAND/CAPACITY

Benefits include increased flexibility, efficiency, safety, and operational validation. These potential benefits are further described below:

Flexibility

- Improve the timely exchange of data to enable aircraft operators to more accurately adjust their departure and arrival times for the most efficient use of available runways, taxiways, and gates.
- Permit taxi operations that support improved flows for takeoff, improving surface efficiency.
- Reduce the effect of weather related delays.

Efficiency

- Enabling more effective scheduling in response to runway, departure fix, and traffic flow management ground-management constraints, with automatic reassessment and update of the departure schedule.
- Assisting in assuring optimized use of available capacity so that all of the available departure throughput capability can be used by aircraft.
- Enhancing the ability to react to changing airport conditions, such as severe weather, by issuing digital pre-departure clearances, including routing revisions, using Data Comm.
- Improving awareness of surface congestion at major hub airports, greatly streamlining the coordination of corrective action, and improving the resilience of the system.
- Reducing fuel burn, operating costs, emissions, and surface congestion related to long departure queues.
- Reducing delays by improving event data quality and adherence to controlled departure times.
- Reducing FAA operating costs through the use of automated flight strips.

Safety

- Enhance safety on the airport surface by improving pilot and controller awareness of surface traffic.
- Surface movement data can be used to support the safety risk management processes by providing a means to calculate the location, type, and duration of an aircraft operation on any part of the airfield. These calculations may provide a basis for determining the probability or likelihood of a hazard incident.

FACILITY REQUIREMENTS

One of the key elements of improved surface operations is departure metering to reduce the departure queue and thereby fuel burn, emissions, and surface congestion. However, effective departure metering requires ample apron area, gates, and hold pads, which many congested airports have difficulty providing. Therefore, improved surface operations could increase facility requirements on the airfield.

Airports should coordinate with their airlines using CDM to determine what infrastructure is needed to support the desired operation.

Historical surface movement data may serve as a baseline for any type of surface study identifying the current use of parallel taxiways, connector taxiways, high-speed exit taxiways, holding areas, and deicing facilities. The data can accurately identify duration of operations, feeding any modeling efforts.

Surface movement data may also be useful to track the exact usage of any pavement area on the airfield as part of pavement management systems, enabling pavement wear and useful life calculations.

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Alternatives

Surface movement data may assist in the development of metrics for the evaluation and selection of conceptual alternatives to accommodate future airport needs over the Master Plan planning horizon. However, traditional metrics on ramps and gates may not meet CDM needs. Alternatives need to be developed that match stakeholder needs and available infrastructure. In addition, to the extent that additional facilities are required to support effective departure metering, there may be additional alternatives to consider for establishing holdpads and potential relocation of existing facilities to accommodate those holdpads. One of the case studies analyzed by the research team was the application of departure metering at John F. Kennedy International Airport, a description of which is included in Appendix B.

Timing and Implementation

The SWIM program currently collects and provides ASDE-X data to industry users. Real-time data is needed for CDM. Real-time and historical data can be obtained through execution of a user agreement.

Improved NextGen Landing Systems

Improved NextGen landing systems include those enabled by PBN and GBAS. PBN technology [RNAV, RNP, WAAS (Wide Area Augmentation System) LP and LPV], previously described, allows for near Category (CAT) I minimums with lateral and vertical guidance. GBAS systems augment the existing GPS, which enables CAT I approach minimums and are expected to ultimately enable CAT II/III precision approach minimums.

PBN procedures are being established at airports throughout the NAS to improve arrival, approach, and departure operations for multiple runway configurations at airports.

The goal of GBAS implementation is to provide GBAS Landing System (GLS) approaches as an alternative to ILS approaches supporting the full range of approach and landing operations. GLS approaches have been implemented at Newark Liberty International Airport and George Bush Intercontinental Airport/Houston as airport funded, non-federal landing systems

GBAS system installation is flexible and can be installed at multiple locations on an airfield including on top of buildings. One system may provide approach capability to all runway ends at an airport and possibly to other airports in proximity (approximately 23 miles) to the installation. However, this concept has not been operationally approved to date. Aircraft using GBAS systems must be equipped with a multi-mode receiver in order to use the system.

Landing system improvements face implementation challenges but will have a significant impact on the capacity of the NAS as well as specific airport environments. The following highlights demand/capacity impacts, facility requirements, alternatives, and timing and implementation considerations of these improvements.

Effects on Airport Planning and Development

Improved landing systems could significantly affect the following major elements of airport planning and development programs:

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DEMAND/CAPACITY

Implementation of PBN-enabled landing systems has already proven to provide access to airports in low-visibility conditions where no access was available previously, reduce noise and emissions, and increase the efficiency of the airspace in multiple locations. Current initiatives are focused on providing improved capacity for closely spaced parallel runways for arrivals and departures.

New PBN-enabled procedures including EoR are applicable to closely spaced and widely spaced parallel runway approaches and enable controllers to clear aircraft on an RNP approach while on the downwind to the airport without the need to use the standard 1,000 feet of vertical separation when the aircraft turns to align with the runway centerline. This change to separation standards allows aircraft to turn to align to the runway much closer to the field as compared to a conventional ILS procedure, reducing track miles and fuel burn. This capability also increases the flexibility to design approach procedures, including the downwind, base, and final segments, in such a way as to minimize overflights of noise sensitive areas.

GBAS is designed to provide CAT I, II, and III approach minimum capability for multiple runway ends. It will be most beneficial in potentially providing access to airports in CAT II/III conditions as one system may serve multiple runway ends. GBAS will aid in capacity enhancement for closely spaced runways as it allows for multiple offset approaches with no additional ground infrastructure.

Implementation of PBN or GBAS procedures may eliminate or reduce the need to protect for ILS and localizer critical areas leading to more efficient use of taxiways and potentially increasing runway capacity.

FACILITY REQUIREMENTS

Implementation of PBN-enabled landing systems can be initiated locally and sponsored by airlines or an airport. The implementation of the procedures is performed by FAA. Development of RNAV SIDs and STARs requires little infrastructure at airports. Development of SIAPs requires an appropriate current aeronautical survey and compliance with airport design standards in Advisory Circular 150/5300-13A.

GBAS is still considered a non-federal system and requires the airport to purchase and install the equipment, and to have it commissioned by FAA. A single system can be procured and installed for approximately $1M to $2M. The siting of a GBAS system is flexible and can even be mounted on existing structures or building on an airfield. Requirements include a clear line of sight to runway ends and satellites for the very high frequency data broadcast (VDB) antenna and the remote satellite monitoring units. Further details can be found in FAA Order 6884.1, *Siting Criteria for Ground-Based Augmentation System*.

ALTERNATIVES

PBN can be used in multiple alternative situations on a case-by-case analysis of the airport or terminal area environment. PBN offers flexible design to avoid obstacles, terrain, and sensitive land use areas, while providing access and unique simultaneous or staggered operations at airports with multiple runways.

Development of PBN procedures is typically done by FAA. However, third-party vendors can develop special procedures, and third-party vendors who are certified may develop public RNP AR procedures. Procedure development costs may range from $30K to over $1M depending upon the complexity and challenges of the airspace environment.

The initial concept of operation for GBAS was for straight-in final approach guidance for CAT I, II, and III operations. The benefits of GBAS greatly increase at a CAT II/III landing system as one installation

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serves several runway ends, eliminating the need for multiple expensive CAT II/III ILS systems. As GBAS is implemented into the FAA inventory and aircraft equipage expands, it will likely replace CAT II/III ILS systems in the long term.

GBAS systems could be used for multiple other applications including offset approaches for closely spaced parallel runways, multiple approaches at varying glide path angles, positive course guidance for continuous descent profiles in the terminal area, and flexible approaches to temporary pavements used during construction. The corrected GBAS signal may also be broadcast for ground vehicle tracking and survey purposes. Whereas this capability exists, it has not been operationally approved to date.

TIMING AND IMPLEMENTATION

EoR safety assessments are still in development but a waivered procedure has been approved for DEN. The DEN procedure will provide data to support concept validation for a proposed national standard. The FAA has scheduled the development of a national standard making it possible to implement the technology at eligible locations throughout the NAS. The FAA will use the Metroplex and single-site processes to deploy the capability. Industry has identified MIA, ORD, CLT, George Bush Intercontinental Airport (IAH), and Fort Lauderdale-Hollywood International Airport (FLL) as candidate locations.

The FAA GBAS program is currently conducting a research and development (R&D) and prototyping effort to reduce the technical risk and validate new requirements associated with meeting the GBAS approach service type capable of supporting approaches to Category III (CAT III) minima. The FAA has delayed its plans for a federal acquisition and implementation of the system. However, the system can be purchased and installed by airports as part of the non-federal navigational aid (NAVAID) program.

The Port Authority of New York & New Jersey purchased and operates the first public-use system to receive FAA operational approval for Newark Liberty International Airport (EWR). The Houston Airport System (HAS) owns and operates the second GBAS to receive FAA operational approval for Houston's IAH. The GBAS systems at both EWR and IAH are currently being used by United Airlines with Boeing 737 and Boeing 787 aircraft.

The Boeing Company has a private-use GBAS installed and approved at its R&D facility at Moses Lake Airport (MWH) in Washington State and another private-use GBAS installed in Charleston, South Carolina (CHS) to support B-787 customer acceptance flights at the Charleston assembly plant.

Honeywell is pursuing the certification of the system for CAT II/III operations. Certification is anticipated in the next one to three years.

Separation Management

Separation management focuses on the enhancement of aircraft separation assurance. Separation management improvements will provide air traffic controllers with tools and procedures to separate aircraft with different kinds of navigation equipment and wake performance capabilities, what is known as a mixed environment.

The elements of this portfolio will achieve success by enhancing current NAS infrastructure through the integration into ATC automation systems of enabling technologies, new standards, and new procedures. The key automation systems impacted by this portfolio are ATOP, TAMR, ERAM, TFDM, and time-based flow management (TBFM).

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Effects on Airport Planning and Development

Separation management could significantly affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

Separation management will enhance the NAS by providing controllers the ability to safely reduce separation between aircraft, resulting in increased efficiency and capacity. Increased airport throughput via TRACON facilities and managed final approach procedures will be of great benefit to airports and the NAS as a whole. Additionally, aircraft on oceanic routes will be provided the ability to ascend and descend to their preferred altitudes and allow controllers to approve requests for direct routing. Wake vortex impacts will be reduced due to reduced but monitored separations along with accurate climatological data to aid in managing separations.

Capabilities in this portfolio will support an increase in capacity by increasing airport throughput as a result of the closer spacing of flights accepted from TRACON airspace and managed on final approach. Automation capabilities will also enable air traffic controllers and pilots through reduced separation between aircrafts to manage increasing traffic levels in oceanic airspace.

This portfolio will provide improved efficiency through the introduction of capabilities that will enable more oceanic flights to ascend and descend to their preferred altitudes. Controllers will also be able to approve additional pilot requests for direct routes and more efficient altitudes.

This technology provides controllers automated information about wake vortex separation requirements for any given aircraft pair, along with accurate wind data that will help predict more accurate and safer separation standards.

FACILITY REQUIREMENTS

The deployment of these systems and implementation of standards and procedures is managed by FAA through various programs. However, if a need can be demonstrated, identification of separation management techniques may be considered as part of the local planning process. Follow-on efforts may be required to harmonize the requirement with FAA implementation requirements.

ALTERNATIVES ADVANCED

ATOP, TAMR, and ERAM systems are customized for each installation based on the requirements of the airspace environment in which they serve. The capabilities of these systems may be used as the basis for capacity-enhancement alternatives.

TIMING AND IMPLEMENTATION

Software is currently being developed and will be released to key sites in the near future. FAA expects operational rollout to be achieved in 2016/2017 for oceanic in-trail climb and descent with trajectory management expected to begin in 2017. En route and ATOP management is expected to be prepared and ready for implementation by 2022. Sector enhancements via ERAM and trajectory-based UAS (unmanned aircraft systems) integration, along with wake turbulences and RECAT procedures are being developed today through 2017 and on.

Multilateration

Multilateration (MLAT) is surveillance capability comparable to secondary surveillance radar enabling air traffic controllers to track aircraft in areas where no radar coverage exists. An MLAT system consists

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of a transmitter, receiving antenna sensors, a central processor, and an optional interrogator system. The sensors and transmitters send out signals interrogating the transponder that, in turn, transmits a response. The response from the transponder is interpreted by the computers to accurately locate an aircraft using triangulation by measuring the "time difference of arrival" (TDOA) of the signal from the transponder at three or more synchronized receiver sites. The altitude of the aircraft is obtained directly from the required Mode C altitude-reporting transponder. The position information is fused with existing radar systems, providing a "target" on the radar screen, enabling air traffic controllers to provide positive control of the aircraft. MLAT was not a part of early NextGen initiatives but has now been adopted into the NextGen Program. MLAT can be deployed in many configurations in en route, terminal, or airport environments. En route and terminal configurations supporting surveillance are referred to as WAM whereas airport installations supporting surface movement, virtual air traffic control towers, and noise monitoring systems are referred to as MLAT.

WAM systems are currently deployed in mountainous airports in Colorado, the state of Alaska, oceanic oil drilling platforms in the Gulf of Mexico, and other locations to provide surveillance in remote areas where radar coverage is not available. These systems provide a "radar-like" environment for aircraft operations.

MLAT systems are used for surface movement systems across the world. FAA incorporates MLAT sensors into the ASDE-X systems to provide ATC with surface movement information across an airfield. This system is deployed at the top 35 major airports in the U.S. and is used on a daily basis for operations.

Another provision planned is the development of an ASSC at nine airports that use ASDE-3/AMASS for surface surveillance and situational awareness. The ASSC system fuses multilateration ground sensor data with ADS-B–equipped aircraft information into an airport surveillance radar/mode select terminal and airport tower controller display (included as part of the ASSC configuration). The FAA recently awarded a contract for ASSC and is scheduled to have the system operational at the nine sites including Anchorage, Andrews Air Force Base, Cincinnati/Northern Kentucky, Cleveland, Kansas City, New Orleans, Pittsburgh, Portland (Oregon), and San Francisco by FY 2017.

MLAT technology is scheduled to be integrated into the precision runway monitoring (PRM) radar system to refresh the aging technology. These systems have been beneficial in supporting simultaneous operations at airports with closely spaced runways. Currently, the PRM at SFO is the only installation scheduled for refresh.

MLAT is used to supplement secondary surveillance radar internationally and will likely be the backup system for ADS-B Out. MLAT systems serve as a major component for surveillance for virtual ATC towers. Virtual tower systems have not been implemented in the U.S. but are currently certified in Norway and Sweden.

MLAT systems are also used by airports as part of noise abatement and revenue tracking systems. These systems are tailored for each installation, providing aircraft tracking on the ground and in the vicinity of an airport.

Effects on Airport Planning and Development

Multilateration could significantly affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

Multilateration technologies will allow for the continuous surveillance of aircraft, especially those around mountainous or precipitous terrain, therefore increasing safety and reliability in aircraft track-

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ing. High costs associated with radar installations will be reduced as cheaper remote sensing units can be installed in less accessible areas. The systems will allow more aircraft to be flown in and out of airports all while being continuously monitored via remote sensing sites. WAM also allows controllers to automate a number of tasks including minimum safe altitude warnings, recording of air traffic events, and tracking aircraft and conflict alerts. An additional benefit of the system is the reduced glare and the ability for controllers to select what they see within a sector and customize color coding of aircraft to aid in organization. Multilateration systems allow for the aircraft to receive its own positions through ground stations signals and could be a backup should the GPS system fail.

FACILITY REQUIREMENTS

MLAT requires the deployment of a network of sensors and integration into the NAS surveillance system. No additional aircraft equipage is required for mode C equipped aircraft.

ALTERNATIVES

MLAT can be configured for multiple types of applications on the surface and in a wide area.

TIMING AND IMPLEMENTATION

MLAT technologies will be rolled out across the system beyond the 2020 timeframe. Operational readiness has been demonstrated at 16 of the 20 en route facilities across the NAS as of the end of 2014; the remaining en route facilities were expected to demonstrate capabilities and readiness by the end of 2015.

FAA costs are significantly reduced, especially in remote areas where there are high capital costs associated with radar installations. Differing sensors and transponders are required depending on the altitude at which aircraft will most likely operate. In the Colorado WAM (case study), it was reported that costs were shared between the state of Colorado and the FAA. The state of Colorado, however, made significant up-front (and at-risk) investments in order to give the program momentum.

Longer-Term NextGen Programs

Longer-term NextGen programs include ADS-B In, and CDTI systems. These technologies are further defined in the following sections.

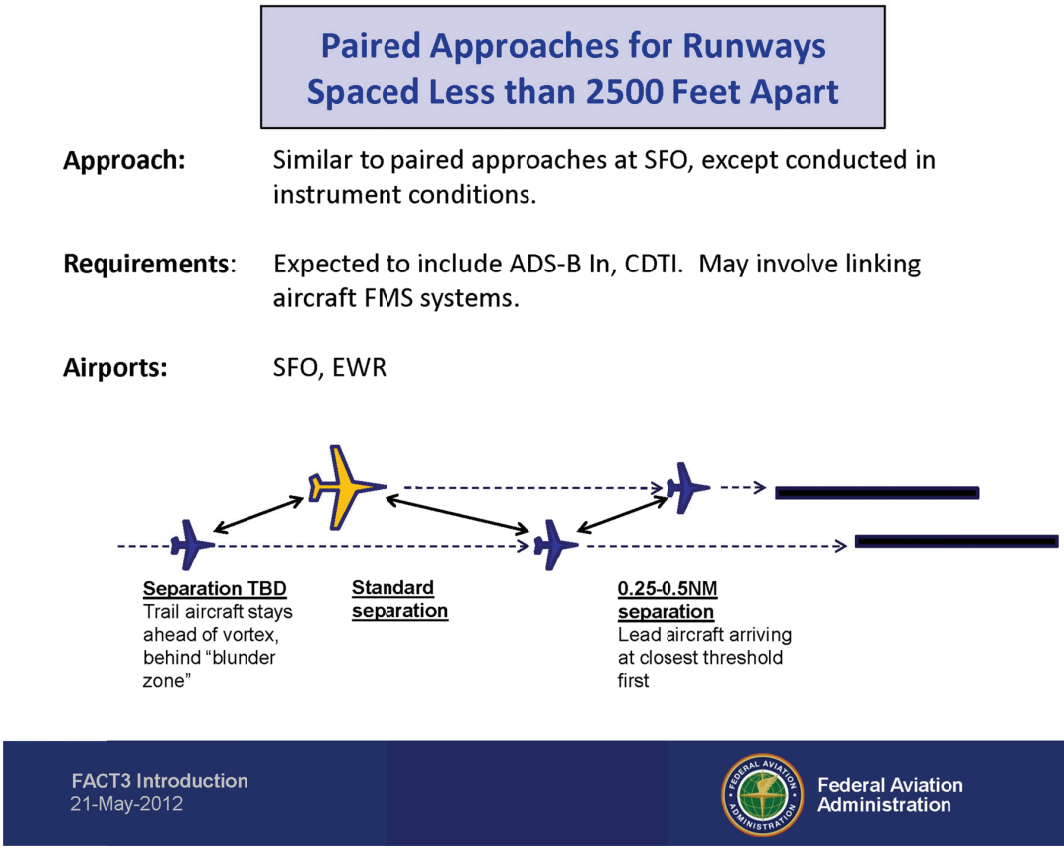
Multiple Runway Operations/Closely Spaced Parallel Operations—Paired Approaches

An advanced concept of paired approaches is currently in the applied research phase. This capability will leverage precision navigation (e.g., from GLS or LPV), ADS-B In, CDTI, linked FMS systems, and 4D trajectory information to support paired approaches by multiple aircraft to closely spaced runways. The advanced paired approach concept is intended to be used in IFR conditions and to support staggered aircraft spacing of ¼ nm on closely spaced parallel runways. Figure 2-7 highlights this concept.

Effects on Airport Planning and Development

The paired approach concept combined with Wake RECAT could affect the following major elements of airport planning and development programs:

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Source: Federal Aviation Administration, FACT3: Airport Capacity Needs in the National Airspace System, January 2015 (https://www.faa.gov/airports/planning_capacity/media/FACT3-Airport-Capacity-Needs-in-the-NAS.pdf)

Figure 2-7. *Advanced paired approaches.*

DEMAND/CAPACITY

The capability would provide a significant capacity increases at airports that (1) have significant volumes of Heavy jet aircraft, (2) already have closely spaced parallel runways that could be used more efficiently under the new rules, or (3) could implement closely spaced parallel runways on available airport property and provide either more efficient independent or dependent parallel operations that could not otherwise have been provided within the available space.

FACILITY REQUIREMENTS

Closely spaced parallel runway operations are currently conducted very efficiently at airports in VMC but experience a significant capacity drop in IMC because they do not meet the current spacing requirements for dependent or independent approaches. The foregoing potential rule changes affecting spacing requirements for CSPO assume that the associated parallel runways would be served by precision instrument approach procedures with vertical guidance, which could include ILS approaches, RNAV/RNP approaches, or RNAV (GPS) approaches such as LPV. This concept will likely require high-update radar (HUR), NextGen-enabling technology, and advanced FMS capabilities.

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ALTERNATIVES

The potential changes in the rules for the minimum spacing between parallel runways required for dependent or independent approaches could open up new opportunities at airports that previously would not qualify for such approaches. Therefore, new alternatives could be considered in the airport planning and development process for such airports. Such alternatives would have to be evaluated in terms of how the spacing between the parallel runways could facilitate the development of passenger terminal facilities between those runways and also how the new capabilities could affect noise exposure in the surrounding communities and operations at other nearby airports.

TIMING AND IMPLEMENTATION

Advanced concepts to will likely be implemented in the long-term horizon as additional research and advanced avionics are required.

Automatic Dependent Surveillance-Broadcast In

ADS-B In provides operators of properly equipped aircraft with weather and traffic position information delivered directly to the cockpit. ADS-B In-equipped aircraft have access to the graphical weather displays in the cockpit as well as text-based advisories, including Notices to Airmen and significant weather activity. The FAA provides three forms of ADS-B In Services: TIS-B, ADS-R, and FIS-B.

Effects on Airport Planning and Development

ADS-B In could significantly affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

Implementation of ADS-B In will increase situational awareness in all phases of flight, improving efficiencies in sequencing and separation, enabling the use of multiple runways (closely spaced and converging) and enhancing capacity in all weather conditions.

FACILITY REQUIREMENTS

There are no airport facilities required for ADS-B In. User requirements for ADS-B In include aircraft equipage and the development of cockpit systems to take advantage of all the capabilities available through the technology.

TIMING AND IMPLEMENTATION

Implementation of ADS-Out is in process. FAA has deployed ADS-B sensors across the NAS and is planning on expanding the network to Mexico in 2016 and 2017. FAA has a regulation requiring ADS-B Out equipage by the year 2020. No formal requirement exists to equip with ADS-B In.

In order to take full advantage of ADS-B, a majority of aircraft must be equipped. Avionics manufactures offer ADS-B In solutions for aircraft operators. However, the full benefits of the technology can only be realized when a substantial proportion of aircraft are operating in the vicinity of similarly equipped aircraft. Over the past decade UPS has used ADS-B In and Out technology for their operations at the Louisville International Airport. The technology enables them to space operations efficiently in the air and sequence aircraft on the ground.

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Cockpit Display of Traffic Information (CDTI)

CDTI is an airborne system which, when combined with ADS-B In and Out, displays neighboring aircraft information, on the ground or in the air, to the flight crew as well as automation functions that, in some cases, provides speed or maneuver guidance to the crew. Benefits to be provided by CDTI system include:

- Airborne Traffic Situation Awareness—providing enhancements in flight crew knowledge of surrounding surface and airborne traffic;
- Airborne Spacing—allowing flight crews to ensure a spacing value from a designated aircraft;
- Airborne Separation—allowing flight crews to ensure separation from a designated aircraft, which relieves the controller from the responsibility for separation between these aircraft; and
- Airborne Self-Separation—allowing flight crews the ability to ensure separation of their aircraft from all surrounding traffic.

CDTI systems were demonstrated in the Safe Flight 21 Ohio Valley trials in the late 1990s. System functionality requirements have been in development as part of industry efforts since that time. Some of the automation functions of the system include:

- Avoid Collisions,
- Avoid Wake,
- Cross,
- De-conflict,
- Follow,
- Hold,
- Merge,
- Pair,
- Pass,
- Separate,
- Space,
- Stagger, and
- Time Departure.

Effects on Airport Planning and Development

CDTI could significantly affect the following major elements of airport planning and development programs:

DEMAND/CAPACITY

According to FAA, implementation of CDTI may increase situational awareness in all phases of flight improving efficiencies in sequencing and separation, enabling the use of multiple runways (closely spaced and converging); and, to the extent that CDTI permits a reduction in in-trail buffers, it can produce slight increases in capacity in all weather conditions.

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FACILITY REQUIREMENTS

A CDTI will be required to take full advantage of the benefits provided by ADS-B In. Aircraft equipage levels of the fleet will drive the various capabilities available through the technology.

ALTERNATIVES

CDTI systems combined with ADS-B In will enable alternatives for the enhancement of capacity and safety to include marginal VFR operations, low-visibility surface movements, MRO, CSPO, offset approach configurations, reduced wake turbulence separation, and efficient separation in all phases of flight. Each alternative will provide an increase in safety and situational awareness for operators and increased capacity during low-level weather conditions.

TIMING AND IMPLEMENTATION

Avionics manufacturers offer CDTI solutions for aircraft operators but equipage is voluntary. This will slow the realization of all the benefits of the system. It is likely that this technology will mature over the next 20 years.

Non-NextGen Technologies and Capabilities for Airports

Multiple technologies and capabilities useful to airports exist today that are not categorized as or included in the NextGen program. These technologies may be emerging toward inclusion of NextGen, part of other FAA programs, or stand alone. Table 2-2 provides a list of known technologies and capabilities by function to be considered as part of airport planning.

Table 2-2. Non-NextGen technologies and capabilities for airports.

FUNCTION	CAPABILITY
Independent Runways	NA
Dependent Runways	NA
CSPO	NA
Performance-Based Navigation (Terminal Area)	NA
Surface Operations & Data Sharing	CDM
	Intelligent Routing and Guidance System/Advanced Surface Movement Guidance and Control System (A-SMGCS)/Airfield Lighting Control and Monitoring System (ALCMS)
	LED Lighting Technology
	Ground Vehicle Tracking
	Traffic Display and Analysis Systems (Non-ATC Automation)

continued

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Table 2-2. Continued

FUNCTION	CAPABILITY
Improved Landing Systems	Synthetic Vision, Enhanced Vision Systems, and Heads-Up Displays
	LED Precision Approach Path Indicator (PAPI)
	LED Approach Lighting
Separation Management	NA
Multilateration (Surveillance)	Virtual ATC Towers
	Virtual Ramp Control
	Noise Monitoring Systems
ADS-B (Surveillance)	Virtual ATC Towers
	Virtual Ramp Control

The following sections describe these technologies and their effects on airport planning in development in further detail.

Surface Operations and Data Sharing

Non-NextGen surface operations and data sharing technology includes CDM, intelligent routing and guidance systems, automated docking systems, LED airport lighting, ground vehicle tracking, and traffic display and analysis systems. The following sections describe each of these technologies and highlight their relevance to the airport planning process.

Collaborative Decision Making

CDM programs established locally at airports are different from FAA-led initiatives. On-airport groups may use guidelines established in the *Collaborative Decision Making Leadership, Strategies, Structure and Guidelines* V 4.0 document and obtain data enabled through the dissemination of TFDM and other SWIM-enabled data to supplement decision making. However, other local factors influencing the day-to-day operations are considered including construction, maintenance, wind and weather conditions, runway use systems, the changing nature of airport environments, and special events.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

Local CDM programs could significantly affect the following major elements of airport planning and development programs:

Demand/Capacity

Benefits include increased flexibility, efficiency, safety, and validation. These are further described below:

Flexibility

- Improve the timely exchange of data to enable aircraft operators to more accurately adjust their departure and arrival times for the most efficient use of available runways, taxiways, and gates.

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- Permit taxi operations to occur that support low-visibility operations for takeoff, improving access during those times.
- Reduce the effect of weather-related delays.

Efficiency

- Enable more effective scheduling that includes runway, departure fix, and traffic flow management ground-management constraints, with automatic reassessment and update of the departure schedule based on the ability of departing flights to meet the designated departure schedule.
- Enhance the ability to react to changing airport conditions, such as severe weather, by issuing digital pre-departure clearances, including routing revisions, using Data Comm.
- Improve awareness of surface congestion at major hub airports, greatly streamlining the coordination of corrective action and improving the resilience of the system.
- Reduce fuel burn and operating costs related to long departure queues.
- Reduce delays by improving event data quality and adherence to controlled departure times.
- Reduce FAA operating costs through the use of automated flight strips.

Safety

- Capabilities in this portfolio enhance safety on the airport surface by improving pilot and controller awareness of surface traffic. This benefit is enabled through data distribution and flight deck capabilities.
- Enhancements to the Aviation Safety Information Analysis and Sharing System (ASIAS) can support NextGen with in-depth analysis of safety data from industry and government sources to identify existing or prospective operational risks that exist in the NAS. These safety analyses have secondary benefits to NextGen key performance areas and may reveal potential improvements for efficiency and capacity.
- Surface movement data can be used to support the safety risk management processes by providing a means to calculate the location, type, and duration of an aircraft operation on any part of the airfield. These calculations may provide a basis for determining the probability or likelihood of a hazard incident.

Validation and the Development of Performance Metric

SWIM data may serve to validate assumptions developed as part of planning efforts and measure the actual performance of the implementation of NextGen concepts. For example, radar data in the terminal area may be used to determine participation on RNAV and OPD procedures, utilization of landing systems-approach concepts, and environmental compliance.

Facility Requirements

Historical surface movement data may serve as a baseline for any type of surface study identifying the current use of parallel taxiways, connector taxiways, high speed exit taxiways, holding areas, and deicing facilities. The data can accurately identify duration of operations feeding any modeling efforts.

Surface movement data may also be useful to track the exact usage of any pavement area on the airfield as part of pavement management systems enabling pavement wear and useful life calculations.

Airports may be asked to support CDM through contracting with outside data supply or analysis contracts.

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Alternatives

Surface movement data may assist in the development of metrics for the evaluation and selection of conceptual alternatives to accommodate future airport needs over the Master Plan planning horizon.

Timing and Implementation

The SWIM program currently collects and provides ASDE-X data to industry users. This data can be obtained through execution of a user agreement.

Intelligent Routing and Guidance Systems

Intelligent routing and guidance systems are being implemented in Europe and some Middle East countries to automatically control surface movements. Supported by the Single European Sky ATM Research (SESAR), the system integrates an advanced surface movement guidance and control system (A-SMGCS), airfield lighting control and monitoring system (ALCMS), surface radar or multilateration and ATC tower automation to allow for the automated movement of aircraft on an airfield during low-visibility conditions.

The A-SMGCS identifies and provides real-time surveillance of taxiing aircraft. The system provides taxi routes for aircraft (conflict-free) through use of optical sensors and warns pilots that a conflict is ahead even without controller intervention. A-SMGCS functionality varies across four levels (Level 1–Level 4). Level 1 provides improved surveillance while Level 4 provides full airport-wide ground movement planning and routing and full conflict resolution.

The ALCMS allows for control and monitoring of ever light on the airfield movement area including land and hold short operations (LAHSO), stop bar lighting, runway guard lights, and safety and centerline lighting. The health of each light may be monitored as well aiding in reducing airfield maintenance effort. The tower automation systems allow controllers to select routing for an aircraft based on automated preferred routes or a custom route. Some systems utilize a touch pad to enable this operation.

When these systems are integrated they may provide guidance in low-visibility conditions using a "follow the green" (FtG) method. The FtG system automatically leads aircraft around the airfield by illuminating ground-based centerline lighting green along a specified path in front of an aircraft. The system can be configured to automatically route aircraft to and from the end of the runway and gates while automatically providing conflict detection and resolution.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

Intelligent routing and guidance systems could significantly affect the following major elements of airport planning and development programs:

Demand/Capacity

An intelligent routing system will aid in increasing the capacity of an airfield in low-visibility conditions by providing enhanced situational awareness to pilot and controller, automating routing and aircraft movement, and automating conflict detection and avoidance. The system reduces pilot and controller workload and the need for the use of "follow me" vehicles operating on the airfield.

Facility Requirements

Requirements of the system are based on the level of automation desired. A fully automated system will require in-pavement lights, circuitry designed to support the desired functionality, ALCMS and A-

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SMGCS hardware and software, integration with surface radar, and integration with ATC automation. The system will also need to have dedicated and backup power.

Alternatives

The system may be tailored to support movement of aircraft by blocks or through the FtG concept. The movement of aircraft may be controlled throughout the entire airport including the apron and ramp areas. Some airports are exploring the use of the system during VMC conditions as it provides for reduced controller workloads.

Timing and Implementation

These systems have been deployed in Germany and are being implemented today in the United Arab Emirates. These systems have not been deployed in the U.S. since FAA has not adopted the concept of the full technology into the NextGen program. Intelligent lighting systems and an ALCMS are available to airports in the U.S. and from various vendors.

Automated Docking Systems

Automated docking systems (ADS) provide guidance to pilots for aircraft gate parking. The system uses laser scanning technology to identify the incoming aircraft and verify the correct model as that selected by the airline or operator. The ADS also provide distance-to-stop positioning, lateral-position assessment, right-left indicators, and continuous closing rate of the aircraft as it proceeds down-centerline. Once docking is complete the system will read “stop” and “ok” to inform a pilot the aircraft is parked and docked correctly within a stand. These systems also provide flight information, pushback time and takeoff time. The ADS can be integrated into an intelligent routing and guidance system making all operations on the airfield automated.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

ADS could significant may affect the following major elements of airport planning and development programs:

Demand/Capacity

The use of ADS may provide an increase in safety and efficiency in the gate areas. Parking and push-back times may be reduced creating additional throughput.

Facility Requirements

Installation of an ADS may vary based on the design of the terminal gate, apron area, and aircraft type. Facility requirements analysis should consider the existing and future fleet mix at the airport, terminal design, and manufacturer’s information to establish the requirements of the system.

Alternatives

ADS may be configured in multiple ways depending upon the gate environment and the aircraft type.

Timing and Implementation

ADS are available today and are being integrated at airports as part of efficient surface movement operations.

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LED Airfield Lighting Technology (Ground-Based Lighting)

Over the past decade, airports have slowly been adopting LED technology to replace runway, taxiway, and obstruction lights. Manufacturers offer multiple LED lights meeting the requirements of the existing airport lighting systems. LED lighting units and fixtures are more expensive to purchase but less expensive to operate. The low operating costs are attributed to the low wattage requirements of an LED and the significantly longer lifespan of the lamps.

When first installed, pilots noticed the LED lamps were much brighter than the incandescent lamps when both were set at the same intensity level. As a result, many pilots, when flying an approach to a runway in low-visibility conditions, were seeing the approach light system as they always have and, as they approached the runway, were blinded by the LED lamps, resulting in a go around. In addition aircraft operating with Heads-Up Display (HUD) systems are not able to see the LED lights as they lack the heat intensity of an incandescent bulb.

FAA and industry are conducting research to solve these problems for the future. Some modifications have included modification of fixture lenses, use of lower intensity lamps, and the incorporation of infrared lights into the fixture.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

LED airfield lighting may affect the following elements of airport planning and development programs:

Demand/Capacity

LED technology will not increase demand or capacity at an airport.

Facility Requirements

Airport practitioners should consider the implementation of LED technology in an effort to reduce operating costs and energy consumption. Facility requirements need to be developed based on the type and application of the lighting systems to be replaced or new installations to ensure compatibility with operations.

Alternatives

Developing alternatives to compare LED and incandescent lighting should be conducted to highlight the benefits and costs associated with the implementation of the technology. In some cases, the wiring and fixtures may need replacing (versus simply plugging a lamp into an existing fixture) to achieve the maximum benefit of the LED technology. This increases the acquisition costs of the technology implementation and extends the break-even time frame further into the future.

Timing and Implementation

LED technology is available today from various airfield lighting manufacturers for implementation. The technology continues to evolve.

Ground Vehicle Tracking

Ground vehicle tracking systems provide improved runway safety by allowing tracking and positional awareness of aircraft and airfield service trucks, including snow removal equipment, aircraft rescue and firefighting (ARFF) vehicles, and construction/maintenance vehicles. The FAA approved ground vehicle tracking systems are based on ADS-B technology and require the installation of certificated transponders in each vehicle to be tracked. The transponders are not required to be permanently installed and can be moved between vehicles. However, programming changes may be required by the vendor. The

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location and identifying features of a vehicle may be transmitted to ATC, airport operations, aircraft cockpits, lap tops, tablets, and cell phones in real time. The systems can be integrated to other on-airport GIS-based systems and can be used to coordinate rescue efforts, lead aircraft in low visibility, and manage ground vehicle fleets: snow removal, mowing, construction, and pavement maintenance. However, FAA-certificated transponders are expensive and are limited to tracking only on the movement area. Non-FAA systems are available that use GPS to track airport vehicles, although they do not provide position information to ATC.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

Ground vehicle tracking airfield may affect the following elements of airport planning and development programs:

Demand/Capacity

Ground vehicle tracking technology will not increase demand or capacity at an airport. It may however, increase the efficiency and safety of operations on the airfield and assist in multiple functions of day-to-day activities.

Facility Requirements

Ground vehicle tracking systems will require additional hardware, software, and transponder equipment to be installed at the airport. The requirements of the system will vary from airport to airport based on the needs, the desired functionality, and the number of vehicles to be tracked. Individual transponders can cost $5,000 to $7,000 each and require expensive annual maintenance contracts.

Alternatives

Multiple alternatives exist for implementation of this technology to solve operational issues such as the management of construction, the movement of aircraft in low visibility, gate operations, emergency response, and maintenance. Alternatives should consider the need for the technology, desired functionality, benefits, and costs.

Timing and Implementation

Ground vehicle tracking systems have been implemented at multiple airports in the NAS and are available today. Early implementation of these systems used GPS technologies as the basis for position information. New technology incorporates ADS-B and multilateration technology to achieve higher resolution of position information.

Traffic Display and Analysis Systems (Non-FAA ATC Automation)

There are multiple non-FAA ATC automation systems capable of displaying air traffic information including marine based radar systems, Internet-based systems, and custom systems. The FAA established a low-cost ground surveillance program in 2007 as an initiative to reduce runway incursions. Several firms were involved in research activities using multiple types of technology as the basis for low cost airport surveillance. In 2013, the FAA cancelled the program after determining it was not viable/cost-effective for sustainment.

Marine radar has been used to validate research, track tall ships near approach runway ends, and as the basis for bird tracking. The most common marine systems are based on the use of S and X radar bands. These systems were originally designed to track large slow moving targets. However, the technology is improving. FAA allows marine-based radar technology for the tracking of birds. These

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systems are called Airport Avian Radar Systems. Advisory Circular 150/5220-25, *Airport Avian Radar Systems,* provides guidance on the implementation of these technologies.

There are multiple Internet-based software applications to track aircraft and provide arrival and departure time information. These applications utilize FAA traffic flow management data and other international data streams to display aircraft, flight, and schedule information to a user.

Several companies offer surface movement analysis systems that incorporate ASDE-X radar data into high-resolution GIS-based mapping coverages for surface operations analysis and management purposes. These systems have been implemented at multiple large airports in the NAS and have been useful in assisting in operational efficiency and safety case analysis.

Surveillance research is ongoing at universities and throughout industry. It is important to note, FAA is the governing authority of systems used for ATC separation or advisory services. FAA certification is required for any system to be used for this purpose. Airports that purchase non-FAA certified surveillance systems with the intent to provide aircraft advisory information to pilots do so at their own expense and risk. Furthermore, the use of these systems may be limited by FAA.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

Traffic display and analysis systems are operational in nature and serve to disseminate information to assist airport operations and provide information to the traveling public. Output from these systems may serve to assist in wildlife management, safety case analysis, providing timely information to the traveling public.

Improved Landing Systems

Non-NextGen improved landing system technology includes synthetic vision systems (SVSs) and LED technology applied to approach lighting and PAPI systems.

Synthetic Vision System, Enhanced Vision Systems, and Heads-Up Display

Synthetic vision was developed by NASA and the U.S. Air Force in the late 1970s and 1980s in support of advanced cockpit research, and in 1990s as part of the Aviation Safety Program. An SVS is a computer-mediated reality system for aircraft to provide pilots with clear and intuitive means of understanding their flying environment. Synthetic vision provides situational awareness to pilots by using terrain, obstacle, geo-political, hydrological, and other databases. A typical SVS application uses a set of databases stored on board the aircraft, an image generator computer, and a display. The navigation solution is obtained through the use of GPS-guided flight management systems, and inertial reference systems. SVSs may depict position, altitude, airframe orientation, terrain, obstacles, airport infrastructure, and other traffic in a 3D perspective eliminating metrological conditions, which reduce visibility.

Enhanced vision systems (EVSs) use actual cameras and other sensors such as forward looking infrared (FLIR) and millimeter wave RADAR to sense the environment outside the aircraft. EVS can look through the weather and allow an aircraft to land in the visual segment of an instrument approach.

HUD systems are common in many commercial aircraft today. These systems depict the runway environment, including approach lighting systems and PAPIs at airports, and allow for lower approach minimums.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

Synthetic vision may affect the following elements of airport planning and development programs:

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Demand/Capacity

SVS may improve access in low-visibility conditions as the runway environment is visible to the pilots. FAA currently allows lower visibility on approach for aircraft equipped with HUD systems depicting airport lighting infrastructure.

Facility Requirements

Implementation of LED technology would need to consider the use of these systems at airports. Some of the SVS, EVS, and HUD avionics have problems detecting LED lighting. FAA is examining the use of infrared lighting to solve these problems.

Timing and Implementation

Avionics manufacturers offer SVSs and EVSs for multiple types of aircraft today. HUD systems are widely used by new commercial aircraft in the fleet. Ultimately, equipage will be a decision made by the aircraft operator.

LED Precision Approach Path Indicator and Approach Lighting

Over the past several years, the FAA has conducted research into replacing the multi-light PAPI system or array, with a PAPI LED system that uses a single light on each unit to orient a pilot vertically on approach. The new configuration still uses a four-light box array to provide guidance to pilots and the same metrics (i.e., red and white lights) are used to signal pilots if they are too low (red) or too high (white) above the glide path.

LED systems enhance light output performance, eliminate warm-up times especially in cold climates, and lower energy costs. The instant light-up of the LED can reduce delay in cold climates and allow a system to be switched on or off in an efficient manner, or they can be operated in standby mode, something not offered by conventional lamp PAPI systems. Additionally, the LEDs increase the life-cycles of the system and reduce maintenance costs since LEDs do not burn out or need to be changed as frequently as traditional lamp-based systems.

Approach light systems (ALSs) provide the basic means to transition from instrument flight to visual flight for landing. The FAA has been researching application of LED technology into current ALS applications.

LED threshold lamps have been developed that produce similar light intensities and colors to incandescent lamps used in a medium intensity ALS with runway alignment indicator lights (MALSR). Testing has occurred in FAA labs, independent labs, and at pilot program airports for temperature, vibration, icing, photometric chromaticity, and electrical characteristics with positive results.

FAA is also working to develop a feasibility study to determine if the LED technology can replicate a MALSR flasher and is looking to implement infrared technology to enable SVS, EVS, and HUD systems to see the lights.

EFFECTS ON AIRPORT PLANNING AND DEVELOPMENT

LED PAPI and approach lighting may affect the following major elements of airport planning and development programs:

Demand/Capacity

LED technology will enable systems to be available on a more frequent basis as the lamps have a much longer useful life as compared to incandescent lighting. When airports plan the acquisition of approach lighting through the FAA non-federal program, they should consider LED technology as way

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to reduce operating costs. FAA estimates a 2.5 times savings for an LED system versus a traditional system.

Facility Requirements

Many airports have aircraft using HUD technology. Implementation of LED approach lighting will need to consider aircraft types, which may drive the requirements of systems to include infrared technology so they can be seen on the HUD.

Timing and Implementation

The FAA has procured PAPI systems from equipment vendors and is in the initial phase of deploying test systems in the NAS. Airports may also purchase these PAPI systems from the vendor as part of the FAA non-federal program.

Multilateration and ADS-B (Surveillance)

Non-NextGen surveillance systems include virtual air traffic control systems and noise monitoring systems.

VIRTUAL AIR TRAFFIC CONTROL TOWERS AND RAMP CONTROL

A virtual air traffic control tower system uses a camera and multi-sensing systems to transfer data to a control staff in a remote tower center (RTC) where a human controller provides direction to aircrews instead of an out-the-window view from a traditional controller. The system simulates an "out-the-window" view that a tower air traffic controller would have and may be configured for a single controller or a small group of controllers to monitor operations at multiple airports in a wide area or a single airport from a single center.

Virtual ramp control functions similarly to a virtual tower in that points throughout the airfield can be monitored from a consolidated center, rather than having sectors of an airport monitored by an individual. The virtual ramp control system simulates an "out-the-window" view for various ramp areas, especially deicing areas, aircraft stands, and remote parking positions. *ACRP Research Report 167: Guidebook for Developing Ramp Control Facilities* discusses the use of virtual systems for tower and ramp control.

Both systems provide video feeds from various optical sensors and thermal cameras to provide real-time position information and consolidate multiple cameras into a single view. Virtual systems can be woven into multilateration, ASDE-X, and A-SMGCS systems to increase safety and efficiency.

Virtual ATC systems are an emerging technology with great potential for the future. Virtual systems are being deployed in Europe and the United Arab Emirates as a cost-effective means to provide ATC services to smaller airports.

In the U.S., the Leesburg Executive Airport has partnered with industry to demonstrate and evaluate remote tower technologies. The program is designed to demonstrate and evaluate the remote tower system for use at non-towered airports. The Virginia Department of Aviation is an advisory partner for the project, and the FAA is providing ATC personnel for testing and the evaluation of human factors in advance of operational trials.

Effects on Airport Planning and Development

Virtual systems may affect the following elements of airport planning and development programs:

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Demand/Capacity

Virtual airport traffic control tower (ATCT) systems may increase capacity and safety at airports that are currently not towered. Virtual ramp control systems may assist in reducing controller workloads, free up space in the ATCT cab, provide a viable a backup system for an ATCT, and serve as a training facility for the airport.

Facility Requirements

Facility requirement consider activity levels, visibility, and line-of-sight issues for ATC operations. Implementation of virtual systems may be required or desired to solve surface movement issues. Implementation of the systems will need to consider the siting of multi-sensor arrays and infrastructure to support the intended functions.

Timing and Implementation

Virtual ATC systems are emerging in Europe today and are a technology listed in the SESAR program. FAA has not included virtual systems into the NextGen program as of the writing of this document, as the concept is still undergoing testing and has a maturing concept of operation. Challenges exist in the U.S. due to the higher volume of GA traffic as compared to Europe. Current development activities are focusing in on this issue as well. It is envisioned that this technology will be embraced in the future.

NOISE MONITORING SYSTEMS

Noise monitoring systems utilize noise monitoring terminals (NTZ) to collect noise data emitted by aircraft and weather data. Noise monitoring systems can also collect radar or flight track data to pair with noise data to evaluate noise and causes across an airport system. Aircraft noise is not limited to flight-based aircraft operations; these systems can also monitor ground-based aircraft noise emitted by taxiing and parked aircraft. Noise monitoring systems are used to determine noise exposure at a community level and estimate the impact across a community which can influence legislation such as curfews and other regulations.

Effects on Airport Planning and Development

Noise monitoring may affect the following elements of airport planning and development programs:

Demand/Capacity

Noise monitoring systems are primarily used for operational and environmental compliance purposes. They do not drive demand or capacity, but the data can be used for multiple planning purposes.

Facility Requirements

Facility requirements may include a noise monitoring system for environmental monitoring, land compatibility, community relations, and operational purposes.

Alternatives

Multiple alternatives exist for the configuration of the system and the method in which it may be hosted. Alternatives will be based on the individual airport requirements for data analysis and community communication. Data sources may include historic FAA radar data (delayed 1 to 3 hours), SWIM, and ADS-B.

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Timing and Implementation

Noise monitoring systems have been deployed at airports since the 1990s. These systems continue to evolve from large-server single-user–based to Internet-based applications, incorporating new surveillance capabilities including multilateration and other trajectory and time-based system technologies.

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3 Incorporating NextGen into Airport Planning and Development

Effects of NextGen on Airport Planning Criteria and Guidelines

NextGen will have both near- and long-term effects on planning criteria and guidelines. The purpose of this chapter is to provide a working-level discussion of the likely changes to planning practices to reflect the expected gradual introduction of NextGen capabilities to airports. It is expected that, as NextGen becomes more widely implemented, additional and more specific guidance will become available from the FAA in the form of revised ACs, orders, and other documents. Not all of the information needed to plan for NextGen is likely to be available from the FAA Office of Airports. As is currently the case, much of the information needed to consider NextGen effects will be distributed by other sources, such as the FAA ATO and Flight Standards Service, RTCA, ACRP, and elsewhere. The appendices to this guidebook will provide additional resources for airport planners.

Airport Master Planning

With a typical time frame of 20 years, airport master plans are the primary vehicle for the facility, operational, and financial planning of major changes to airports, such as new runways, taxiways, terminals, or other significant additions to the airport infrastructure. For example, the required infrastructure related to airfield capacity improvements is heavily dependent on the standards for separations and clearances between runways, taxiways, obstructions, and other factors likely to change as a result of forthcoming NextGen capabilities. NextGen has already affected runway capacity through reductions in wake turbulence aircraft separations, improved arrival and departure capabilities, and optimization of the terminal airspace; it is anticipated that similar changes will continue to occur as the subsequent phases of this program unfolds. Although these changes are expected to have positive effects on airport planning, it is possible that some unexpected negative impacts might occur. Other airport planning considerations that will be affected by NextGen include:

- Obstacle clearances—The final-approach obstacle clearance surface (OCS) associated with certain advanced NextGen PBN instrument approach procedures (e.g., RNAV LNAV and RNAV RNP 0.3) is significantly wider and potentially more restrictive near the runway threshold than the OCSs associated with certain existing instrument approach procedures, as shown in Figure 3-1. In planning for these future PBN approaches, there may be a need for additional clearance of obstacles close in to the airport. Alternatively, under procedures outlined in FAA Order JO 7400.2K, *Procedures for Handling Airspace Matters*, the FAA could conduct an aeronautical study to determine whether obstacles under these wider OCS areas would have a substantial adverse effect on aeronautical operations (i.e., a determination of whether or not those obstacles would be a hazard to air navigation).

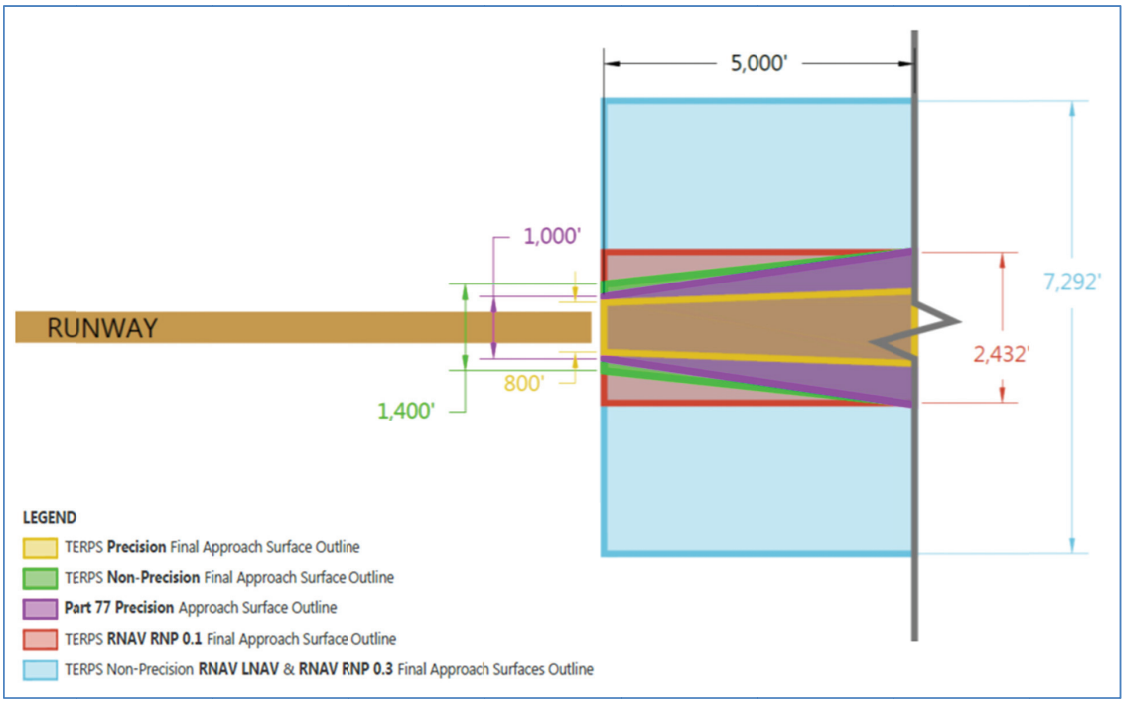
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- Potential environmental effects of new or relocated airfield facilities or enhanced flight procedures enabled by NextGen (positive and negative).
- The potential ability to eliminate on-airport facilities and equipment, such as ground-based NAVAIDs and ATCTs, may provide airport operators the ability to develop other revenue generating facilities. In addition to eliminating the physical infrastructure associated with these facilities, the elimination of certain NAVAID-critical areas and ATC line-of-sight restrictions could enhance operations on the airfield.
- Airport capital, operating, and maintenance budgeting and planning for investments to leverage the opportunities associated with certain NextGen technologies and procedures.

Increasing airfield capacity (including arrivals, departures, and ground movements) can have the potential to delay other capital expenditure (Capex) projects. This is one of the more positive aspects of NextGen as it relates to airport owners/operators and how they need to map out their future improvements to accommodate demand. It can result in major cost savings, or at least the deferment of major capital programs such as a new runway.

Surface operations with CDM are also important considerations in an airport master plan, particularly with respect to the planning and operations of the gates, hardstands, and ramp control.

Forecasting future activity is a typical element of a master plan. In a master plan study, NextGen should be viewed in terms of its ability to allow aircraft operational enhancements to occur. This can



Source: FAA Order 8260.3C, *United States Standard for Terminal Instrument Procedures (TERPS)*; FAA Order 8260.58A, *United States Standard for Performance Based Navigation (PBN) Instrument Procedure Design*; and Title 14, *Code of Federal Regulations*, Part 77, "Safe, Efficient Use, and Preservation of the Navigable Airspace."
Note: Cross section distance reference is for display purposes and does not correlate to a requirement for instrument procedure design.

Figure 3-1. *Comparison of final approach obstacle clearance surfaces.*

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take place in several forms. In some cases, for example at an airport in a mountainous environment, NextGen could enhance the aircraft operating environment to the point where more operations can occur because NextGen has increased access. In other cases, such as a high-density airport with significant peaking activity, NextGen could allow for closer spacing of aircraft operations, thus allowing for more activity to occur. Forecasting could also include estimates of future aircraft equipage trends (e.g., by airlines and by GA), which would help assess the types of PBN procedures that could be implemented.

During the master planning process, the defining of facility requirements and identification of alternatives should consider the potential capacity benefits and deferment of capital investments that could result from implementing NextGen technologies and operations. It is in this portion of the master plan that operations requirements are established to meet existing and future demand and in which many alternatives are considered before a final plan is established. For example, in some cases, successful NextGen implementation could result in an airport's ability defer the building of an additional runway and still meet operational needs. However, the master planning process should also consider the potential that community groups opposed to airport development may claim that future projects will not be needed because of long-term NextGen developments. Airport planners need to provide realistic assessments of the extent to which future NextGen capabilities will and will not substitute for facility development.

Because of the uncertainty about the timing and benefits of NextGen operational improvements, it may be prudent for a master planning process to include a scenario-based approach to integrating NextGen into planning. For example, mature, near-term technologies could be assumed for the main planning scenario, while longer-term less mature enhancements could be included as an alternative scenario so that multiple possible outcomes could be considered in the evaluation of the planning alternatives. Such an approach could be integrated into the master planning process of evaluating multiple future forecast levels, to consider the overall tradespace of possible future outcomes.

Airport Environmental Planning

NextGen has an array of benefits, including improved safety, increased capacity, enhanced fuel efficiency, and reduced delays. However, the implementation of NextGen is undertaken through a series of discrete projects, and any given project may make improvements under only some of these categories. As such, the environmental impacts of NextGen improvements are also best examined on a discrete basis. As such, this discussion on the environmental considerations associated with NextGen projects is structured based on the intended purpose and need of a specific NextGen project.

The typical environmental impact categories for NextGen projects are aircraft noise exposure and emissions, although other categories may be affected depending on a project. A secondary concern is that a project may disproportionately impact a low-income or minority community and be subject to the Executive Order on Environmental Justice.

Projects That Enhance Capacity

Increased capacity and delay reduction are effectively two sides of the same coin. The central question in the environmental analysis of a NextGen project is whether aircraft activity (i.e., demand) is likely to increase as a result of a capacity-enhancement project, that is, would the airlines schedule more flights as a result of the project? The answer to this question is probably "yes" for a capacity-enhancement project at a congested airport with latent demand where the project would relax a system constraint. At non-congested airports, where there is not latent demand or where the project would not relax a system constraint, the capacity-enhancement project is more likely to result in a reduction in aircraft

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delay, not an increase in demand. As such, a NextGen efficiency gain may not necessarily improve airport capacity in terms of decisions by airlines to schedule more flights.

This careful categorization of delay reduction versus capacity enhancement affects the potential extent of environmental impacts. In the case of projects that enhance capacity, there is a much broader range of environmental impacts from the newly enabled flight operations that need to be considered. Enhanced capacity includes obvious impacts like additional aircraft noise, emissions, and ground transportation impacts; but it also could include less-obvious impacts like the need to accommodate greater volumes of aircraft deicing fluid, which could require expanded runoff-mitigation procedures and facilities.

Projects That Reduce Delays/Fuel Consumption

While delay reduction generally entails decreasing the magnitude of environmental impacts, for environmental impacts that are defined by geospatial factors (i.e., air quality from non-reactive plumes and aircraft noise exposure) changes that move those impacts, or concentrate those impacts need to be properly investigated. As an example, a given airspace change might not affect the number of flights occurring, but those flights could be concentrated over a noise sensitive area. In this circumstance, there could be an adverse impact in that area. Likewise NextGen projects that result in a concentration of flight tracks are also likely to result in a concentration of aircraft noise impacts. Figures 3-2 and 3-3 provide an example of flight tracks more widely dispersed vs. more concentrated as a result of PBN implementation (Source: Kristi McKinney, Port of Oakland, ACI-NA "Airports at Work Conference," April 2016).

Typically geospatial impacts are evaluated relative to a threshold of significance that FAA has established for the respective environmental categories. However, environmental impacts could be deemed significant even if the adverse impact falls below an FAA-defined threshold of significance, but the

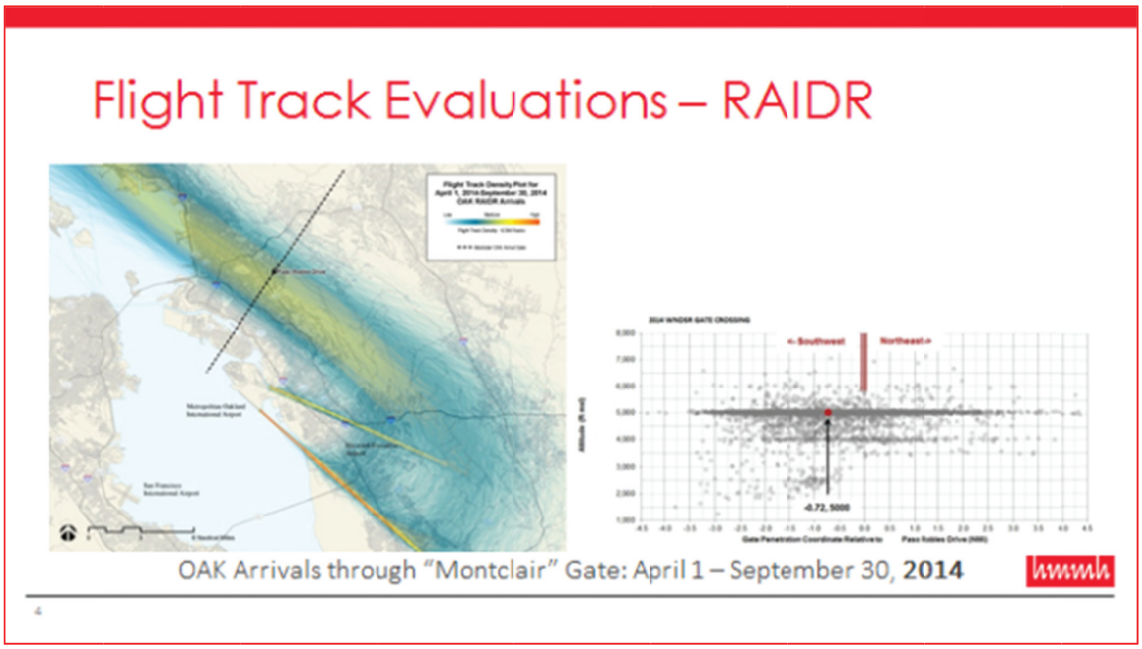


Figure 3-2. *Flight tracks before PBN implementation at Oakland International Airport.*

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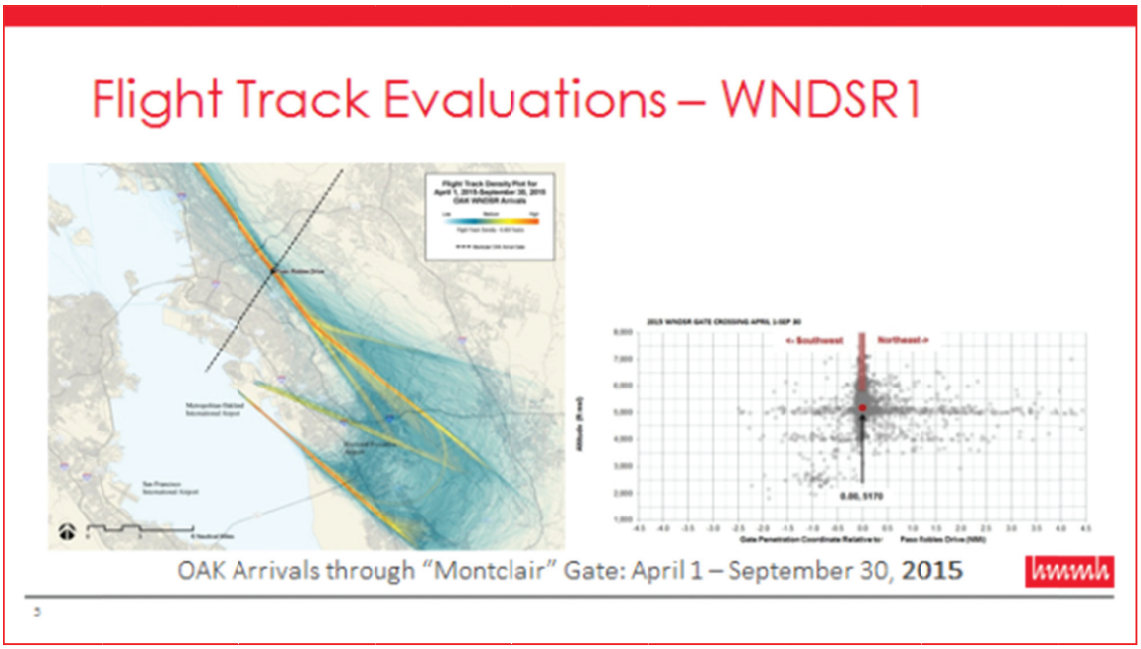


Figure 3-3. *Flight tracks after PBN implementation at Oakland International Airport.*

adverse impacts disproportionately affects economically or socially disadvantaged communities. Under Executive Order 12898, "Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations," a NextGen-related action that does not create a significant environmental impact (e.g., an increase of greater than 1.5 DNL inside the 65 DNL) could be deemed significant if the increases are disproportionately focused on low-income or minority communities. This is an important consideration because some neighborhoods in close proximity to airports could be characterized as being comprised of low-income or minority populations.

Projects That Improve Reliability

Many of the benefits of NextGen are associated with retaining near VMC operational capacity irrespective of meteorological constraints or nearby airspace congestion. Enhancing reliability with such improvements would likely be environmentally benign if the improvements applied to operations throughout the year. However, such improvements often target a very specific set of circumstances and, like the geospatial consideration associated with delay reduction, such reliability projects could also have impacts driven by geospatial considerations. As an example, perhaps a cross wind runway is used for 2% of annual aircraft operations, but a NextGen improvement allows that runway to now be used in a specific set of IMC and that runway would now be used for 4% of annual operations. While this is a small fraction of annual operations, this is a doubling of activity on this one runway and has the potential to cause a significant adverse environmental impact.

Projects Affecting Physical Facilities

Some NextGen facilities may impact physical facilities on the ground. Such impacts could range from building a new ATCT to enabling the abandonment of functionally obsolete air navigation facilities. These types of projects need to consider all the environmental impact categories defined in FAA Order 1050.1F that would normally be associated with a standard construction review for any other facility

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on the airport. Beyond the typical impact categories, one that may present a challenge for NextGen projects is the U.S.DOT's Section 106 regulation regarding historic structures. A structure only needs to be 50 years old to be eligible for listing in the National Register of Historic Places, meaning that coordination with a State Historic Preservation Officer could be required for the removal of impacted facilities that are more than 50 years old.

In considering future airport development plans, it is important that airport owners/operators effectively balance the airside and landside capacities commensurate with demands.

Environmental Review for NextGen Projects

Whether a NextGen project is being environmentally reviewed by the Office of Airports or the ATO, both offices are required to follow the NEPA (National Environmental Policy Act) and FAA's corresponding guidance, Order 1050.1F. However, while the Office of Airports frequently manages NEPA in the course of daily business, ATO staff is less likely to encounter NEPA on a daily basis. As such, there is the possibility of non-uniform application of Order 1050.1F between these two offices.

The topic most often expressed by airport operators during the NEPA process is the degree to which the FAA has involved airports as a stakeholder for airspace actions. In several cases, airports reported feeling that the FAA did not sufficiently engage key airport planning staff when conduction their environmental review, nor did they provide sufficient opportunity for public involvement. As a result, the FAA has updated its policies and guidelines for implementing NextGen projects (in particular PBN projects), in terms of how it engages airport operators and representatives of the public in the environmental review process. In addition, airports have been made increasingly aware of the importance to proactively reach out to ATO personnel and express an interest in being made aware of all potential NextGen projects.

It is also worth noting that in the FAA Modernization and Reform Act of 2012 Congress created a new legislative categorical exclusion Section 213 (c) (1) that states:

> (c) COORDINATED AND EXPEDITED REVIEW
> *1) In General—Navigation performance and area navigation procedures developed, certified, published, or implemented under this section shall be presumed to be covered by a categorical exclusion (as defined in section 1508.4 of title 40, Code of Federal Regulations) under chapter 3 of FAA Order 1050.1E unless the Administrator determines that extraordinary circumstances exist with respect to the procedure.*

This legislative categorical exclusion has since been incorporated in FAA Order 1050.1F, and is noteworthy because it may result in a less rigorous environmental review of a NextGen project than otherwise would have been expected. There has been significant controversy and pushback from the airport industry on this new legislative categorical exclusion, and there have been attempts by industry organizations such as RTCA to more clearly define and quantify "extraordinary circumstances."

State and Regional/Metropolitan Airport System Planning

Even before NextGen emerged as an FAA program, the potential of applying advanced communication, navigation, and surveillance (CNS) technologies to the benefit of aviation for state, regional, and metropolitan areas was obvious. The core concepts of using advanced CNS capabilities were tested in the FAA Capstone Project in conjunction with the state of Alaska Department of Transportation and Public Facilities, Division of Statewide Aviation. The lack of a usable instrument-navigation infrastructure and radar coverage, combined with the harsh weather conditions, caused Alaska to have a high rate of accidents and to frequently experience unreliable flight completion performance to rural communities without road access.

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Under phase 1 of the Capstone Project, a fleet of small commercial aircraft evaluated safety benefits of technologies during day-to-day operations in Alaska. The aircraft were fitted with IFR capable GPS receivers, a universal access transceiver (UAT) data-link system that enabled ADS-B and FIS-B including real-time weather, a multifunction display depicting terrain, other ADS-B aircraft, and weather graphics and text data. This bundle of avionics was installed on 200 aircraft used for commuter, charter, and mail flights in southwest Alaska.

Phase 2 of the program recognized that the minimum en route altitudes (MEAs) of the airways in southwestern Alaska were typically above the altitudes that the GA fleet of aircraft could safely fly without encountering icing conditions, even in the summer. Legacy en route NAVAIDS required land-based sites where power was available and maintenance crews had access. This land-based system resulted in a traditional IFR route structure that was not optimized for the terrain or typical small aircraft users in southeast Alaska. The technology that changed this and put the routes and approaches where they were needed, at low altitudes over the fjords, was the introduction of the WAAS and associated approaches. Capstone worked with the WAAS program office to help provide the WAAS signal to the Phase II Capstone equipment. Certification and initial installations of Capstone Phase II WAAS avionics took place in 2002.

On March 13, 2003, Special Federal Aviation Regulation (FAR) Part 97 was approved by FAA, authorizing en route IFR navigation in Alaska based solely on satellite technology without reference to ground radio navigation stations. In 2006, the FAA integrated the Alaskan Capstone project into the national ADS-B program.

A similar early success of NextGen technologies was initiated by the Colorado Department of Aviation to address reductions in capacity during IMC weather at several high mountain airports. This effort was chosen as one of the case studies in this project and is more fully described in Appendix B.

In 2005, the FAA, at the request of the state of Colorado Department of Transportation's (CDOT's) Division of Aeronautics, conducted an analysis of these delays and cancellations. IMC at those airports can reduce aircraft acceptance rates from 12 to 17 flights per hour, to only 4 per hour because of the lack of radar coverage and the need to impose procedural separation. The FAA study determined that the lack of surveillance contributed to reduced capacity during IMC, and identified multilateration as the preferred solution for providing surveillance to the Colorado mountain airports. In September 2006, the FAA and the CDOT's Division of Aeronautics signed a Memorandum of Agreement for a project to plan, install, test, and commission a surveillance system for the Colorado mountain airports.

WAM began initial operations on September 12, 2009, at Denver Center serving the Yampa Valley-Hayden, Craig-Moffat, Steamboat Springs, and Garfield County Regional-Rifle Airports. The WAM capability provide these airports with improved safety, efficiency, and capacity by allowing controllers to see aircraft that are outside radar coverage, saving time and money that would otherwise be lost due to flight delays and cancellations or diversions to other airports. On December 8, 2009, the FAA approved the next phase of the Colorado WAM. The phase 2 allows for the development and implementation of air traffic separation services, using multilateration and ADS-B surveillance for en route air traffic operations in and out of the mountain areas.

Another regional planning success is documented more fully in Appendix B, the "Puget Sound Regional Council—Preparing Busy Airports for NextGen Technology."

This project represents the first NextGen study focused on a system of regional GA airports in the NAS and was funded via airport improvement program (AIP) grants. From the FAA's perspective, the Puget Sound Regional Council (PSRC) was an ideal organization to fulfill the objectives of this grant. It already served as a forum for communication with airports in the region. There were also clear economies of scale to be achieved by studying the impact of NextGen on many closely related airports at the same time. This project was probably the most comprehensive study to date of NextGen

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implementation at smaller GA airports from a system perspective. In addition, a number of NextGen technologies and operational improvements were being concurrently evaluated at the Seattle-Tacoma International Airport (SEA).

The FAA New England Office of Airports has long been active in encouraging regional planning activities, supporting regional efforts during the 1990s, funding the "New England Regional Airport System Plan," (NERASP) published in 2006 (http://www.faa.gov/airports/new_england/planning_capacity/airport_system_plan/), and the "New England Regional Airport System Plan, General Aviation—Phase 1," published in 2015 (www.pvdairport.com/documents/planning%20docs/neraspgasummary brochure.pdf). The latest document provides a broad inventory of the needs of New England's GA airports but also identifies areas where NextGen applies, particularly in the importance of maintaining OFAs to achieve the lowest possible landing minimums. The conclusions and observations of the report[1] cited a survey of 175 business aviation operators contacted during the study; 50% felt that improved instrument landing capability was required throughout the region.

While much of the FAA funding for planning activities is directed to individual airport planning activity, there is increasing interest in using regional planning studies to help address congestion in the NAS. In December 2009, the Government Accountability Office published report GAO-10-120, "Regional Airport Planning Could Help Address Congestion If Plans Were Integrated with FAA and Airport Decision Making." The report highlighted the potential for NextGen capabilities to relieve pressure on congested large airports and recommended that the Secretary of Transportation direct the FAA to create a review process for regional airport system plans (RASPs) and use its current authority to give priority to funding airport projects that are consistent with RASPs.

Although the application of NextGen capabilities to state, regional, and metropolitan planning has long been recognized, organizational issues arising from the difficulty of coordinating planning among different airports within a region, each having different sponsors and planning objectives, has inhibited the widespread adoption of regional planning. Nevertheless, the examples cited here illustrate the importance of, and potential benefit from, applying NextGen capabilities to state, regional, and metropolitan airport planning initiatives.

Airspace Redesign Studies (e.g., OAPM or Metroplex)

Since the 1990s, there have been multiple airspace redesign studies and efforts conducted in the NAS, including the development of satellite-based GPS overlay SIAPs, the National Airspace Redesign Program, RNAV and RNP SIAPs, RNAV SIDs, and RNAV STARs.

Initial implementation of RNAV and RNP occurred through multiple initiatives and grassroots local efforts lead by airlines, airport authorities, RTCA, FAA National Flight Procedures Division, and the FAA's PBN Office (formerly the RNAV/RNP Office) to address capacity, efficiency, and environmental-related problems. Other programs were established on a federal level by the ATO. Table 3-1 highlights some of these early programs.

These early efforts drove the FAA to begin developing and implementing RNAV (LNAV, LNAV/VNAV, LPV, and LP) and RNP SIAPs at airports throughout the NAS. The implementation of RNAV SIAPs is managed by the FAA National Flight Procedures Office located at the Mike Monroney Aeronautical Center (MMAC) in Oklahoma City, OK. This group is responsible for the development, publication, and maintenance of all flight procedures in the NAS. IFPs are developed on a 56-day publication cycle. Each procedure published is reviewed on a biannual basis to ensure safety with new RNAV implementation as part of the ongoing activities associated with this office. These efforts lead to a sharp increase in published PBN-based procedures available to pilots for use. However, in many cases, procedures

[1] NERASP GA Phase 1 Conclusions and Observations, www.pvdairport.com/corporate/planning.

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were developed without local coordination, thereby leading to unusable procedures or situations in which airports were not compliant with FAA airport design standards.

Today airspace redesign occurs through multiple efforts including FAA Metroplex, airport master planning, airport systems planning, land use and aircraft noise exposure studies, and locally initiated efforts. The following sections provide further detail on these processes.

Optimization of Airspace Procedures in the Metroplex

The early terminal area PBN initiatives identified in Table 3-1 were initiatives established from a single airport perspective. As PBN implementation matured, the concept of implementing PBN for multiple airports in a metropolitan area was suggested by industry. This term became known as a Metroplex and is defined as a geographical area covering several airports serving a major metropolitan area and multiple aviation stakeholders. In response to industry demands, the FAA established the OAPM or Metroplex.

Table 3-1. Early NextGen implementation initiatives.

PROGRAM	LEAD	AIRPORT	ACHIEVEMENT	APPROXIMATE IMPLEMENTATION DATE
RNP Approach Development	Alaska Airlines	Multiple Alaskan Airports	Development and Implementation of RNP Approach Procedures at Alaskan Airports	1996
GPS Overlay Program	FAA National Flight Procedures Division	NAS-Wide Airports	Development of very high frequency omnidirectional radio range (VOR) Overlay GPS-Based Approaches at NAS-Wide Airports	Mid-1990s
Denver RNAV Implementation	United Airlines	Denver International Airport	Implementation of RNAV SIDs and STARs Improving Capacity, Efficiency and Noise Compliance	2012
Las Vegas Optimization	Initiated by McCarran International Airport then transitioned to MITRE and FAA RNAV/RNP Office	Las Vegas Area Airports	Implementation of RNAV SIDs and STARs	2007

continued

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Table 3-1. Continued

PROGRAM	LEAD	AIRPORT	ACHIEVEMENT	APPROXIMATE IMPLEMENTATION DATE
National Airspace Redesign	FAA	U.S. En Route Airspace	Development of Low-Altitude ("T") PBN Routes & High-Altitude ("Q") PBN Routes) and Efficient PBN En Route Structure	Mid-2000s
Boston Overflight Noise Study	FAA and Massport	Boston Logan International	Implementation of RNAV SIDs and STARs and GPS Charted Visual Procedures for Noise Abatement	2009
New York/ New Jersey/ Philadelphia Airspace Redesign	FAA and Airlines	New York John F. Kennedy, La Guardia, Teterboro, Newark, Philadelphia	Redesigned Terminal Airspace, Arrival Fixes, Departure Fixes, Implementation of RNAV SIDs and STARs	2007
Atlanta RNAV Implementation	FAA RNAV/RNP Office, Delta Airlines	Atlanta Jackson Hartsfield International	First Successful FAA ATO RNAV/ RNP Initiative	2006
John Wayne Orange County Noise Abatement Departure	FAA, Orange County, Alaska Airlines	John Wayne Orange County	Implementation of DUUKE RNAV SID for Noise Compliance (Currently Named the STREL RNAV SID)	2009
Minneapolis/ St. Paul RNAV Implementation	Minneapolis/ St. Paul Metropolitan Airports Commission, Northwest Airlines	Minneapolis/ St. Paul International Airport	Development of RNAV Departure for Runway 35 to Improve Noise. Initiative Lead to the Development of Multiple RNAV SIDs and STARs by FAA	2010
Seattle Greener Skies	Boeing, Alaska Airlines	Seattle International Airport	Dual Simultaneous RNP Approaches on Dependent Closely Spaced Runways	2010

Source: Ricondo & Associates.

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The intent of the Metroplex program is to safely improve the overall efficiency of the NAS by increasing efficiencies at metropolitan areas with multiple airports and complex air traffic flows. The goal is to improve the way aircraft navigate these complex areas to make flight routes and airport access more efficient. Designing the airspace based on NextGen capability provides the potential to reduce aircraft operational costs, fuel burn, and emissions, and improve on-time performance at the Metroplex and between the departure and arrival airports.

The advent of the Metroplex initiative represented the first time that satellite airports were considered in the process. The program provides a more structured and repeatable approach to implementation of primarily PBN-based SIDs, STARs, and in some cases RNP approaches.

The process includes a Study Team Phase (approximately 3 months) and Design and Implementation Phase (approximately 24 months or longer). The process includes subject-matter experts (SMEs) from other areas during the study team phase to get a different perspective on how to improve operations other than local, legacy-based procedures. These SMEs often include airline technical performance representatives that are needed to ensure that the PBN procedures developed are actually flyable by a diverse fleet of aircraft. Typically, the lead carrier at the airport will take on this responsibility, but it can be shared. Lessons learned in this regard are often captured and applied to subsequent Metroplex efforts.

The FAA initially identified 21 Metroplexes in the NAS where near-term benefits could be realized through the implementation of NextGen. Through industry coordination, this initial list was prioritized and reduced to 12, representing Phase I of the program. Table 3-2 highlights the status of the Metroplex program.

Table 3-2. FAA Metroplex program snapshot.

METROPLEX	DAILY AVERAGE BY FISCAL YEAR		PROJECTED ANNUAL BENEFITS			
	2014 TOTAL OPERATIONS	2014 SCHEDULED FLIGHTS	FUEL SAVINGS (GALLONS OF FUEL IN MILLIONS)	VALUE OF FUEL SAVINGS (FUEL COSTS IN MILLIONS)	CARBON SAVINGS (METRIC TONS OF CARBON IN THOUSANDS)	METROPLEX PHASE
Atlanta	3,097	2,329	2.2	$6.3	18.8	Implementation
Charlotte	2,636	2,026	3.3	$9.4	28.0	Implementation
Cleveland-Detroit	2,341	1,435	3.4	$9.7	28.9	Evaluation
D.C.	2,760	2,001	2.1	$6.1	18.2	Post Implementation
Denver	2,757	1,530	0.6	$1.8	5.4	Design
Houston	2,609	1,615	1.8	$5.3	15.7	Complete
Las Vegas	2,044	859	2.6	$7.5	24.8	Design
North Texas	4,046	2,080	2.6	$7.5	22.4	Complete
Northern California	3,220	1,824	1.3	$3.7	11.0	Post Implementation
Phoenix	1,796	1,049	1.4	$3.9	11.6	Evaluation
South Central Florida	6,735	2,602	5.4	$15.5	46.1	Design
Southern California	5,897	2,557	2.7	$7.8	23.3	Evaluation

Source: FAA webpage (www.faa.gov).

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Airport participation on PBN design and implementation initiatives has varied widely. In early projects, now known a "legacy projects," airports led the initiative and the public was involved throughout the design and implementation of the technology. However, the new initiatives have been, or are led, strictly by the FAA. In some instances, the airports have been omitted entirely from the process or have been engaged at the last minute. This lack of engagement has resulted in misunderstanding of NextGen, mistrust from the airport stakeholders, and in some cases, opposition from some local surrounding airport communities.

Professional and industry groups including the RTCA and the AAAE have established technology and NextGen working groups. These working groups have highlighted these problems and are calling for reform. In October 2014, the RTCA NAC released its *Blue Print for Success to Implementing PBN*, which details the airports role as part of implementation process of PBN procedures.[2] The FAA has concurred with many of the recommendations established in the PBN *Blue Print for Success to Implementing PBN* document and is in the process of revising its implementation process.

The Metroplex Process

The current Metroplex process takes approximately three years to complete but has widely been considered beneficial. It has been extensively applied and refined in numerous PBN procedure design projects across the U.S. The current process directly incorporates the EA process as part of NEPA in the development of the PBN procedures. It also includes characterization of current-day operations to establish a sound baseline for PBN procedure design and to assess the benefits of PBN procedures. It includes as well a formal process for deploying PBN procedures. Steps are summarized in Table 3-3 below.

The Metroplex process conducts extensive analysis to characterize baseline operations and to support the design and benefits analysis of the PBN procedures. Analyses may include the following:

- Characterizing STAR traffic flow conflicts with other flows;
- Characterizing the level segments of arrival or departure flights;
- Quantifying the number of aircraft in each arrival traffic flow to the airport, and major and satellite airport arrivals;
- Analyzing the traffic loading of airspace sectors;
- Comparing the tracks and published routes;
- Conducting conceptual airspace and procedure design and simulation;
- Analyzing traffic management initiatives and playbook route usage;
- Analyzing annual traffic counts and aircraft type;
- Developing traffic density charts; and
- Evaluating the fleet mix from PBN capability report.

The Metroplex process may also include outreach to impacted stakeholders via regularly scheduled briefings for each milestone in the design and development phase.

[2] RTCA, *Blueprint for Success to Implementing Performance-Based Navigation*, Report of the NextGen Advisory Committee, October 2014.

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Table 3-3. Summary of FAA Metroplex program IFP design process.

STEP	DESCRIPTION
Study and Scoping 3 months	The Study Team conducts meetings with facility and industry representatives to identify issues with the legacy procedures and airspace, and to propose solutions. The Study Team produces conceptual designs of proposed procedures and a high-level assessment of the benefits, costs, and risks of the procedures.
Design 6–9 months	The Design and Implementation (D&I) Team conducts integrated airspace and procedure design based on the findings of the Study Team. The D&I Team includes representatives of the lead aircraft operator. Additional analyses, including human-in-the-loop simulations, may be conducted to support this work.
Evaluation 12–18 months	The D&I Team conducts operational modeling, safety management system (SMS) analyses, and environmental reviews with representatives of the lead carrier. The Evaluation phase may also continue design analyses. Evaluation includes a project kickoff, completing the design of the procedures, and validating the final operations. The Evaluation phase includes the SMS process: facilitating the safety risk management (SRM) panel, completing the SMS process, and creating and completing the final SMS documentation. Evaluation includes drafting the complete EA as required by NEPA, including the purpose and need, alternatives, affected environment, and the environmental consequences section with a goal of achieving the final EA finding of no significant impact (FONSI).
Implementation 9–15 months	The D&I Team works with the representatives of the lead aircraft operator to conduct all steps for implementation, including flight inspections, publishing procedures, and planning and executing training. Implementation includes developing a procedure implementation plan, a training plan, flight checks, stakeholder coordination, training, and procedure implementation.
Post-Implementation Monitoring and Evaluation 2–3 months	The D&I Team reviews the benefits and impacts of the implemented procedures and modifies them as needed.

Airport Master Planning

Airspace redesign analysis as part of the master planning process is a prudent way to address potential airspace improvements that could lead to increased capacity, efficiency, environmental compliance, and safety.

The master plan process can be augmented to identify the establishment of existing airspace structure, procedures, aircraft type, and equipage levels as part of the inventory. Requirements and concepts can be developed based on the examination of the multiple types of GPS SIAPs usable to the fleet operating at the airport. Final alternatives should consider airport design compliance, obstacle removal, aeronautical survey requirements, and identification of preferred airspace design and procedures to benefit the airport.

The master planning process allows for close coordination with FAA airports' personnel in the planning and implementation of NextGen technologies. Implementation of new SIAPs will require a formal request to the National Flight Procedures Office. This request can be done through the master plan and airport layout plan approval process or independently through the FAA IFP Gateway. Appendix E provides more detail on the IFP Gateway webpage.

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Airport System Planning

NextGen airspace redesign from a system perspective is an approach recently taken by the PSRC. This project, financed in part through FAA AIP funds, explored how NextGen technologies and programs could be applied to the GA system of airports, not considered in a Metroplex study, surrounding SEA.

The study was split into two phases. Phase I focused on nine general aviation airports within King, Pierce, Snohomish, and Kitsap counties. The analysis identified existing conditions and constraints, requirements and needs, benefits of implementing NextGen initiatives, conceptual development and evaluation, and recommended improvements required to prepare these airports for the implementation of NextGen technologies.

Phase II focused on NextGen technologies and programs that could be applied to enhance safety, improve efficiency, provide enhanced access to the airport system, and reduce congestion in the Puget Sound region airspace. Some of the concepts considered included development of RNP approaches, development of GPS terminal area routes to allow access to and from the smaller airports without interfering with SEA, and implementing RNAV-based approaches at multiple airports.

This approach provides benefits to multiple smaller airports that may be omitted from airspace projects associated with larger airports. It is an approach that can also be applied into any type of system plan within a state or a high-density metropolitan area to address the issues and needs of the users in a coordinated manner.

Land Use and Environmental Planning

NextGen technologies can be used to address multiple issues involving land use and environmental issues. These efforts are typically associated with an EA, Environmental Impact Statement (EIS), or a FAR Part 150 Study and involve the identification of PBN concepts for the development of alternatives to mitigate impact on surrounding communities. Typical capabilities involved include optimized approach profiles, development of procedures with flight paths over compatible land use areas, and development of procedures to allow aircraft to climb to an established altitude prior to turning over noise sensitive areas.

Concepts and alternatives developed as part of these planning efforts typically are forwarded to the FAA for approval and implementation. Implementation can be a slow process. However, many airports have been able to harmonize theses effort with the Metroplex initiative and other FAA-led airspace projects to have the approved airspace concepts implemented in a more expeditious manner.

Local Initiatives

Airports may work locally with their operators to establish airspace initiatives. Procedure requests can be made to the FAA through the FAA IFP Gateway or through coordination with the Regional and National Flight Procedure Office. At minimum, a lead operator is needed to move a procedure forward. The lead carrier is typically responsible for confirming that the NAV database has been updated and for reviewing aeronautical charts for errors. In this role, a lead aircraft operator is instrumental in moving a PBN procedure forward and achieving consensus that the design is current and ultimately flyable by many aircraft. This is distinct from design of conventional (non-PBN) procedures that do not require extra evaluation of flyability. Development of new procedures is a process that may take two to three years and requires a current aeronautical survey and adequate airport facilities meeting airport design standards. Obstacle identification and mitigation efforts should also be considered as part of this process to ensure the maximum benefits can be realized through the development of new procedures.

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Airport Financial and Strategic Planning Issues

The uncertainty about the timing of NextGen implementation and its potential benefits to airports has been a factor in many airports' reluctance to incorporate NextGen into their existing financial and strategic planning processes. As a result, until recently, many airport operators have chosen to delay engaging with the NextGen planning and implementation process.

This section will review the operating and capital financial issues, as well as the likely impacts of NextGen on airport strategic planning. Recognizing that the current understanding of financial and strategic impacts of NextGen may change as programs mature, it is essential that airports maintain ongoing awareness of FAA policies and other issues affecting financing of NextGen projects. Also, in addition to the direct financial impacts, there may be ancillary financial needs stemming from FAA- or airline-sponsored NextGen projects. An example might be an airport's decision to undertake environmental analyses of proposed flight procedures initiated by others, even though the airport operator has no direct role in the development or implementation of the procedures. Most airport strategic planning will be performed in updates to master plans, or in specially dedicated strategic planning processes.

As a general matter, NextGen navigation and ATC infrastructure capital and operating costs will be primarily funded by a combination of the FAA and the aircraft operators using the procedures. Many programs, like ADS-B Out, will require ground-based and aircraft-based capital equipment that will be funded by FAA, airlines, and other aircraft operators. The costs of developing most NextGen ATC procedures will generally be funded by the FAA ATO. Some airline-specific ATC special use procedures may be developed by airlines. Operating costs for these capabilities will, likewise, be usually funded by FAA and aircraft operators. However, there are a number of circumstances under which airports could face costs for Next Gen capabilities as outlined below:

1. The first is a case in which the airport may desire a capability that FAA does not support, like a GBAS installation to enable lower minimums and precision approaches to multiple runways, or the provision of airport vehicle tracking capability using core technologies like FAA's ADS-B or Multilateration, which FAA will permit an airport operator to install, but will not pay for. In those cases, airports need to identify funding streams for the project, which may include AIP grants if the item is eligible, user fees, and internally generated capital or bond issues (especially if the capability is part of a major capital improvement project, like a new or rehabilitated runway).
2. The second case is one in which the provision of a new capability triggers the need for the airport to make investments in airfield facilities or other enabling projects.
    a. One example is that NextGen surface management procedures such as departure metering may require the airport to invest in additional apron areas and hold pads to facilitate the efficient departure metering.
    b. Another example is that a NextGen improvement that reduces approach minimums could trigger a need for an airport operator to undertake obstruction mitigation to meet the requirements of the new, lower minimum approach. Many airport operators have been surprised to find that a new approach, having lower minimums, has required them to spend unanticipated funds to maintain the more stringent obstacle clearance surfaces for the new approach. The obstacle clearance criteria are not a new requirement of NextGen: they are the same criteria that would apply to a new approach supported by land-based NAVAIDS. However, because the airport operator does not need installation of a new piece of equipment to achieve the lower minimum approach, some airports have mistakenly assumed that they would not assume any costs to get the new approach.

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3. A third case is one in which the airport wishes to take advantage of revised separation standards as part of its capital program and construct a runway that would not provide a cost-beneficial improvement in capacity under current separation rules. It is possible over the long term that significant reductions in separation standards might enable construction of a runway that would otherwise be of limited utility, given current separation standards. Although the longer-term potential for significant reductions in runway separations is presently remote, airport operators that are considering long-term capital development in master plans need to give consideration to this potential. At the same time, some may believe that future NextGen capacity improvements would defer or eliminate the need for planned capital programs. The likely realistic potential for dramatic improvements in capacity through NextGen is remote and uncertain. However, airport planners need to give fair consideration to this issue within the scope of their present master planning time horizon and the maturity of evolving NextGen technologies, as follows:
   a. The useful life of a master plan is about 10 years, which is within the foreseeable range of CSPO efforts.
   b. The ability to use "paired approaches" to significantly improve throughput at airports with closely spaced parallel runways is possibly something that can be achieved in the mid-2020s and beyond.
   c. However, it may not be appropriate for a master plan to be speculative on future, undefined potential CSPO enhancements (e.g., simultaneous operations on parallel runways spaced 700 feet apart) or spending substantial time/resources on such efforts beyond a basic exploration of the benefit tradespace.
4. A fourth case is one in which an initiative, which may be undertaken by FAA and air carriers, can be facilitated or improved if the airport operator assumes an active role in the development. One important example of this is in cases where FAA has undertaken NextGen-enabled changes to flight procedures, which have generated community opposition on environmental grounds. Several controversial examples at major airports have drawn airport operators into the public environmental discussion, even though the proposed changes in flight procedures have not been airport proposals and they have no formal role in the environmental review process. In such cases, airport operators may find that they need to assume costs for modeling, public participation processes, litigation, or other expenses in excess of their normal role in the FAA's development of flight procedures. Other cases might involve NextGen initiatives, like improved surface operations, where the active participation of the airports by undertaking revisions to taxiways, constructing remote holding aprons, engaging consultants, or otherwise making investments in capital or operating expenses can greatly improve the success of the program.
5. A fifth case is one in which there is a need for airport planners to consider the broad potential that NextGen improvements in aircraft traffic flows, on-time performance, and airfield capacity could flow through the airports systems and affect the requirements for gates, terminal capacity, curbs, support facilities, parking, and roadways. In many cases the improvements from NextGen projects will tend to reduce delays, particularly during instrument conditions and will increase the airport's throughput to become nearer to the visual meteorological capacity of the airport. Given that airport terminals, curbs, roads, and support facilities are normally designed for a VMC design day, serious impacts on the non-airside facilities at most airports would not be expected. However, airport planners need to consider the possible effects of NextGen improvements on all airport facilities and to be prepared to modify non-airside facilities, if needed.

In summary, the vast majority of NextGen expenses will accrue to FAA, air carriers, and other aircraft operators. With the exceptions noted above, airport operators should not face significant financial issues from NextGen programs. They must, however, be prepared to assess the longer-term strategic issues that may be affected by NextGen, particularly in their long-term planning horizon.

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4 Applicability of NextGen to Medium and Large Airport Planning and Development

This section will clarify the traditional airport size classifications with respect to the types of NextGen capabilities and issues that are likely to be of most import to the larger facilities. The guidance for medium and large airports will consider the typically greater availability of staff and consultant resources at medium and large airports.

As summarized in Appendix C, the impact that NextGen will have on a particular airport will depend upon its size and complexity, location, layout, type and level of air service, and role in the NAS. Because NextGen technologies and operational improvements are designed to increase system capacity, reduce aircraft delays, reduce aircraft emissions, and enable more reliable flight schedules, the biggest beneficiaries of NextGen are likely to be the users and operators of busy medium and large airports.

For medium and large airports, much of the focus of near-term NextGen improvements has been on reducing aircraft emissions, reducing aircraft separations, reducing the required spacing between parallel runways for conducting independent and dependent instrument approaches, and reducing congestion on the airport surface and in the surrounding airspace.

This chapter identifies the various NextGen capabilities that could enhance operations at medium and large airports and how these capabilities may influence planning and development initiatives at these airports. It differentiates between the NextGen initiatives that have global application at most medium and large airports from those that have application only under certain operating conditions and environs.

Definition of Medium and Large Airports

The National Plan of Integrated Airport Systems (NPIAS) categorizes public use airports by type of activities, including commercial service, primary, cargo service, reliever, and GA.[1] For purposes of this guidebook, medium and large airports generally include the following NPIAS Classifications:

- Large Hub Primary Commercial Service (1% or more of annual passenger boardings).
- Medium Hub Primary Commercial Service (at least 0.25% but less than 1% of annual passenger boardings).

While medium and large private-use and military airports could benefit from NextGen technologies, their application is excluded from these discussions. Table 4-1 presents the overall NPIAS airport classifications. The opportunities to enhance the NAS through the implementation of NextGen technologies and operational improvements can vary significantly among airports, regardless of an airport's size, role, and classification. There are numerous factors that will influence the direct and indirect ben-

[1] Federal Aviation Administration, National Plan of Integrated Airport Systems, http://www.faa.gov/airports/planning_capacity/passenger_allcargo_stats/categories, accessed January 7, 2016.

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Table 4-1. FAA airport classifications.

Categories of Airport Activities

Airport Classifications		Hub Type: Percentage of Annual Passenger Boardings	Common Name	
Commercial Service: Publicly owned airports that have *at least 2,500* passenger boardings each calendar year and receive scheduled passenger service **§47102(7)**	**Primary:** Have *more than 10,000* passenger boardings each year **§47102(16)**	**Large:** 1% or more	**Large Hub**	Medium and Large Airports[1/]
		Medium: At least 0.25%, but less than 1%	**Medium Hub**	
		Small: At least 0.05%, but less than 0.25%	**Small Hub**	Small Airports[1/]
		Nonhub: More than 10,000, but less than 0.05%	**Nonhub Primary**	
	Nonprimary	**Nonhub:** At least 2,500 and no more than 10,000	**Nonprimary Commercial Service**	
Nonprimary (Except Commercial Service)		Not Applicable	**Reliever §(47102(23))** **General Aviation (47102(8))**	

[1]Medium and Large Airports generally include the following NPIAS Classifications:

- Medium Hub Primary Commercial Service (at least 0.25% but less than 1% of annual passenger boardings).
- Large Hub Primary Commercial Service (1% or more of annual passenger boardings).

Small Airports generally include the following NPIAS Classifications:

- Nonhub Primary Commercial Service (>10,000 annual passenger boardings, but less than 0.25% of total passenger boardings in the United States).
- Non-primary Commercial Service (no more than 10,000 annual passenger boardings).
- General aviation/reliever airports.

Source: NPIAS, Federal Aviation Administration.

efits associated with each of the various NextGen initiatives for any given airport or system of airports. Therefore, the application of NextGen technologies and operational improvements, and the resulting benefits, will be unique for each airport.

While each airport's ability to leverage NextGen operational improvements is unique, the information contained herein is intended to distinguish the NextGen initiatives that are most likely to be applied at the various airport types and sizes, based on, but not limited to, the following criteria:

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- Benefit/Cost,
- Aircraft Equipage Limitations,
- Aircraft Fleet Mix and Performance Characteristics,
- ATC and Flight Crew Training Requirements,
- Airfield Configuration,
- Airspace Constraints and Airport Dependencies,
- Surrounding Terrain and Obstacles, and
- Operational Needs.

Each of these criteria is discussed in greater detail in Appendix C.

FAA NextGen Technologies and Initiatives Applicable to Medium and Large Airports

The deployment of FAA's NextGen initiatives and supporting technologies has been, and will continue to be, mainly focused on medium and large airports. This section provides an overview of the applicability of the FAA's NextGen initiatives and support technologies at medium and large airports and describes the factors that may determine the applicability and benefits of their deployment. It also describes direct and indirect effects on planning and development at medium and large airports.

While the basic approach and reference sources are the same for all sizes of airports, this section will emphasize those NextGen issues affecting efficiency and capacity, long-term infrastructure needs, surface management, noise concerns, and other interests typical of airport planning practitioners at medium and large commercial service airports. For each of the NextGen technologies affecting these issues, this chapter will discuss the following airport implications:

- Facility impacts,
- Operational impacts,
- Environmental impacts, and
- Financial/business impacts.

PBN and Improved Landing Systems

Because PBN procedures can take many forms, this chapter will concentrate on those PBN procedures of greatest interest to airport planning practitioners at medium and large airports: (1) RNAV SIDS and STARs, (2) RNAV and RNP instrument approach procedures, and (3) RNAV-enabled departure separations such as ELSO, EDO, and UDOS.

Facility Impacts

RNAV SIDS and STARs—RNAV SIDS and STARs have no direct impact on-airport facilities.

RNAV and RNP Instrument Approach Procedures—RNAV and RNP instrument approach procedures are satellite-based approach procedures and do not rely on ground-based navigation aids such as the

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ILS localizer and glide slope antennas. At most medium and large airports that already have RNAV and RNP approach procedures, ILS approaches still exist.

Ultimately, FAA is planning that RNAV and RNP approach procedures may replace many ILSs, in which case there would be no need to have the associated localizer and glide slope antennas on the airfield. Some ILS installations may continue to serve as backups to guarantee adequate instrument approach procedures, but the transition away from ILS could take many years. However, other components of low-visibility ILS approaches, such as approach lights, obstacle-free areas, or touchdown zone lighting would continue to be required, depending on the desired minimums.

RNAV-Enabled Departure Separations—In order to maximize the effectiveness of RNAV-enabled departure procedures, there may be a need to provide additional facilities such as hold pads or parallel taxiways. ELSO, EDO, and UDOS all require adequate holdpads and parallel taxiways to facilitate the departure staging and sequencing necessary to achieve the benefits of multiple divergent headings provided by these PBN-enabled departure separation procedures.

Operational Impacts

RNAV SIDS and STARs—The implementation of RNAV SIDS and STARs may provide the ability to enhance the operational throughput at a medium or large airport that is constrained by the capacity of its surrounding terminal airspace. For example, with the increased precision provided by RNP, coupled with the benefits associated with RNAV, additional independent arrival and departure routes to and from medium and large airports may be established, thereby increasing airspace capacity by allowing greater segregation of arrivals and departures by origin/destination and reducing conflicts with flight paths to and from nearby airports in the region.

RNAV and RNP Instrument Approach Procedures—RNAV, RNP, and GBAS-based instrument approaches could increase operational capability and reduce landing minimums at airports that currently cannot accommodate ILS approaches because they can't meet the critical-area siting requirements for either the ILS localizer or glideslope antenna. In addition, if these PBN-based instrument approaches could replace ILS approaches, there could be operational improvements at medium and large airports that currently have operational constraints to prevent aircraft from encroaching upon localizer or glideslope critical areas.

GBAS is not currently programmed by FAA at any airport. At the request of airlines, FAA is conducting a benefit/cost to investigate investing with facilities and equipment (F&E) funds in GBAS, but there has been no determination yet. RNP approaches are implemented primarily at medium and large airports unless there are specific airspace where terrain constraints justify implementing the RNP approach at a smaller airport.

The FAA has been implementing the LP, LPV, and LNAV/VNAV approaches to enhance operations during inclement weather at medium and large airports that have a substantial volume of GA and regional aircraft operations to supplement existing instrument approaches and/or to provide redundancy.

RNAV-Enabled Departure Separations—ELSO, EDO, and UDOS all are designed to provide additional departure throughput at airports with limited departure headings (e.g., at airports at which departures from different parallel runways must fly to the same waypoint for noise abatement purposes). These NextGen programs provide the benefits of multiple departure headings by either (1) requiring less divergence between departure headings off the end of the runway or (2) moving the point where divergence occurs out from the runway end to the nearest waypoint.

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Environmental Impacts

The possibility of significant environmental impacts could arise from implementation of RNAV SIDS and STARs, RNAV and RNP instrument approach procedures, and/or RNAV-enabled departure separations. There could be significant concerns over increased noise exposure at some medium and large airports because of the associated concentration of flight paths over a narrow geographic area characteristic of all three of these PBN procedures. The ATO environmental screening process does not normally consider noise impacts below the FAA threshold of significance [day-night average noise level (DNL) 65]. However, even noise impacts below that DNL 65 value can provoke significant adverse community reaction.

On the plus side, these PBN procedures following precise flight paths can be designed to avoid noise sensitive areas and possibly provide multiple departure headings. In addition, RNAV and RNP approaches also have the potential to reduce aircraft emissions and aircraft noise exposure.

The FAA's plans call for PBN procedures to be implemented at many if not most medium and large airports. The FAA has also undertaken the implementation of PBN procedures through its Metroplex initiative. Airport planners at airports affected by these Metroplex implementations should be aware that PBN procedures implemented as part of a Metroplex project will entail a full EA and public process, while single-site PBN procedures adopted as local initiatives may not be afforded that level of public process unless the airport intercedes and requests an EA by citing the potential for significant adverse public opposition on environmental grounds [CEQ regulations at 40 CFR 15056(c) 1 and 2].

Financial/Business Impacts

RNAV SIDS and STARS—Because RNAV SIDs and STARs do not require additional infrastructure at an airfield, they have no impact to an airport's capital expenditures.

RNAV and RNP Instrument Approach Procedures—Because RNAV and RNP approaches do not require additional infrastructure on the airfield, the primary increases in expenditures would be for the costs of surveying obstacles, obstacle removal, lighting, visual guidance, and weather systems. However, as with traditional ground-based instrument approaches, the establishment of RNAV and RNP instrument approaches requires an evaluation of obstacles to ensure adequate obstacle clearance for aircraft utilizing these procedures, which may require capital expenditures for obstacle removal. If the approaches allow lower landing minimums, installation or upgrade of runway or approach lighting may be required.

RNAV-Enabled Departure Separations—Because ELSO, EDO, and UDOS all require adequate holdpads and parallel taxiways to be effective, they could have an impact on an airport's capital expenditures if additional holdpads and parallel taxiways would be required to optimize their operation.

Surface Operations and Data Sharing

The key elements of the FAA Surface Operations and Data Sharing NextGen program are (1) advanced electronic flight strips, (2) surface departure management, (3) surface surveillance event data distribution to users via SWIM, (4) the SWIM Surface Visualization Tool, and (5) traffic flow management system, and (6) TBFM.

The sharing of data is intended to enable the establishment of CDM efforts between FAA and industry stakeholders and between stakeholders at medium and large airports. CDM is enabled through the dissemination of traffic flow data management and other SWIM data to the CDM group to support improved real-time decision making, tools and procedures to more easily respond to changing conditions, and advanced technological solutions and decision making for all stakeholders.

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Surface operation and CDM data sharing technology could track the movement of surface vehicles and aircraft at medium and large airports, incorporating the movement data into the airport surveillance infrastructure and sharing the information with controllers, pilots, and airline operations managers. As such, at constrained airports, improved surface operations could substantially increase facility requirements on the airfield, or at the terminal if new gates are required.

Facility Impacts

One of the key elements of improved surface operations is surface departure management, more widely known as "departure metering," which is intended to reduce the departure queue and thereby fuel burn, emissions, and surface congestion. However, effective departure metering requires ample apron areas, gates, and hold pads, which some space-constrained medium and large airports may have difficulty providing. At such constrained airports, improved surface operations could substantially increase facility requirements on the airfield.

Ramp control towers are also an important means of improving surface operations and data sharing. At some airports, ramp control towers are operated by the lead carrier. Recently implemented ramp control towers have been operated by the airport rather than the airlines. Las Vegas McCarran International Airport was one of the first airports to take over the management of its gates and the ramp operations, which has resulted in significant efficiencies in terms of common use of the gates and centralized management of ramp operations.

Surface movement data may also be useful to track the exact usage of any pavement area on the airfield as part of pavement management systems enabling calculations of pavement wear and useful life.

Operational Impacts

Improved surface operations will improve safety, efficiency, and flexibility on the airport surface by implementing new traffic management capabilities for pilots and controllers using shared surface movement and en route data. The capabilities address surface movement and the exchange of information between controllers, pilots, and air traffic managers that occur from before the aircraft pushes back from its gate up to the departure of the aircraft from the airport and, for landing traffic, from exiting the runway to arriving at the terminal gate.

Surface operations and data sharing at medium and large airports also provide opportunities to improve airport and airfield operational monitoring and vehicle operational data analyses. These data could facilitate facility planning studies, such as detailed traffic studies and defining vehicular staging and storage requirements. EAs could also leverage the data to conduct air quality analyses and assessments of movement area activity to support SMS analyses.

Environmental Impacts

Surface departure management or departure metering has the objective of reducing fuel burn and carbon emissions by holding departing aircraft at their gates with their engines off as long as possible without losing their sequence in the departure queue, or allowing the departure queue to dry up, which could result in the loss of departure slots. In addition, improved surface operations may reduce the number of aircraft taxiing stops and starts, which could further reduce fuel burn and emissions.

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Financial/Business Impacts

To leverage the benefits of surface operations and data sharing applications, medium and large airports would need to include the acquisition costs of these technologies into their airport capital, operating, and maintenance budgets. Ground vehicle tracking and the associated traffic display and analysis systems will also require investment into the associated technologies, including GPS receivers/transmitters, graphics, and database systems. Two-vehicle tracking methodologies are available to airports. AC 150/5210-25 describes GPS-based tracking through a Runway Incursion Warning System. AC 150/5220-26 describes vehicle tracking at airports served by multilateration and vehicles equipped with ADS-B transponders. If either method is chosen, additional capital and operating costs would be required.

Wake Turbulence Recategorization

As previously described, the FAA's recategorization of wake turbulence separation standards is being implemented in three phases. Prior to the implementation of Phase I in 2014, aircraft wake turbulence classifications were based strictly on aircraft weight. The FAA's Wake RECAT efforts are now focused on actual wake turbulence characteristics generated by each specific aircraft type. Phase I included the classification of each aircraft type into one of six wake turbulence categories and adjusted the minimum separations between aircraft arrivals and/or departures accordingly. Phase II will ultimately establish wake turbulence separate criteria for individual aircraft-type pairs, in lieu of the six-category classification system established for Phase I.

Due to the complexities associated with the planned establishment of aircraft wake turbulence separations by individual aircraft types, it is anticipated that Phase II would be adopted at more congested airports that need to optimize the capacity of the airfield and/or airspace. When considering such implementation at an airport, FAA will evaluate whether Phase I or Phase II Wake RECAT is the better option given the fleet mix at the airport. Phase II is not necessarily better, but rather it's a different approach that may be a better fit at some airports depending on the makeup of their fleet mixes.

Facility Impacts

Wake RECAT should have no significant effect on medium and large airport facilities. The expected increase in capacity with the implementation of Wake RECAT could possibly defer the need to increase the capacity of the airfield through capital improvements such as new runways and/or taxiways. On the other hand, increasing the capacity of the airport in all weather conditions may affect the capital planning for airside or terminal infrastructure.

Operational Impacts

These new separation minima could result in significant improvements in maximum arrival and departure throughput at medium and large airports that have significant numbers of Heavy jets and B757s in their aircraft-fleet mix. For example, FedEx has experienced an increase in airfield capacity of 20% at Memphis. However, airport planners should be cautioned that the benefit of increased throughput is heavily dependent on the fleet mix at their individual airports. At many medium and large airports the capacity increase due to Wake RECAT is expected to be on the order of 2%–4%.

One factor that could limit the achievable benefits of Wake RECAT at medium and large airports is the presence of departure airspace restrictions. These departure airspace restrictions could be due either to noise abatement procedures, conflicts with nearby airports, or surrounding obstacles or mountainous terrain.

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Environmental Impacts

Wake RECAT is not expected to have significant adverse environmental effects even at airports where it could result in a significant increase in arrival and departure capacity. On the contrary, the potential reduction in in-trail separation between aircraft, and the corresponding reduction in aircraft delays, could shorten aircraft traffic patterns on arrival and reduce aircraft hold times for departure, which could ultimately (1) reduce aircraft noise exposure over certain communities caused by arrivals, and (2) reduce aircraft emissions caused by departures.

Financial/Business Impacts

For medium and large airports that may benefit from Wake RECAT, the capacity of the airfield could be enhanced, thereby reducing aircraft operational delays and potentially deferring the need for other capital improvements necessary to enhance the capacity of the airfield. Otherwise, Wake RECAT is not expected to have any financial/business impacts at medium and large airports.

Closely Spaced Parallel Runway Operations

NextGen offers the ability to enhance the airfield capacity of airports with multiple runways through wake turbulence avoidance and/or potential changes in the minimum separation standards for aircraft landing on parallel runways during IMC. Wake turbulence avoidance procedures could benefit medium and large airports with either intersecting or closely spaced parallel runways (<2,500 feet of lateral separation). As with Wake RECAT, however, the application of wake turbulence avoidance procedures would be limited to medium and large airports that serve substantial numbers of Heavy jet and B757 aircraft operations.

Similarly, closely spaced parallel runway operations are of greatest interest to airport planning practitioners at medium and large airports that have sufficient land area and traffic volumes to justify such procedures. Medium and large airports with dual parallel runways and a lateral separation between 2,500 feet and 4,300 feet would have the best opportunity to improve arrival capacity during IMC.

Through the use of high-resolution color monitoring displays with alert algorithms (i.e., the final monitor aid or FMA), and without HUR, the minimum lateral spacing between dual parallel runways required for simultaneous independent approaches can be reduced to (1) 3,600 feet for straight-in instrument approaches, and (2) 3,000 feet for 2.5 degree-offset instrument approaches.

Under current ATC rules, aircraft arrivals to parallel runways with a lateral separation between 2,500 feet and 3,600 feet would still be dependent, although the diagonal spacing between such dependent parallel instrument approaches was recently reduced from 1.5 nautical miles to 1.0 nautical mile. Other instrument approach enhancements, such as triple dependent/independent parallel operations, dual independent operations with offset approaches, and RPAT, are candidates for application at medium and large airports.

Facility Impacts

The recent and potential changes in the rules for the minimum spacing between parallel runways required for dependent or independent approaches, and the types of instrument approach procedures that can use such parallel runways, could open up new opportunities at medium and large airports that previously would not qualify for such approaches. Therefore, new alternatives could be considered in the airport planning and development process for such airports. Such alternatives would have to be evaluated in terms of how the spacing between the parallel runways could facilitate the develop-

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ment of airport facilities between those runways and also how the new capabilities could affect noise exposure in the surrounding communities and operations at other nearby airports.

For example, with the reduction in spacing between parallel runways required to accommodate simultaneous independent parallel instrument approaches, medium and large airports considering such operations will have less difficulty fitting such parallel runways on their existing property or acquiring additional property for a new runway.

For long-term planning, the spacing between the inboard-most parallel runways for independent approaches may be insufficient to accommodate the planned terminal and landside facilities for meeting future demand. At best, a spacing of 3,600 feet between two parallel runways limits the options for developing passenger terminal and landside facilities between those runways. Therefore, for long-term planning of a new airport at a "greenfield site" (i.e., undeveloped land in a city or rural area) for example, airport planning practitioners should consider providing wider-than-the-minimum spacing between the parallel runways to accommodate the needed passenger terminal gates and landside (roadways and parking) facilities, in addition to planned air cargo and GA facilities. Having such facilities between the runways greatly reduces the risk of runway incursions due to aircraft having to cross one or more active runways to get to or from their parking position.

Operational Impacts

For medium and large airports that may benefit from wake turbulence avoidance procedures or reduced aircraft and/or runway separation standards for instrument approaches to closely spaced parallel runways, the capacity of the airfield could be increased. Such procedures would reduce aircraft operational delays and potentially defer the timing of the need for other capital improvements necessary to enhance the capacity of the airfield.

The NextGen CSPO program includes a number of technologies and operational improvements that affect the operation and capacity of parallel runway layouts in IMC. The primary focus has been on reducing the spacing between parallel runways required for conducting either simultaneous independent parallel instrument approaches or dependent parallel (staggered) instrument approaches. These changes have enabled dual or triple approach procedures in IMC at airports that previously could not conduct such procedures. The rules for conducting simultaneous independent visual parallel approaches, which are conducted in VMC, have not been affected.

Enabling dual or triple approaches in IMC can significantly increase the arrival capacity of a parallel runway operation, sometimes by as much as 50%–100%. The actual capacity increase depends upon how many parallel runways an airport has and how those parallel runways are operated for arrivals and departures. One factor affecting the capacity increase is whether the parallel runways are dedicated to either arrivals or departures (i.e., "segregated operations"), or whether there are "mixed operations" (i.e., both arrivals and departures) on one or more of the parallel runways. A second factor that affects the capacity increase is whether the arrival and departure runways are dependent or independent.

Environmental Impacts

The recent CSPO-enabled reductions in the minimum parallel runway spacings and aircraft separations required for independent and dependent parallel runway operations could have significant environmental impacts in cases where (1) the change enables the construction of a new parallel runway not previously possible, or (2) the change enables a new or modified flight procedure that affects the arrival or departure paths over the surrounding community depending on runway or flight procedure locations. In either case, there could be significant environmental impacts due to overflights of neighborhoods that previously were not overflown. In such cases, airport planning practitioners would

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almost certainly have to undertake either an EA study or an EIS study in order to obtain the necessary environmental approvals for implementing either (1) a new CSPO-enabled parallel runway, or (2) operating independent or dependent parallel approaches on an existing pair of parallel runways in a way that was not previously possible without CSPO.

In the case of existing parallel runways, the procedure change would be an ATO action. ATO would normally be responsible for NEPA, not the airport. The decision on the NEPA type in this case is complex. A CSPO procedure change (where that was the entire project) may not require a full EIS. While a new runway will almost certainly require an EIS, a CSPO change to existing runways might not trigger an EIS unless there is a significant change in flight procedures or runway use required to enable achieving the benefit of the project.

In some cases, the increased capacity afforded by the new CSPO could have a positive impact on the environment. For example, these enhancements could shorten aircraft flight paths on arrival and aircraft hold times for departure, which would ultimately reduce aircraft emissions and potentially reduce aircraft noise exposure to surrounding communities.

Financial/Business Impacts

At medium and large airports where a new parallel runway is enabled by the recent changes in requirements, there would be significant capital expenditures required for the planning, design, and construction of the new runway. At airports with existing parallel runways that would gain enhanced capability from the recent changes in requirements, there would be no capital expenditures required, and investment in additional capacity enhancements might be deferred by the enhanced capability of the existing layout.

Multilateration

MLAT systems are used for surface movement systems across the world. FAA incorporates MLAT sensors into the ASDE-X systems to provide ATC with surface movement information across an airfield. This system is deployed at 35 major airports in the U.S. and is used on a daily basis for operations. Both ASDE-X and ASSC include MLAT and ADS-B. ASDE-X is deployed at 35 hub airports, but it is not the top 35. ASSC is coming to eight airports.

MLAT technology had been expected to replace the PRM radar system to refresh the aging technology. These systems have been beneficial in supporting simultaneous operations at airports with closely spaced runways. However, the FAA has not determined MLAT to be a replacement for PRM, although it is a likely technology candidate. Because ADS-B also provides 1-second position updates, it also was once considered as a replacement for PRM. Integration/fusion software of these various systems remains to be developed, at a later date. For now, FAA is not proceeding with a PRM replacement.

MLAT is also anticipated to be the backup surveillance system for ADS-B Out.

Facility Impacts

An MLAT system consists of a transmitter, receiving antenna sensors, a central processor, and an optional interrogator system. The sensors and transmitters send out signals interrogating aircraft transponders which, in turn, transmit a response. The response from the transponder is interpreted by the computers to accurately locate an aircraft using triangulation by measuring the TDOA of the signal from the transponder at three or more synchronized receiver sites. The altitude of the aircraft is obtained directly from the required Mode C altitude-reporting transponder. En route and terminal config-

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urations supporting surveillance are referred to as WAM, whereas airport installations supporting surface movement, virtual air traffic control towers, and noise monitoring systems are referred to as MLAT.

Operational Impacts

The position information provided by MLAT and ADS-B is fused with existing radar systems, providing a "target" on the radar screen, enabling air traffic controllers to provide positive control of the aircraft and properly equipped surface vehicles. MLAT systems allow for the aircraft to receive its own positions through ground station signals and could be a backup should the GPS system fail. No additional aircraft equipage is required for mode C equipped aircraft.

Environmental Impacts

MLAT systems are also used by airports as part of noise abatement systems. These systems are tailored for each installation providing aircraft tracking, noise, and other data services.

In addition to enhanced operational safety, a benefit of MLAT used for WAM could be to reduce aircraft diversions, thereby reducing aircraft emissions and noise exposure in the region. The use of MLAT could also enhance the ability for an airport noise and operations monitoring system (ANOMS) to correlate aircraft noise exposure with actual aircraft operations. Implementation of ANOMS is normally feasible only at airports that have a significant aircraft noise exposure problem or noise abatement program that would justify the significant investment in manpower and equipment required to operate such a system.

Financial/Business Impacts

MLAT systems are also used by airports as part of revenue tracking systems. These systems are tailored for each installation providing aircraft tracking and other data services. Airports can purchase new MLAT systems or expand existing MLAT systems through the acquisition of additional sensors. The same is true for ADS-B systems.

To the extent that the MLAT sensors would be installed on the airport, there would still be a cost for installation and maintenance.

ADS-B In and Cockpit Display of Traffic Information

ADS-B In provides operators of properly equipped aircraft with weather and traffic position information delivered directly to the cockpit. ADS-B In–equipped aircraft have access to the graphical weather displays in the cockpit as well as text-based advisories, including Notices to Airmen and significant weather activity.

ADS-B In and CDTI are intended primarily to improve operational safety in both flight and on the ground. However, they can be of benefit to airports by (1) ultimately enabling equivalent visual approaches at lower minimums, thereby increasing capacity and reducing emissions and (2) reducing the risk of certain runway incursions between aircraft taxiing on the ground or between aircraft and ground vehicles.

FAA has a mandated timetable for replacing most of its current radar aircraft surveillance systems with satellite-based systems utilizing ADS-B Out technologies. In particular, FAA is requiring all aircraft operating within controlled airspace to be equipped with ADS-B Out by 2020. However, there is no mandated timetable established for the implementation of ADS-B In procedures. Concepts and time-

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frame for ADS-B applications remain an active discussion between FAA and industry, through industry organizations such as RTCA.

CDTI is an avionics system that, when combined with ADS-B In and Out, displays neighboring aircraft information, on the ground or in the air, to the flight crew as well as automation functions that, in some cases, provide speed or maneuver guidance to the crew.

Facility Impacts

There are no airport facilities required for ADS-B In. User requirements for ADS-B In include aircraft equipage and the development of cockpit systems to take advantage of all the capabilities available through the technology under which pilots of suitably equipped aircraft with proper training would be responsible for their own separation.

Operational Impacts

The primary benefit of ADS-B In is to enhance operational safety both in flight and on the ground. Aircraft equipped with ADS-B In equipment, and associated cockpit displays of traffic, weather, and other advisories, can maintain uniform separation from other aircraft, thus potentially increasing the capacity of the airfield at airports without radar surveillance systems. However, the capacity benefits would be marginal and difficult to quantify.

In the past, there were projections that CDTI, when combined with ADS-B In and Out, would provide the capability to conduct "equivalent visual separations" in IMC, under which pilots with proper and operating suitably equipped aircraft would be responsible for their own separation, including wake turbulence separation, much like they are under current visual approach procedures conducted in VMC. Recently, however, this notion has not been promoted; instead, CDTI-enhanced visual approaches are a long-term possible concept, but there are other higher priorities that are more critical to fund and implement in the near term. Nevertheless, it is expected that with further advances and implementation of enhanced vision and synthetic vision systems, that there will come a time when visual approach type procedures could be conducted with ADS-B In combined with enhanced CDTI displays.

Environmental Impacts

There should be no significant environmental impacts to airport operators from the implementation of ADS-B In and CDTI.

Financial/Business Impacts

There should be no significant financial/business impacts to airport operators from the implementation of ADS-B In and CDTI.

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5 Applicability of NextGen to Small Airport Planning and Development

While the FAA's current NextGen initiatives will enhance the overall NAS, the direct and indirect benefits to most small commercial service and GA airports are likely to be focused on improving access during its initial implementation. For these airports, much of the focus has been on lowering approach minimums and improving airfield capacity during inclement weather conditions; particularly through the implementation of satellite-based augmentation system (SBAS) or WAAS instrument approaches where ground-based NAVAIDS are not available or practical to install. Although other NextGen technologies and operational improvements have been and will continue to be implemented at smaller airports, their effectiveness and/or applicability may be restricted due to a variety of factors. These factors include, but are not limited to, the types of aircraft serving the airport, aircraft equipage limitations, ATC training, funding constraints, low demand levels, or the inability to demonstrate a positive cost-benefit ratio. Nonetheless, many NextGen initiatives are scalable and therefore could be implemented at small airports with the use of alternative technologies and/or procedures. Appendix C provides a summary of the various NextGen technologies and operational improvements and their potential application to different sizes of airports.

This chapter identifies the various NextGen capabilities that could enhance operations at small airports and how these capabilities may influence planning and development initiatives at these airports. It differentiates between the NextGen initiatives that have global application at most small airports from those that have limited application under certain operating conditions and environs. It is important to understand that the success of NextGen at these airports is as much a function of the equipage of the fleet operating at the airport as it is of the size or type of service the airport has. Any given small airport may be used by aircraft with Nav/Com capabilities ranging from minimal (no electrical system) to superior (full flat screen avionics suite). So, simply characterizing the impact of NextGen by airport size or type of service is difficult. Despite the difficulty of characterizing NextGen capabilities in terms of airport size or classification, there are some general trends for smaller airports that are worth identifying.

Definition of Small Airports

The opportunities to enhance the NAS through the implementation of NextGen technologies and operational improvements can vary significantly among airports, regardless of an airport's size, role, and classification. There are numerous factors that will influence the direct and indirect benefits associated with each of the various NextGen initiatives for any given airport or system of airports. Therefore, the application of NextGen technologies and operational improvements, and the resulting benefits, will be unique for each airport.

However, the opportunities to realize the benefits of NextGen at small commercial service and GA airports can be differentiated from those for large and medium airports that have a significant composition of air carrier and cargo aircraft, which is what this chapter is intended to illustrate.

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The NPIAS categorizes public use airports by type of activities, including commercial service, primary, cargo service, reliever, and GA airports.[1] For the purpose of this document, small airports generally include the following NPIAS Classifications:

- Nonhub Primary Commercial Service (>10,000 annual passenger boardings, but less than 0.25 percent of total passenger boardings in the U.S.; 380,000 boardings in 2014).
- Non-primary Commercial Service (no more than 10,000 annual passenger boardings).
- GS/reliever airports.

While private-use and military airports could benefit from NextGen technologies, their application is excluded from these discussions. Table 5-1 presents the overall NPIAS airport classifications, while highlighting those that are considered as small airports.

As of September 2014, there were 3,283 airports contained in the NPIAS, and nearly 90 percent of these airports are classified as GA and/or reliever airports. GA airports either have no scheduled commercial service or less than 2,500 annual passenger boardings. Due to the unique physical constraints and demand characteristics, the operational complexities and aircraft fleet mix associated with these airports can vary significantly. Many GA airports within urban areas have a high concentration of corporate jet activity, some with performance characteristics and avionic platforms similar to, and sometimes more advanced than, modern air carrier aircraft. Other municipal and rural airports may serve small piston and turboprop aircraft with varied, perhaps limited, avionics capabilities.

In 2012, the FAA published a report titled *General Aviation Airports: A National Asset*. This report was the result of an extensive evaluation of the various roles of public use GA airports throughout the U.S. As a result, the FAA now classifies public use GA airports in the following manner:

- National—Supports the national and state system by providing communities with access to national and international markets in multiple states and throughout the U.S.
- Regional—Supports regional economies by connecting communities to statewide and interstate markets.
- Local—Supplements local communities by providing access primarily to intrastate and some interstate markets.
- Basic—Supports GA activities such as emergency services, charter or critical passenger services, cargo operations, flight training, and personal flying.
- Unclassified—no clearly defined role.

While these classifications are not intended to replace the NPIAS airport classifications for GA airports, they are intended to provide consistent evaluation criteria for prioritizing project funding commitments, including NextGen initiatives. Therefore, to better correlate the diverse range of GA airports with NextGen technologies, this guidebook further segregates GA airports in accordance with the four primary GA airport asset classifications. Table 5-2 summarizes the resulting subcategories of small airports as it relates to NextGen opportunities.

[1] Federal Aviation Administration, National Plan of Integrated Airport Systems, http://www.faa.gov/airports/planning_capacity/passenger_allcargo_stats/categories, accessed January 7, 2016.

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Table 5-1. *FAA airport classifications.*

Categories of Airport Activities

Airport Classifications		Hub Type: Percentage of Annual Passenger Boardings	Common Name	
Commercial Service: Publicly owned airports that have *at least 2,500* passenger boardings each calendar year and receive scheduled passenger service **§47102(7)**	**Primary:** Have *more than 10,000* passenger boardings each year **§47102(16)**	**Large:** 1% or more	**Large Hub**	Medium and Large Airports[1/]
		Medium: At least 0.25%, but less than 1%	**Medium Hub**	
		Small: At least 0.05%, but less than 0.25%	**Small Hub**	Small Airports[1/]
		Nonhub: More than 10,000, but less than 0.05%	**Nonhub Primary**	
	Nonprimary	**Nonhub:** At least 2,500 and no more than 10,000	**Nonprimary Commercial Service**	
Nonprimary (Except Commercial Service)		Not Applicable	**Reliever §(47102(23))** **General Aviation (47102(8))**	

[1]Medium and Large Airports generally include the following NPIAS Classifications:

- Medium Hub Primary Commercial Service (at least 0.25% but less than 1% of annual passenger boardings).
- Large Hub Primary Commercial Service (1% or more of annual passenger boardings).

Small Airports generally include the following NPIAS Classifications:

- Nonhub Primary Commercial Service (>10,000 annual passenger boardings, but less than 0.25% of total passenger boardings in the United States).
- Non-primary Commercial Service (no more than 10,000 annual passenger boardings).
- General aviation/reliever airports.

Source: NPIAS, Federal Aviation Administration.

FAA NextGen Technologies and Initiatives Applicable to Small Airports

The deployment of FAA's NextGen initiatives and supporting technologies has been, and will continue to be, limited at small airports. While a few of the FAA's NextGen initiatives have global application at most airports, the deployment of many of these initiatives at small airports may be inhibited due to

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Table 5-2. Subcategorization of small airports for NextGen opportunities.

SUBCATEGORY	NPIAS CRITERIA	TYPICAL AIRCRAFT FLEET MIX COMPOSITION
Limited Commercial Service		
Nonhub Primary Commercial Service	10,000 Annual Passenger Boardings—<0.25% of Total U.S. Passenger Boardings	Air Carrier and Business Jets.
Non-primary Commercial Service	2,500–10,000 Annual Passenger Boardings	Limited Air Carrier and Business Jets.
GA		
National	No Scheduled Service or <2,500 Annual Passenger Boardings	Very High Levels of Activity with Many Jets and Multiengine Propeller Aircraft; Averaging About 200 Based Aircraft, Including 30 Jets.
Regional	No Scheduled Service or <2,500 Annual Passenger Boardings	High Levels of Activity with Some Jets and Multiengine Propeller Aircraft; Averaging About 90 Based Aircraft, Including Three Jets.
Local	No Scheduled Service or <2,500 Annual Passenger Boardings	Moderate Levels of Activity with Some Multiengine Propeller Aircraft; Averaging About 33 Based Propeller-Driven Aircraft and No Jets.
Basic	No Scheduled Service or <2,500 Annual Passenger Boardings	Moderate to Low Levels of Activity; Averaging About 10 Propeller-Driven Based Aircraft and No Jets.

Sources: NPIAS, Federal Aviation Administration; General Aviation Airports: A National Asset, Federal Aviation Administration, May 2012.

a variety of factors. This section provides an overview of the applicability of the FAA's NextGen initiatives and support technologies at small airports and describes the factors that may inhibit or preclude their deployment. It also describes direct and indirect effects on planning and development at small airports.

Improved Landing Systems

With the exception of RNP approaches and GBAS, each of the various improved landing systems has global application at small airports. GBAS is currently being programmed for large- and medium-hub airports with no planned implementation at small airports. Due to the limited use of RNP by smaller regional airlines and some GA aircraft, coupled with the airfield lighting, obstruction clearance and development requirements, the deployment of low minimum RNP approaches at small airports is typically cost prohibitive. Therefore, the establishment of RNP approaches at small airports will likely be limited to the rare occasion in which an airport has a very high volume of RNP-equipped aircraft, particularly (1) those airports that are part of a congested airspace system (Metroplex) that includes other medium- or large-hub air carrier airports and (2) those airports surrounded by mountainous terrain where conventional instrument approaches or radar coverage cannot be implemented. (See Table 5-3.)

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Table 5-3. Improved landing systems application to small airports.

DESCRIPTION	LIMITED (NONHUB) COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/REGIONAL GA AIRPORTS	LOCAL/BASIC GA AIRPORTS
LPV Approach	X	X	X
LP Approach	X	X	X
LNAV/VNAV Approach	X	X	X
LED PAPI[2]	X	X	X
LED ALS[2]	X	X	X
RNP Approach	Limited Benefits to Small Airports Within a Constrained Airspace System		
GBAS	Not Generally Applicable to Small Airports		

Notes:

[1]Per the NPIAS, limited commercial service airports consist of nonhub airports with less than 0.25% share of the annual passenger boarding at U.S. airports (approximately 380,000 enplanements in FY 2014).

[2]Pending final approval by the Federal Aviation Administration.

The FAA has been implementing the LP, LPV, and LNAV/VNAV approaches to enhance operations during inclement weather where ground-based navigational facilities are not available. These procedures have also been established at airports that are served with ground-based navigation facilities to supplement existing instrument approaches and/or provide redundancy. While some older aircraft may not have the avionics to support LP, LPV, and LNAV/VNAV approaches, the FAA has implemented them at most public use airports throughout the country. However, these less equipped aircraft are often less likely to operate during IMC. It is also anticipated that small GA aircraft will opt to upgrade their avionics packages as they install ADS-B Out equipment prior to 2020, as mandated by the FAA.

The FAA has determined that the replacement of incandescent lights of PAPI and MALSR systems with LED lamps would have a return on investment of approximately two years. The FAA is currently conducting a feasibility study for LED replacement. The feasibility study will also determine their ability to support enhanced flight vision system (EFVS) operations. While not all small airports are equipped with MALSR systems, the FAA has made considerable investment to install PAPIs at commercial service and most GA airports.

Effects of Improved Landing Systems on Planning and Development at Small Airports

The establishment of PBN (LPV, LP, LNAV/VNAV, and RNP) approaches will not only improve the ability to operate during inclement weather, but also reduce aircraft emissions and aircraft noise exposure by reducing the number of aircraft diversions to nearby airports. By providing a more precise approach path and descent profile, some PBN approaches could also enhance the capacity of the airspace, thereby potentially increasing the capacity of an airfield that is impaired by the capacity of surrounding airspace due either to other airports or surrounding terrain. Since PBN approaches do not require additional infrastructure at an airfield, they have no impact to an airport's capital expenditures, unless the procedure lowers minimums enough to trigger additional obstacle clearance requirements. (See Table 5-4.)

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Table 5-4. Improved landing systems application—small airport planning considerations.

DESCRIPTION	AIRFIELD/AIRSPACE CAPACITY/ACCESS ENHANCEMENTS	OBSTACLE EVALUATION/ OBSTRUCTION MITIGATION PROGRAMS	ENVIRONMENTAL (NOISE & AIR QUALITY)	AIRPORT CAPITAL PLANNING
LPV Approach	X	X	X	
LP Approach	X	X	X	
LNAV/VNAV Approach	X	X	X	
RNP Approaches	X		X	
LED Approach Lighting Systems	X		X	X
LED PAPI				X
GBAS	Not Applicable to Small Airports			

When establishing LPV, LP, and LNAV approaches to an airport, the scope of obstacle evaluation and obstruction mitigation programs for the airport may increase. As with traditional ground-based instrument approaches, the establishment of LPV, LP, and LNAV/VNAV approaches requires an evaluation of obstacles to ensure adequate obstacle clearance for aircraft utilizing these procedures. FAA Order 8260.3C, *United States Standard for Terminal Instrument Procedures (TERPS)*, and FAA Order 8260.5, *United States Standard for Performance-Based Navigation Procedure Design*, establish the minimum obstacle clearance criteria for the development of instrument approach procedures at airports. Due to the unique accuracy characteristics associated with the LPV, LP, and LNAV/VNAV procedures, the obstacle clearance requirements prescribed in TERPS can be much different from those associated with instrument approaches served with traditional ground-based NAVAIDS. For LNAV/VNAV approaches, the lateral extent of the obstacle clearance surfaces (OCS) can be greater than those associated with existing approach capabilities. In addition, the slope of the final approach surface associated with OCS for LPV approaches can be more restrictive than existing approach procedures designated at the airport. Therefore, evaluations of obstructions are required, regardless of the presence of existing ground-based procedures, even if existing ground-based procedures have lower minimums. Airport operators with existing LPV, LP, and LNAV/VNAV approaches may also consider obstruction removal programs to allow the lowering of approach minimums at the airport, thereby further enhancing the capacity of and access to their airfields.

In addition to the direct and indirect airfield capacity and access benefits associated with improved landing systems, the potential approval of LED PAPI and ALS could reduce an airport's operating budget. If implemented, LED ALS may provide a cost-effective means for a small airport to install an approach lighting system that it otherwise could not afford. In these instances, lower landing minimums could be achieved, thereby improving the ability to land during inclement weather. By reducing the number of aircraft diversions to nearby airports, aircraft emissions and aircraft noise exposure could also be reduced with the installation of a new LED ALS.

Airspace Routing with Performance-Based Navigation

PBN may provide the ability to enhance the operational throughput at small airports that are constrained by the capacity of its surrounding terminal airspace. With the increased precision provided by RNP, coupled with the benefits associated with area navigation (RNAV), dedicated arrival and

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departure routes to and from small airports could be established, thereby allowing small aircraft to be segregated from heavy aircraft that are destined to or originating from other nearby airports. This may allow both airports to optimize the throughput of their airfields. (See Table 5-5.)

Table 5-5. PBN application to small airports.

DESCRIPTION	LIMITED (NONHUB) COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/REGIONAL GA AIRPORTS	LOCAL/BASIC GA AIRPORTS
RNAV	X	X	X
RNP	Limited Benefits to Small Airports Within a Constrained Airspace System		

Notes:
[1]Per the NPIAS, limited commercial service airports consist of nonhub airports with less than 0.25 percent share of the annual passenger boarding at U.S. airports (approximately 380,000 enplanements in FY 2014).

Similar to RNP instrument approach procedures, the ability to utilize RNP in the terminal airspace by smaller regional airlines and GA aircraft is typically cost prohibitive. However, the growing prevalence of GPS systems in regional/commuter and GA aircraft has made it feasible to establish dedicated STARs and SIDs to small airports. Through the use of RNAV, the ability to modify SIDs and STARs without the capital investment necessary to establish ground-based NAVAIDS is enhanced significantly. Ultimately, the FAA intends to replace all ground-based navigation with RNAV procedures.

Effects of PBN on Planning and Development at Small Airports

NextGen allows better sequencing for aircraft and the ability to route traffic in narrower corridors. High precision GPS technologies also reduce cancellations and diversions. Airports could experience reduced delays from the traffic that would otherwise be diverted. As with enhanced landing systems, PBN could allow some small airports to optimize the capacity of their airfields, thereby reducing delays and deferring capital development costs for new runways. With the improved reliability of aircraft movements, controllers will be able to reduce delay across the entire system and at the local level. (See Table 5-6.)

The NextGen systems will allow aircraft to be continuously monitored and tracked via new flight management technologies, therefore increasing situational awareness for both pilots and controllers alike. Aircraft operators may experience a reduction in fuel cost, diversions, aircraft emissions, and aircraft operating expenses as a result of increased efficiency (direct routing) across the NAS. Weather rerout-

Table 5-6. PBN application—small airport planning considerations.

DESCRIPTION	AIRFIELD/ AIRSPACE CAPACITY ENHANCEMENTS	OBSTACLE EVALUATION/ OBSTRUCTION MITIGATION PROGRAMS	ENVIRONMENTAL (NOISE & AIR QUALITY)	AIRPORT CAPITAL PLANNING
RNAV	Primary Benefits to Small Airports Within a Constrained Airspace System or Obstacle Rich Environment			
RNP Approaches	Limited Benefits to Small Airports Within a Constrained Airspace System			

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ing will also be improved, air traffic controllers having the ability to open and close departure and arrival routes in real time.

Multilateration

MLAT is utilized by ASDE-X and WAM systems that enhance aircraft tracking in mountainous terrain and improve coverage and separations services within terminal airspace environs. Although ASDE-X equipment has only been installed at the nation's busiest airports, MLAT could ultimately support virtual ATCTs, which are currently being evaluated by the FAA and have been adopted in Europe, and virtual ramp control. Therefore, MLAT would benefit certain small airports through the establishment of WAM or the potential development of virtual ATCTs, should they be approved for use in the U.S. by the FAA. The leveraging of MLAT for WAM would be particularly beneficial at airports with airspace that will not require ADS-B Out equipped aircraft. WAM could also support the tracking of aircraft noise events through the use of an ANOMS, where other traditional radar tracking is not available. The use of ASDE-X and virtual ramp control is not applicable at small airports because of cost considerations. (See Table 5-7.)

WAM has been beneficial to small airports in remote mountainous regions where radar coverage is limited due to the surrounding terrain. Although the installation costs associated with the remote sensors is cheaper than a full radar installation, the establishment of WAM to improve operations in mountainous regions would likely need to benefit multiple airports to be cost effective, which was the case in the case study of the WAM application in the state of Colorado. Each situation would need to be evaluated individually. WAM has also been employed overseas to enhance the tracking of aircraft on independent approaches on closely spaced parallel runways and could be adopted for this type of use in the U.S.

Virtual ATCTs have been employed at airports in Europe, and the FAA has recently initiated pilot programs to assess their applicability to airports in the U.S. Should the technology be adopted by the FAA, virtual ATCTs could be employed at airports that currently do not have an ATCT, or to replace other towers at low volume facilities that currently have a contract ATCT. The virtual ATCTs consist of a suite of cameras and remote sensors that includes HD video cameras, a pan-tilt-zoom (PTZ) camera, signal light gun, and microphones.

Table 5-7. Multilateration application to small airports.

DESCRIPTION	LIMITED (NONHUB) COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/REGIONAL GA AIRPORTS	LOCAL/BASIC GA AIRPORTS
WAM	Potential Benefits to Small Airports in Mountainous Regions or Within the Terminal Airspace of Small Airports		
Virtual ATCTs	FAA May Consider Virtual ATC Towers in the Near Future (Potential Benefit to All Small Airports)		
ANOMS	X	X	N/A
Virtual Ramp Control	Not Applicable to Small Airports		
Surface Movement (ASDE-X)	Not Applicable to Small Airports		

Note:

[1]Per the NPIAS, limited commercial service airports consist of nonhub airports with less than 0.25 percent share of the annual passenger boarding at U.S. airports (approximately 380,000 enplanements in FY 2014).

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Effects of Multilateration on Planning and Development at Small Airports

While MLAT applications in remote areas with mountainous terrain have proved effective for enhancing operational safety, they have also been proven to increase capacity at small airports where the installation of a traditional radar system cannot supply adequate coverage in mountainous terrain. It can also improve the operational safety and capacity of the terminal airspace environs, particularly at airports with closely spaced parallel runways. The capacity benefits would primarily be achieved during periods of low visibility and/or inclement weather. While the remote sensors may not necessarily be installed on the associated airport, there would still be a cost for installation and maintenance. In addition to enhanced operational safety, the primary benefit would be to reduce aircraft diversions. The use of MLAT could also enhance the ability for ANOMS to correlate aircraft noise exposure with actual aircraft operations. (See Table 5-8.)

Table 5-8. Multilateration application—small airport planning considerations.

DESCRIPTION	AIRFIELD/AIRSPACE CAPACITY & SAFETY ENHANCEMENTS	AIRPORT DEVELOPMENT CONSTRAINTS	ENVIRONMENTAL (NOISE & AIR QUALITY)	AIRPORT CAPITAL PLANNING
WAM	X		X	
Virtual ATCTs		X		X
ANOMS		X		X
Virtual Ramp Control	Not Applicable to Small Airports			
Surface Movement (ASDE-X)	Not Applicable to Small Airports			

The potential implementation of virtual ATCTs could have a marginal benefit on the overall capacity of the airfield. However, operating and capital costs for FAA could be reduced, while also enhancing the development opportunities on the associated airport. The infrastructure associated with virtual ATCTs in Europe typically consists of HD video cameras, a PTZ camera, a signal light gun, and microphones. In addition to eliminating the need to preserve airport property to accommodate a fully functional ATCT, controller line-of-sight impacts could be mitigated, thereby providing greater flexibility to develop other structures on the airfield. However, the siting of cameras and/or remote sensors on the airfield would need to be assessed and identified on the ALP drawings.

Surface Operations and Data Sharing

Of the various surface operations and data sharing applications, small airports could benefit from LED lighting technologies, ground vehicle tracking, traffic display and analysis systems, and SWIM. Other surface operations and data sharing applications, such as CDM, intelligent routing and guidance systems, and automated docking systems are not applicable at lower-activity airports due to high implementation costs and limited operational benefits. (See Table 5-9.)

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Table 5-9. Surface operations and data sharing application to small airports.

DESCRIPTION	LIMITED (NONHUB) COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/REGIONAL GA AIRPORTS	LOCAL/BASIC GA AIRPORTS
LED Lighting Technology	X	X	X
Ground Vehicle Tracking	X	X	X
Traffic Display and Analysis Systems	X	X	X
SWIM	X	X	X
CDM	Not Applicable to Small Airports		
Intelligent Routing and Guidance Systems	Not Applicable to Small Airports		
Automated Docking Systems	Not Applicable to Small Airports		

Notes:
[1]Per the NPIAS, limited commercial service airports consist of nonhub airports with less than 0.25 percent share of the annual passenger boarding at U.S. airports (approximately 380,000 enplanements in FY 2014).

Airports of all sizes have benefitted from the installation of LED lighting technologies on the airfield, particularly for taxiway and runway lighting and signage. Such an installation includes elevated runway and taxiway edge lights and flush-mounted taxiway centerline and hold bar lights. LED-lighted wind cones and obstruction lights are also viable options for small airports. As with LED PAPIs and approach light systems, the LED lighting technologies may have a higher initial capital investment cost, but total life-cycle costs are lower due to the durability and lower power consumption and reoccurring maintenance costs than traditional lighting systems. The FAA has also approved solar LED technologies for airfield signage, wind cones, runway guard lights, and obstruction lights with wireless controls at small airports.

While increased data sharing has enhanced CDM capabilities for movement of traffic on the airfields of large airports, small airports could benefit as well. Ground vehicle tracking systems, coupled with traffic display and analysis systems not only enhance operational safety on the airfield, but also offer airport management and other vehicle operators the ability to track vehicles and equipment on the airfield in real time. While the FAA-supported technology at larger airports having MLAT installations for ground vehicle tracking currently requires expensive certificated ADS-B transponders on each vehicle tracked, smaller airports can benefit from runway incursion warning systems (RIWS) described in AC 150/5210-25. Vehicles equipped with RIWS can be tracked on the airport surface and have the benefit of providing alerts for possible runway incursions. These systems also allow for the mining of historical data to support operational studies and opportunities to enhance operational safety and efficiency on the airfield.

If able to leverage the data sharing of surface operations, airport operators would need to be integrated with SWIM. This application will primarily consist of airport GIS systems, combined with PBN, ADS-B, Data Comm, and weather data that will serve to reduce system error and increase efficiencies in all domains.

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Effects of Surface Operations and Data Sharing on Planning and Development at Small Airports

To leverage the benefits of surface operations and data sharing applications, small airports would need to include the costs of the acquisition of these technologies into their airport capital, operating, and maintenance budgets. The conversion of LED lighting technology should be considered when airfield lighting systems are upgraded or new installations are developed. Ground vehicle tracking and the associated traffic display and analysis systems will also require investment into the associated technologies, including GPS receivers/transmitters, graphics, and database systems. The technologies may also facilitate fleet and equipment planning and management activities by airport management and other vehicle operators. (See Table 5-10.)

Surface operations and data sharing at small airports also provide opportunities to improve airport and airfield operational monitoring and vehicle operational data analyses. These data could facilitate facility planning studies, such as detailed traffic studies and defining vehicular staging and storage requirements. EAs could also leverage the data to conduct air quality analyses and assessments of vehicular traffic noise impacts. As NextGen systems mature, more data will be available to smaller airports and/or aircraft operators.

Table 5-10. Surface operations and data sharing application—small airport planning considerations.

DESCRIPTION	VEHICULAR OPERATIONAL MONITORING AND DATA ANALYSIS	FACILITY PLANNING (TRAFFIC STUDIES AND VEHICLE STORAGE)	ENVIRONMENTAL ASSESSMENTS (AIR QUALITY/ NOISE ASSESSMENT)	AIRPORT CAPITAL, OPERATING, AND MAINTENANCE BUDGET PLANNING
LED Lighting Technology				X
Ground Vehicle Tracking	X	X	X	X
Traffic Display and Analysis Systems	X	X	X	X
SWIM	X	X	X	
Collaborative Decision Making	Not Applicable to Small Airports			
Intelligent Routing and Guidance Systems	Not Applicable to Small Airports			
Automated Docking Systems	Not Applicable to Small Airports			

Wake Turbulence Recategorization—Single Runway Operations

As previously described, the FAA's Wake RECAT separation standards are being implemented in three phases. Prior to the implementation of Phase I in 2014, aircraft wake turbulence classifications were based strictly on aircraft weight. The FAA's Wake RECAT efforts are now focused on actual wake turbulence characteristics generated by each specific aircraft type. Phase I included the classification of each aircraft type into one of six wake turbulence categories and adjusted the minimum separations between aircraft arrivals and/or departures accordingly. Phase II will ultimately establish wake turbulence separate criteria for individual aircraft-type pairs, in lieu of the six-category classification system established for Phase I.

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Since wake turbulence hazards are greater with heavy aircraft, the recategorization of wake turbulence separation standards are much more beneficial at large airports than small airports. In fact, the direct benefits associated with Phase I Wake RECAT at small airports have mostly been limited to those airports that also serve significant volumes of Heavy jet and B757 aircraft that produce significant wake turbulence hazards. Due to the complexities associated with the planned establishment of aircraft wake turbulence separations by individual aircraft types, it is anticipated that Phase II would be adopted at more congested airports that need to optimize the capacity of the airfield and/or airspace. In addition to some of the limited (nonhub) commercial service airport airports, Wake RECAT Phase II could benefit some GA airports with a high concentration of large corporate jet aircraft operations. Smaller GA and utility airports are not anticipated to benefit directly from wake turbulence recategorization. (See Table 5-11.)

Table 5-11. Wake RECAT application to small airports.

DESCRIPTION	LIMITED (NONHUB) COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/REGIONAL GA AIRPORTS	LOCAL/BASIC GA AIRPORTS
RECAT Phase I	Limited to Non-Hub Airports That Serve Heavy Aircraft Operations or Part of a System of Airports with a Significant Share of Heavy Aircraft Operations in the Terminal Airspace	Potential Benefit to General Aviation Airports That Are Part of a System of Airports with a Significant Share of Heavy Aircraft Operations in the Terminal Airspace	
RECAT Phase II	Potential Benefit to Small Airports That Are Congested or Part of a System of Airports with Congested Terminal Airspace		

Notes:

[1]Per the NPIAS, limited commercial service airports consist of nonhub airports with less than 0.25 percent share of the annual passenger boarding at U.S. airports (approximately 380,000 enplanements in FY 2014).

Effects of Wake RECAT on Planning and Development at Small Airports

For small airports that may benefit from Wake RECAT, the capacity of the airfield could be enhanced, thereby reducing aircraft operational delays and potentially deferring the need for other capital improvements necessary to enhance the capacity of the airfield. Given the relatively low risks associated with the wake turbulence generated by GA and small regional passenger aircraft, the overall benefits of Wake RECAT are likely to be marginal at small airports and difficult to quantify or factor into airport planning analyses. Another factor that can limit the achievable benefits of Wake RECAT is the presence of departure airspace restrictions that limit the number of departure headings possible at an airport. These departure airspace restrictions could be due either to noise abatement procedures or surrounding obstacles or terrain. (See Table 5-12.)

Table 5-12. Wake RECAT application—small airport planning considerations.

DESCRIPTION	AIRFIELD/AIRSPACE CAPACITY ENHANCEMENTS	ENVIRONMENTAL (NOISE & AIR QUALITY)	AIRPORT CAPITAL PLANNING
RECAT Phase I	Marginal Benefits and Difficult to Quantify		
RECAT Phase II	Marginal Benefits and Difficult to Quantify		

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Dependent Runway Operations

NextGen offers the ability to enhance the airfield capacity of airports with multiple runways through wake turbulence avoidance and/or potential changes in the minimum separation standards for aircraft landing on parallel runways during IMC. Wake turbulence avoidance procedures could benefit small airports with either intersecting or closely spaced parallel runways (<2,500 feet of lateral separation). As with single runway operations, however, the application of wake turbulence avoidance procedures would be extremely limited for small airports, primarily benefiting only those airports that may serve Heavy jet and B757 aircraft operations. (See Table 5-13.)

Although NextGen provides additional opportunities to enhance instrument operations on closely spaced parallel runways, their application at small airports is also limited. This is primarily due to the limited number of small airports configured with parallel runways served by instrument approach procedures, the additional technological requirements such as high-resolution color monitoring displays necessary to support reduced aircraft separation, and the controller staffing and training requirements. The financial feasibility of enhancing instrument operations on closely spaced parallel runways at small airports with low occurrences of IMC would also be difficult to demonstrate.

Table 5-13. Dependent runway application to small airports.

DESCRIPTION	LIMITED (NONHUB) COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/REGIONAL GA AIRPORTS	LOCAL/BASIC GA AIRPORTS
Wake Turbulence Avoidance Procedures			
Parallel Runways (<2,500′ Spacing)	Limited to Small Airports That May Serve Heavy Aircraft Operations		N/A
Wake Turbulence Mitigation for Departures (Upwind Runways)	Limited to Small Airports That May Serve Heavy Aircraft Operations		N/A
Enhanced Instrument Operations—CSPO			
Dual Independent Parallel Operations (>3,600′ Spacing)	Limited to Small Airports with Closely Spaced Parallel Runways Served by Precision Instrument Approach Procedures		N/A
Dual Dependent Parallel Operations (2,500′–3,600′)	Limited to Small Airports with Closely Spaced Parallel Runways Served by Precision Instrument Approach Procedures		N/A
Triple Dependent Parallel Operations (>3,900′ Spacing)	N/A to Small Airports		
Dual Independent Parallel Operations with Offset (>3,000′ Spacing)	N/A to Small Airports		
Dependent Parallel Operations (>4,300′ Spacing)	N/A to Small Airports		
RPAT	N/A to Small Airports		

Notes:

[1]Per the NPIAS, limited commercial service airports consist of nonhub airports with less than 0.25 percent share of the annual passenger boarding at U.S. airports (approximately 380,000 enplanements in FY 2014).

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Small airports with dual parallel runways and a lateral separation between 2,500 feet and 4,300 feet would have the best opportunity to improve arrival capacity during IMC. Through the use of high-resolution color monitoring displays with alert algorithms (i.e., FMA), the minimum lateral separation between dual parallel runways operating with simultaneous arrivals can be reduced from 4,300 feet to 3,600 feet with today's surveillance radar. Although aircraft arrivals to parallel runways with a lateral separation between 2,500 feet and 3,600 feet would still be dependent, NextGen also offers the ability to enhance dependent parallel (staggered) instrument approach procedures by reducing the diagonal separation between arriving aircraft, thereby enhancing the arrival capacity of the airfield during IMC. Other instrument approach enhancements, such as triple dependent/independent parallel operations, dual independent operations with offset approaches, and RPAT, are not likely to be employed at small airports.

Effects of Dependent Runway Operations on Planning and Development at Small Airports

For small airports that may benefit from wake turbulence avoidance procedures or reduced aircraft and/or runway separation standards for instrument approaches to closely spaced parallel runways, the capacity of the airfield could be increased. Such procedures would reduce aircraft operational delays and potentially defer the timing and the need for other capital improvements necessary to enhance the capacity of the airfield. These enhancements could also reduce aircraft flight paths on arrival and aircraft hold times for departure. (See Table 5-14.)

Given the relatively low risks associated with the wake turbulence generated by GA and small regional passenger aircraft, the overall benefits of wake turbulence avoidance procedures are likely to be marginal at small airports and difficult to quantify or factor into airport planning analyses. However, the potential reduction in minimum separation between aircraft arrivals on closely spaced parallel runways during IMC may allow the implementation of closely spaced parallel runways and provide either more efficient independent or dependent parallel operations that could not otherwise have been provided within the available space. To be effective, however, the parallel runways would need to be served by precision instrument approach procedures with vertical guidance. This could include ILS approaches, RNAV/RNP approaches, or RNAV (GPS) approaches such as LPV approaches.

As with large airports, the recent and potential changes in the rules for the minimum spacing between parallel runways required for dependent or independent approaches, and the types of instrument approach procedures that can use such parallel runways, could open up new opportunities at small airports that previously would not qualify for such approaches. Therefore, new alternatives could be considered in the airport planning and development process for such airports. Such alternatives would have to be evaluated in terms of how the spacing between the parallel runways could facilitate the development of airport facilities between those runways and also how the new capabilities could affect noise exposure in the surrounding communities and operations at other nearby airports.

ADS-B In

Although a timetable has not yet been established for ADS-B In procedures, the FAA does have a timetable for replacing its current radar aircraft surveillance systems with satellite-based systems utilizing ADS-B Out technologies. While the FAA is requiring all aircraft operating within controlled airspace or within 30 miles of certain airports to be equipped with ADS-B Out by 2020, no mandate or deadline has been set for requiring the ADS-B In option, which would provide aircraft operators with access to traffic, weather, and other advisory information. The TIS-B primarily benefits GA aircraft, while the FIS-B benefits all aircraft types that are equipped with ADS-B In. Although there is no current mandate to equip aircraft with ADS-B In capabilities, ADS-B will be required for aircraft flying in Class A, B, and C

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Table 5-14. Dependent runway application—small airport planning considerations.

DESCRIPTION	AIRFIELD/ AIRSPACE CAPACITY ENHANCEMENTS	ENVIRONMENTAL (NOISE & AIR QUALITY)	AIRPORT CAPITAL PLANNING	INSTRUMENT PROCEDURE DEVELOPMENT PLANNING
Wake Turbulence Avoidance Procedures				
Parallel Runways (<2,500′ Spacing)	Limited to Small Airports that May Serve Heavy Aircraft Operations			N/A
Wake Turbulence Mitigation for Departures (Upwind Runways)	Limited to Small Airports that May Serve Heavy Aircraft Operations			N/A
Enhanced Instrument Operations—CSPO				
Dual Independent Parallel Operations (>3,600′ Spacing)	X	X	X	X
Dual Dependent Parallel Operations (2,500′–3,600′)	X	X	X	X
Triple Dependent Parallel Operations (>3,900′ Spacing)	N/A to Small Airports			
Dual Independent Parallel Operations with Offset (>3,000′ Spacing)	N/A to Small Airports			
Dependent Parallel Operations (>4,300′ Spacing)	N/A to Small Airports			
RPAT	N/A to Small Airports			

airspace, around busy airports, and above 10,000 feet, which will open up opportunities for developing improved procedures because of ADS-B's much greater accuracy compared with today's conventional radar surveillance systems.

Effects of ADS-B In on Airport Planning and Development

The primary benefit of ADS-B In is to enhance operational safety both in flight and on the ground. Aircraft equipped with ADS-B In equipment, and associated cockpit displays of traffic, weather, and other advisories, can maintain uniform separation from other aircraft, thus potentially increasing the capacity of the airfield at airports without radar surveillance systems. However, the capacity benefits would be marginal and difficult to quantify.

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6 Role of Airports in NextGen Development, Implementation, and Community Outreach

This chapter provides guidance on how airports can most effectively engage with community, FAA, and industry stakeholders on the development and implementation of various NextGen technologies and operational improvements at their airports.

NAS-Wide vs. Local NextGen Programs

The various NextGen technologies and operational improvements differ according to how much they can be influenced by airports and their stakeholders, depending on whether they are NAS-wide programs or local (terminal airspace/airport) programs.

NAS-Wide NextGen Programs

NAS-wide NextGen programs that are intended to upgrade the capabilities of the NAS, such as ERAM, ADS-B Out, and Data Comm, are primarily the responsibility of the FAA in collaboration with aviation industry participants such as the airlines, aircraft and avionics manufacturers, industry associations, etc. In this chapter, these programs are referred to as "NAS-wide NextGen." Airports are likely to have little influence on these nationwide programs; nevertheless, airports can have an influence on their implementation through the following activities:

- Support the development and implementation of NAS-wide NextGen technologies and operational improvements.
- Encourage airlines and other aircraft operators to equip their aircraft and undergo whatever pilot training may be required for participation in these NAS-wide NextGen programs.
- Participate in aviation industry programs that support the implementation of NextGen.
- Understand how each of these programs could ultimately impact their airport, stakeholders, and surrounding communities.
- Participate in FAA-sanctioned technology demonstration programs with local airport stakeholders (e.g., airlines, operators, avionics manufacturers, etc.).

Terminal Airspace/Airport NextGen Programs

NextGen programs that are ultimately implemented in the terminal airspace or at individual airports, such as PBN, surface operations and data sharing, closely spaced parallel runways, and Wake RECAT, can be influenced and possibly even initiated by airports and their stakeholders. In this chapter, these programs will be referred to as "NextGen/PBN" projects.

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Airports and their stakeholders may stand to realize direct benefits from these programs, such as increased capacity, reduced fuel burn and emissions, and the ability to defer capital expenditures on capacity enhancements. However, surrounding communities could also be adversely impacted by additional overflights and increased noise exposure often caused by the high degree of flight track concentration that the greater navigation precision of NextGen/PBN provides over traditional vectored procedures.

PBN is by far the NextGen program most likely to impact communities and require stakeholder engagement and community outreach in order to be approved. PBN projects can vary from individual single-site implementations of RNAV and RNP procedures to region-wide, multi-airport Metroplex implementations of STARs and SIDs. The scale of FAA's airport and community involvement will vary according to the type of NextGen/PBN project under consideration. FAA's current process is to conduct an EA, with traditional integrated noise modeling (INM) and public process, for NextGen/PBN procedures considered as part of a Metroplex study. Other NextGen/PBN procedures initiated in other than a Metroplex study will not necessarily involve an EA. It is crucial for airport staff to be aware of FAA's internal plans for changes in airspace procedures affecting their airport environs, especially for non-Metroplex procedures that will not trigger an EA by FAA.

If an airport believes that a proposed procedure may have significant adverse impacts on environmental grounds, it should consider requesting supplemental analysis, a full EA or filing of a formal objection with FAA on the grounds of the potential for significant impacts, including controversy.

While it may seem unconventional for an airport to consider objecting to a proposed FAA airspace procedure, it is worth noting that the environmental screening process used by FAA's ATO are significantly less rigorous than those required by the Office of Airports for AIP/passenger facility charges (PFC) funded projects, and that they may not identify public reactions prior to internal ATO approval of the changed procedure.

Airports, Stakeholders, and Their Roles in NextGen/PBN Implementation

This section will review the possible roles of airports and their stakeholders, including surrounding communities, in the development and implementation of NextGen/PBN projects at airports. In particular, guidance will be provided on the roles of each of the following types of organizations:

- Airports;
- Community/Public;
- Community Groups and Non-Governmental Organizations (NGOs); and
- Local, State, and Federal Governments, and Elected Officials.

Airports

Airport engagement is critical during all phases of NextGen/PBN projects, including planning, design, and implementation. Airports are most familiar with local conditions, constraints, and relationships, making them a key partner for effective community/public engagement. Airports also can provide input on how to best conduct public engagement efforts as well as who specifically should be included in these efforts. However, there have been several instances in which airport staff were unaware of proposed NextGen/PBN procedures until late in the procedure development process and needed to insert

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themselves into the process to avoid what, in their opinion, would have been a problematic public issue. It is crucial for airport planning staff to actively monitor the plans being considered by their local ATO tower and TRACON. In addition, contact with the airport's major air carriers' ATC specialists and the FAA Office of NextGen will assist airport staff awareness of developing NextGen procedures affecting their airport. Developing relationships with ATO Tech Ops reps and ATO Planning and Requirements staff in the Service Center is also useful.

Moreover, it is important to ensure that (1) airport staff members are trained on NextGen programs and (2) qualified staff members are identified to work with FAA through the implementation of NextGen operational improvements and technologies.

Most large airports have technical experts on their staff who could facilitate implementation of NextGen/PBN procedures. Airport technical representation could come from one or more of the following departments or divisions:

- Airport Planning and Development,
- Airport Operations,
- Airport Noise Abatement Office, and
- Community Affairs/Public Relations, and Airport Owner (e.g., city, county, state, or authority) departments such as planning and development, legal, and zoning.

In addition, airport technical staff may have long-standing local partnerships or consulting relationships that can assist with coordinating airport improvement projects and may have developed successful working relationships with local and regional air traffic facilities (e.g., ATCT and TRACON), airlines, local planning organizations, and surrounding communities.

Airports also can provide information about prior and ongoing plans and programs that could affect, or be affected by, the proposed NextGen/PBN project, such as the following:

- Existing or proposed noise abatement flight procedures, airfield operational constraints, and modifications of airport standards;
- Airport Master Plans, Part 150 Studies, and prior operational studies;
- EIS/EA and prior agreements with surrounding communities; and
- Existing and planned infrastructure, development programs, and zoning ordinances.

In considering future airport development plans, it is important that airport owners/operators effectively balance the airside and landside capacities commensurate with demands.

Community/Public

Some NextGen/PBN projects require specific public engagement activities as defined by NEPA in order to undertake an EA of the proposed NextGen/PBN procedures. Even when such community engagement may not be a regulatory requirement, it is recommended that the community be recognized as a critical, non-technical stakeholder in NextGen/PBN development and implementation.

Potential outreach efforts should include residents or communities that will be (or will likely perceive to be) impacted by the proposed NextGen/PBN project. In this case, "impacted" can be defined by individual perceptions that often go beyond statutory definitions of significant impacts in the federal regulations. Concentrations of operations associated with NextGen/PBN procedures may not increase noise exposures at the levels defined as "significant" under NEPA but can still be perceived as having an impact by surrounding residential communities. In some recent cases, these concerns have resulted in project delays, increased cost, and in severe cases, termination of NextGen/PBN implementation.

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In addition to engaging communities that would be impacted, residents or communities who will benefit should be engaged. For example, there may be areas that will benefit from less noise, lower emissions, or reduced overflight activity.

Community Groups and NGOs

As a subset of the general public, community groups and non-governmental organizations are potentially key entities to engage. Community groups could include formal or informal organizations based on neighborhood or a particular concern (e.g., airport noise or aircraft overflights). NGOs can be based on geographic area or special interest. An example of an NGO that has long been following airport noise issues is the National Organization to Insure a Sound Controlled Environment (N.O.I.S.E.). Many airports have long-established community forums for addressing noise issues, such as the SFO Airport/Community Roundtable, the O'Hare Noise Compatibility Commission, and the Boston Logan International Airport Citizens Advisory Committee. These types of organizations are logical places to review proposed flight track changes.

Engaging these groups provides an opportunity to educate local leaders on a proposed NextGen/PBN project to gain a better understanding of local interests and potential concerns that can then be integrated in the NextGen/PBN development and implementation plans. These groups and the individuals involved are often trusted thought leaders within their community who can be helpful in gaining public understanding of, and support for, the proposed NextGen/PBN project.

Local, State, and Federal Governments, and Elected Officials

Government officials at the local, state, and federal levels should be engaged early in the process starting with an effort to educate them about the NextGen/PBN project and the benefits it is intended to provide to the airport, community, and the region. Smaller NextGen/PBN projects that are limited in scope are likely to remain strictly a "local" interest. However, for larger projects such as Metroplex/PBN programs, or even limited projects at larger airports, concerns may be of such a magnitude that outreach may need to be escalated to the state or federal level. Airports should take care to make sure that all parties understand (1) which of the recommended changes were sponsored by the FAA, airlines, or other parties and (2) which were sponsored by the airport (if any).

Outreach Guidance

Why Community Outreach Is Needed

NextGen/PBN projects can provide significant benefits in terms of increased capacity, reduced aircraft delays, reduced fuel burn and carbon emissions, and modernization of infrastructure. However, they also can raise significant concerns regarding impacts on the community with respect to aircraft noise, flight paths, and structures/development in the vicinity of the airport.

Moreover, NextGen/PBN projects can have a wide range of impacts on the community depending upon their purpose and scope. Metroplex projects likely would entail greater needs and opportunities for community outreach than less complex airspace actions such as a single-site NextGen/PBN project. Similarly, NextGen/PBN projects that might impact noise-sensitive or historic community areas, or which may affect development in the vicinity of the airport (including temporary issues such as cranes impacting airspace), may require greater outreach than even larger projects with fewer such concerns.

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The issues of aircraft noise exposure, and the linkage of noise exposure to changes in ATC procedures, are not new, but the flightpath flexibility and concentration afforded by NextGen/PBN procedures has raised new noise issues and concerns as outlined below:

- Transforming the ground-based system to the satellite-based system has the potential to raise the issue of who is exposed—and the level of their exposure—at every location where NextGen/PBN procedures are being implemented.
- It is highly likely that the operational or efficiency benefits accrue primarily to airlines and that the associated environmental benefits, such as reduced carbon emissions, will be heavily discounted by those that experience increased overflights and noise as a result of NextGen/PBN procedures.
- While the number of people exposed to aircraft noise may be the same or even reduced with new NextGen/PBN procedures, the ability to fly more precise flight tracks can result in increased concentrations of flights over some areas (and thus increased noise levels) and/or "new" noise exposure areas over other areas.

For real estate development in the vicinity of the airport, issues that may need to be addressed include but are not limited to height of structures, lighting and glare impacts associated with proposed development, and unmanned aerial vehicle activity.

Because of these new concerns associated with NextGen/PBN procedures, FAA and the industry have recognized that increased community outreach and attention to community concerns with NextGen/PBN implementation are very important to the success of NextGen/PBN implementation. Some of these new outreach processes are so recent that they have not yet been formalized. However, greater community outreach may help in the successful implementation of some of the NextGen/PBN procedures being developed.

Preliminary Findings from ACRP Project 01-28

ACRP Project 01-28, "NextGen—Guidance for Engaging Airport Stakeholders," had the objective to develop guidance to airports to engage with the FAA and other airport stakeholders on NextGen development and implementation. That report, *ACRP Report 150: NextGen for Airports, Volume 2: Engaging Airport Stakeholders* provides the most detailed guidance on engaging airport stakeholders and communities. Below is a list of selected findings from that research that are of most interest to the airport planning practitioner:

- PBN, Surface Operations, and Multiple Runway Operations represent the NextGen initiatives that are most likely to benefit from airport-focused stakeholder engagement tools in the short term (i.e., within five years).
- NextGen initiatives are being driven by technology, specifications, and process improvements, the details and effects of which are not fully understood by many airport managers, staff, consultants, and community members.
- Complex terminology, inconsistent definitions, and the differences between NextGen programs and portfolios as reported by the FAA contribute to a lack of understanding about NextGen initiatives.
- Despite the abstract nature of many NextGen initiatives, several can have significant impact on airport stakeholders. Safety, capacity, operational efficiency, and economic growth are expected to increase.

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- Operational cost, greenhouse gas emissions, and fuel usage are expected to decrease. Noise is expected to increase in some areas and decrease in others. Complaints and concerns are focused on areas where noise increases are perceived.
- Over the past several months there has been a notable increase in negative press, driven by residents (and the elected officials who represent them) who experience increased or more frequent noise impacts.
- The information most required by stakeholders includes maps of flight tracks and the configuration of airspace around an airport, associated noise impacts, statistical information on aircraft operations, fuel usage, emissions, separation, and obstacle clearance criteria. General information on NextGen and the priorities of its various initiatives (such as expected site-specific implementation dates), cost/benefit information, and maps of airport/airspace configuration are also needed. Most expect to get this information from FAA publications or regional staff members, although some have received helpful information from regional agencies.
- Different types of stakeholders require different types of information on the impacts of NextGen. For example, the FAA and airlines require conceptual plans and technical details. Residents and businesses require information on the noise impacts of proposed flight track changes. Project funding considerations are of interest to airlines, the airport, and the FAA. All stakeholders need information on the timeline for implementing specific procedures in their area.
- Some airports have successfully engaged the FAA, residents, airlines, elected officials, and other stakeholders to address noise, safety, and operational efficiency (in descending order of the number of responses received). This engagement has predominantly been bi-directional, meaning in forums that allow dialog to take place. Periodic, scheduled meetings that involve two or more stakeholders are a preferred method of engagement. Participants in these meetings appreciate general, informative presentations, but they require specifics with regard to timing, impacts, and funding, as relevant. If this information is not available, frustration and ultimately a lack of trust can ensue.
- The tools that can cost-effectively convey this information include material to support briefings, easy to navigate websites with airport specific statistics, maps, and customizable information as well as kits that include brochures and fliers, press releases, infographics, and posters.
- To be effective, information must be easy to access, at the appropriate technical level for the audience, and as specific as possible to the particular airport and current NextGen Initiative(s) being implemented. While there is a need for generalized information on NextGen as a whole, most (two-thirds of those who responded to the question) feel that information must be specific to their organization or airport. This presents a challenge as to how to prepare tools or guidance at a national level that can help individual airports develop the local information they require.
- A few respondents have noted that productive two-way dialog seems to occur when the airport and/or an operator is the proponent of a new procedure. Airports and community members report that lack of local input to the procedure development process can lead to a less than optimal design that doesn't adequately take into account operational, land use, and noise impact factors. The result can be frustration, anger, and (in at least one case) law suits.

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Key Enabler of NextGen Implementation—Airports GIS

As of January 2015, Airports GIS is the authoritative method of acquiring survey data for all FAA lines of business. Airports GIS receives survey data about obstacles and NAVAIDS through a web feature.

The FAA uses the Airports Surveying GIS (Airports GIS) to collect data voluntarily submitted by an airport and other aeronautical data to meet certain requirements of NextGen planning and implementation. The Office of Airports GIS program is used to develop electronic airport obstruction charts and electronic airport layout plans. The FAA's goal is to integrate multiple versions of the airport data: preliminary, current, planned, and temporary. Additional plans are to share data with other FAA systems such as iOEAAA and eNASR.

To support the implementation of NextGen, the FAA has requested that airports move to the defined Airports GIS system and use the web application features to access required airport data and submit changes; the web application will automatically notify the proper FAA office of the changes. FAA has issued three ACs to provide guidance to airports for the collection and submission of aeronautical data and airport-related data.[1] The airports GIS system identifies the geographic location and characteristics of natural or constructed features or boundaries on the earth's surface. The airport data are used to develop and implement WAAS/Localizer Performance with Vertical Guidance (LPV) approaches, PBN approaches, obstruction analyses, and flight deck airport moving map functionality.

The central database for airport GIS information enhances sharing of accurate safety-critical data, such as runway endpoints or location of NAVAIDS, and non-safety-critical data, such as the location of a building on an airfield. The airport GIS system provides users and procedure developers with current airport data, facilitating the FAA and airports to collaborate more effectively on airport planning, design, operations, and maintenance decisions.

Local Partnerships with ATC, Airlines, and Community Leaders

Although most large airports maintain ongoing forums where they interact with various stakeholders and constituencies, there are circumstances in which a key interest is not a normal participant. The difficulty of communicating effectively with large, distributed organizations is well known and planners need to be aware that merely having airline, ATC, or community participation in an established forum is not a guarantee that the proper parts of those organizations are aware of plans.

For example, an airline station manager may be the ideal participant in developing a snow and ice control plan or in meetings discussing ramp safety or foreign object debris (FOD) control, but may not adequately represent the views of their flight dispatch or operations organizations on ATC matters. Most airport contacts with airlines occur with their station managers or their property departments and may not be effective at gaining the airline's perspective on ATC NextGen issues. Likewise, airports often look to FAA tower or Airports District Office (ADO) or Regional Office personnel as conduits to

[1] The three advisory circulars are (1) AC 150/5300-16A, *General Guidance and Specifications for Aeronautical Surveys: Establishment of Geodetic Control and Submission to the National Geodetic Survey*; (2) AC 150/5300-17C, *Standards for Using Remote Sensing Technologies in Airport Surveys*; and (3) AC 150/5300-18B, *General Guidance and Specifications for Submission of Aeronautical Surveys to NGS: Field Data Collection and Geographic Information System (GIS) Standards*.

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other FAA offices. In many cases these contacts will direct the planner to the appropriate FAA office. However, while they may be the ideal participants in construction coordination, operationally oriented meetings, airport certification, land use planning, or AIP grant issues, they may be less helpful in understanding longer-term planning within the various ATO offices. They may not be aware of long-term plans within FAA to deploy new airspace procedures until just a few months before they commence implementation. Airport planners should think critically about what parts of airlines or FAA need to be queried about NextGen plans for their airport and aggressively pursue contacts with those offices.

Airport planning practitioners should also understand the roles of various offices in NextGen planning and implementation (NextGen, Flight Standards, ATO Program Management Office, ATO Mission Support Services). These offices develop the plans, develop CSPO, implement new technologies, and implement PBN, respectively. Further details of these organizations and their role in NextGen implementation can be found in Appendix D of this guidebook. Below is a brief overview.

- The Office of NextGen is structured under the ATO of the FAA. The mission of this office is to manage the implementation of NextGen technologies and capabilities into the NAS. The Assistant Administrator for NextGen and the Deputy Assistant Administrator for NextGen both work directly with the Chief of Staff, Office of Chief Scientist for NextGen, and Office of Chief Scientist for Software. The FAA Office of Airports (ARP) serves as a liaison in facilitating the implementation of the NextGen integration with airport planning initiatives and airport development activity.
- It would also be helpful for airport planning practitioners to become familiar with the FAA IFP Information Gateway, which is described in more detail in Appendix E of this guidebook. This Gateway is the centralized IFP data portal, providing a single source for aeronautical charts, the IFP Production Plan, IFP Coordination, and IFP Documents. The site provides a user the ability to look up, and be notified of, changes in existing and proposed IFPs by airport, state, region, and FAA service area.

As with airlines and ATC, planners need to understand the breadth of engagement that may be needed with community leaders as NextGen issues arise. Many airports have long-established relationships with community groups focused on environmental, economic, or other issues. Many others depend on the public affairs or government affairs staff at their airport to act as liaisons and interface with community groups, elected officials, or local business leaders. While using those resources to help guide outreach efforts, planners should be aware that broader outreach may be necessary due to the potential for reactions outside of the traditionally affected areas that some NextGen procedures can produce.

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A ACRP NextGen Initiative Projects

Each year, the ACRP Oversight Committee (AOC) has a conversation about identifying an emphasis area of unmet airport research needs. In 2013, at its January meeting, the AOC identified a need for research on the topic of NextGen at airports and provided funding to accomplish that objective. A workshop of invited industry practitioners was held by TRB in February 2013, and several problem statements were drafted. Over the next several months, these problem statements were refined, and the AOC allocated a total of more than $2 million for five of these problem statements at their July 2013 meeting. The five projects were:

- ACRP Project 01-27, "NextGen—A Primer,"
- ACRP Project 03-33, "NextGen—Airport Planning and Development,"
- ACRP Project 01-28, "NextGen—Guidance for Engaging Airport Stakeholders,"
- ACRP Project 03-34, "NextGen—Understanding the Airport's Role in Performance-Based Navigation (PBN)," and
- ACRP Project 09-12, "NextGen—Leveraging NextGen Spatial Data to Benefit Airports."

ACRP Project 01-27, "NextGen—A Primer," is seen as the flagship project in the initiative and is intended to provide the highest level overview of NextGen, focused specifically on providing three products:

- An "Executive Primer" for airport directors, CEOs, and other senior executives. This is a high-level introduction to those issues that an airport's senior management needs to be informed of. It will give an overview of what NextGen is, what it means for the CEO's airport, the possible funding or development implications of NextGen and, most importantly, any risks to the airport that the NextGen programs may pose. Of particular interest in the executive primer, is the potential for public controversy related to NextGen activities. The Executive Primer will present information at a basic level of detail.
- A Practitioners' Resource Guide focused on providing somewhat more detailed information needed by airport staff who will have their job functions affected by NextGen. This guide provides information at an introductory level, giving a reasonable in-depth treatment, while not going into the much greater detail with which the other three projects treat their subjects.
- A Public Information Toolkit designed to help airports communicate with community groups, local media, business leaders, elected officials, and others not generally familiar with aviation issues. The Public Information Toolkit will also provide information that can be used with local aviation interests, pilots, and others with a direct stake in NextGen issues at their airport. The toolkit will standardize, elevate, and simplify messages for generic industry-wide and public distribution. The goal is to raise general awareness of NextGen and the role airports have in its implementation. Messages will focus on community effects; show how airports fit into NextGen; what NextGen means to communities; and how the roles of airports, communities, and other stakeholders may change.

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This project, **ACRP 03-33, "NextGen, "Airport Planning and Development"** is intended to develop a guidebook to help airport industry practitioners understand and incorporate NextGen capabilities into planning for all categories of airports, including:

- A description of how specific NextGen capabilities affect airport planning and development;
- Methods to assess the improvements in safety, capacity, efficiency, and environmental performance related to NextGen capabilities and their economic impacts;
- Case study examples of NextGen projects highlighting best practices and lessons learned for incorporating NextGen into the airport planning process; and
- A comprehensive list of published documents applicable to incorporating NextGen into airport planning.

ACRP Project 01-28, "NextGen—Engaging Airport Stakeholders Guidebook" is intended to develop guidance for airports to engage with the FAA and other airport stakeholders on NextGen development and implementation life cycle to include, but not be limited to, planning, environmental, review, design, and deployment. This will be done at a more detailed level than the public information toolkit of the "NextGen—A Primer" project and will include:

- Tools to create proactive communication (including key messaging), collaboration and cooperation plans and strategies specifically tailored for a variety of factors, including airport category, stakeholder role, and type of NextGen technology. This effort includes a description of different engagement methods (e.g., a guidebook, social media, website, workshops and forums, community roundtables, town hall meetings).
- Methods that will enable airports to proactively engage with and to solicit and consider input from stakeholders about the environmental and economic benefits as well as costs of NextGen implementation.
- A continuous engagement strategy that will consider the important balance between enhanced community engagement and efficient NextGen implementation including managing schedule and costs.
- Example applications of tools, methods, and strategies for a prioritized list of NextGen initiatives and stakeholder scenarios.

Case studies of a representative spectrum of airports that provide lessons learned and best practices of stakeholder engagement. The case studies are based on scenarios that include consultation with the FAA, airport management, airline representatives, and, where appropriate, members of the community.

ACRP Project 03-34, "NextGen—Understanding the Airport's Role in Performance-Based Navigation (PBN)" is an in-depth study of the details of NextGen PBN as it relates to airports.

PBN is a critical near-term component of the NextGen program. Design and implementation of PBN will have significance for airports of all sizes. Development of PBN procedures is currently underway, or will be underway shortly, in a number of communities. Involvement by airport operators is essential for successful implementation; and potential opportunities exist for realizing operational and environmental benefits as well as improvements to safety, reliability, and efficiencies of air services to the community.

To maximize their productivity, airport operators need to have an understanding of the FAA design and implementation procedures, and have the means to identify and monitor metrics of expected benefits and impacts of these procedures so they can report back to their communities. To help implement that program, the aviation community needs comprehensive and understandable information

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concerning PBN, presented in a usable and accessible format—describing implementation requirements, related benefits, and potential costs.

The objective of this research was to prepare a user-friendly "airport PBN resource guide" that presents essential, relevant material for use by airport operators, planners, managers, and others. The guide includes background information, an overview of federal policy (including recent legislation), accurate measures of the tradeoffs resulting from implementing PBN, and an identification of potential implications to stakeholders. Trade-off measures encompass both quantitative and qualitative approaches to measure change in noise exposure, air quality, safety-related incidents, and other factors. In addition, the resource guide describes practices and methods airport operators can use to work with their surrounding communities, the FAA, and other stakeholders in all phases of PBN development and implementation. The guide addresses the following primary issues:

- Safety enhancements;
- Improved efficiency of operations at the airport and in the surrounding airspace; and
- Environmental outcomes, particularly relating to noise and emissions.

ACRP Project 09-12, "NextGen—Leveraging NextGen Spatial Data to Benefit Airports." The objective of this project in the NextGen initiative was to develop a guidebook for airport operators that identifies the benefits that can be derived from spatial data that is to be collected in support of the FAA's NextGen effort, and provides guidance on how airports can maximize use of this data. The project includes:

- An executive summary that demonstrates the benefits of the spatial data to be collected, and an overview of NextGen technologies for airport executive staff;
- General information on return on investment including maintenance of data;
- Benefits of data sharing with stakeholders including identification of the risks and challenges;
- Identification of additional data that may be available and beneficial to airports;
- Discussion of how data may be monetized to generate revenue for the airport, including:
  - Legal issues, if any;
  - Limitations to selling/sharing data;
  - Liabilities;
  - Uses of data that can be used to achieve operational objectives;
  - Uses of data that can be used for non-operational purposes (e.g., community engagement, marketing);
- Benefits that the airport may achieve from the integration of spatial data with NextGen technologies;
- Case study examples that illustrate best practices;
- Identification of relevant technology trends that will impact airports in the future;
- Expected timelines for airports to address spatial data requirements related to NextGen technologies;
- List of additional resources; and,
- Glossary of terms.

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Collaboration Among ACRP NextGen Initiative Projects

Taken together, the five NextGen Initiative projects are intended to provide comprehensive guidance on NextGen as it affects airports from the highest level concerns of the airport director or CEO to the detailed needs of working-level airport staff charged with implementation of NextGen-related programs.

One vision of the information and results exchanged among the various projects is represented graphically in Figure A-1. This graphic is admittedly an ACRP 03-33–centric view of the ACRP NextGen Initiative. The information exchanged during the research occurred primarily at the February 2015 Workshop in Irvine, California, where all five NextGen Initiative project research teams were present. However, certain information was also exchanged following the workshop, particularly in the areas of survey results and available literature.

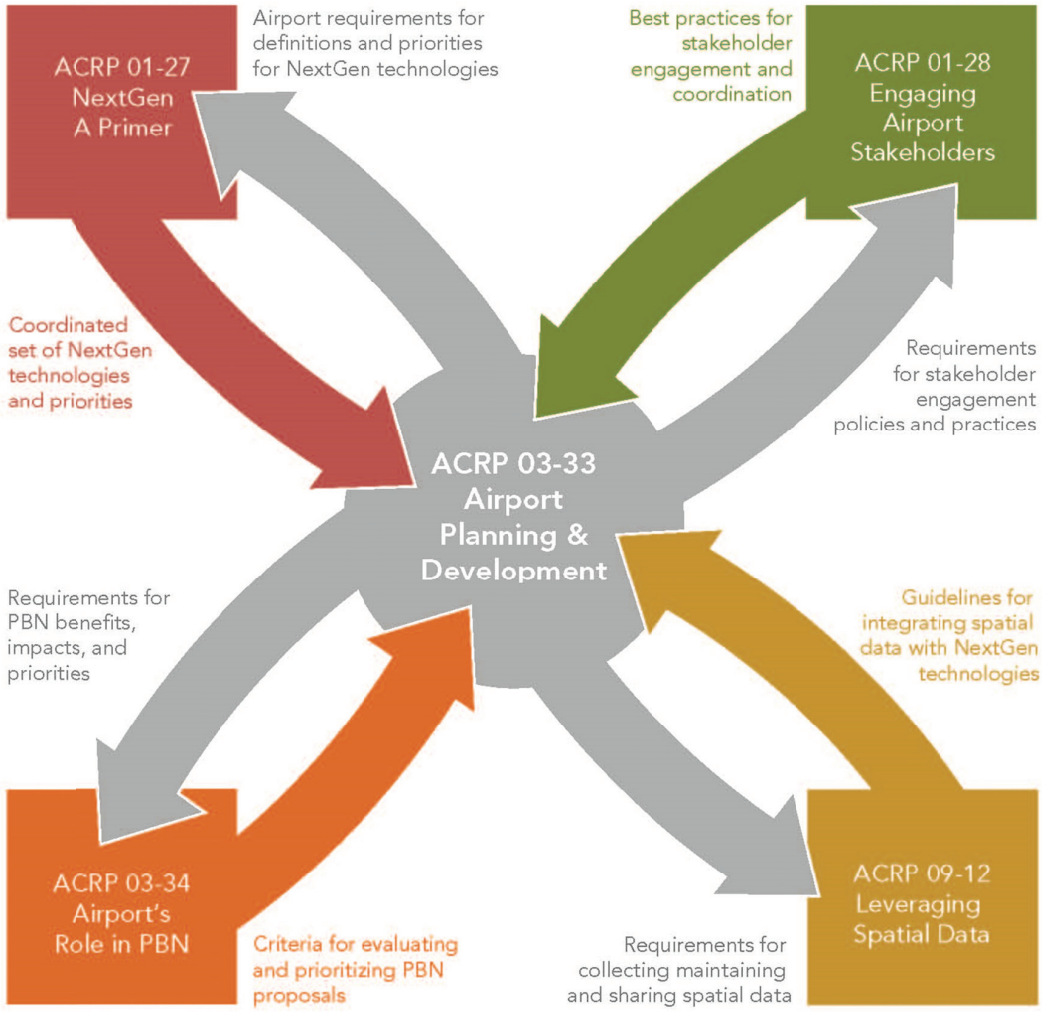


Figure A-1. *ACRP NextGen Initiative—information and results exchange.*

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B Best Practices and Lessons Learned from Airport Case Studies

Case Study Selection Criteria

The goal of this team's case study research is to identify, select, and study projects that reflect the various types and sizes of airports in the NAS with direct relevancy to FAA NextGen program initiatives and airport planning. The research team's approach for this task was to:

- Develop several screening criteria for the selection of candidate case studies,
- Identify a preliminary set of candidate case study airports or projects to be considered based on the initial screening criteria,
- Evaluate and weigh each candidate study,
- Select final list of case study projects, and
- Conduct the case studies for the guidebook.

The following sections describe each step of this methodology through the identification of six case study projects selected by the panel, as well as a summary of each case and a discussion of best practices and lessons learned.

Screening Criteria for the Identification of NextGen Projects at Airports

As part of the literature review phase of the project and through team discussions, several criteria were identified as a means to identify candidate case studies for the final project. These criteria include:

- Incorporation of NextGen into airport system, master planning, or environmental planning studies;
- Performance-based navigation;
- Multiple runway operations;
- Reduced airspace interaction through the de-coupling of adjacent airports;
- Surface operations management;
- Time-based flow management;
- Separation management; and
- Low-visibility approaches.

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Case Studies Selected

Six cases were selected based on the criteria established as part of this research effort as follows:

- Colorado Wide Area Multilateration Implementation,
- Friedman Memorial Airport Required Navigational Performance Approach,
- The New York John F. Kennedy International Airport Collaborative Decision Making,
- Newark Liberty International Airport Ground-Based Augmentation System,
- San Francisco International Airport—Simultaneous Offset Instrument Approach, and
- The Puget Sound Regional Council—Preparing Busy Airports for NextGen Technology.

Table B-1 highlights the relevance of each case as related to the criteria established. These six case studies represent over 15 airports of all sizes throughout the country with relevance to multiple NextGen capabilities.

Case Studies Summary of Best Practices and Lessons Learned

Colorado Wide Area Multilateration Implementation

Many of the commercial airports in Colorado are located in high mountainous terrain, serving tourist destinations including ski areas. Historically, operations in low-visibility conditions to and from these airports were constrained due to the lack of radar coverage available below 10,000 feet MSL. During low-visibility weather events, arrivals and departures at these airports would drop from a range of 12 to 17 per hour to just 4 to 5 per hour. This often resulted in the cancellation or delay of flights, lost revenue, and inconvenienced travelers.

Most of the flights cancelled originated at the Denver International Airport. When flights were cancelled, travelers would resort to renting cars and driving to their final destination, increasing the usage of an already overcrowded Interstate and state highway road system. CDOT made solving this problem a high priority and dedicated significant resources to it. CDOT leadership also developed a "laser-focused" strategy, with tenacity, to solve this problem.

The Colorado WAM project demonstrates multiple best practices and lessons learned including:

- Collaborative working process between CDOT, local communities, businesses, airports, airlines, and FAA. CDOT identified the many stakeholders that would benefit from solving this problem and their strategy included actively reaching out to them, educating them, enlisting their service, and even raising necessary funds from them.
- The relative ease of implementing the technology and the utilization of existing infrastructure. The CDOT team included the right experts in the subject matter, who devised a way of using largely existing technology in a unique manner to tailor a solution to this problem.
- Airports, airlines, passengers, and businesses have benefitted in measurable ways through the implementation of this technology. The system has proven itself in saving money and time and increasing reliability and safety.

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Table B-1. Case Study selection matrix.

	COLORADO WIDE AREA MULTILATERATION IMPLEMENTATION	FRIEDMAN MEMORIAL AIRPORT (SUN) RNP APPROACH	THE NEW YORK JOHN F. KENNEDY INTERNATIONAL AIRPORT (JFK) COLLABORATIVE DECISION MAKING	NEWARK LIBERTY INTERNATIONAL AIRPORT (EWR) GBAS	THE SAN FRANCISCO INTERNATIONAL AIRPORT (SFO) – SIMULTANEOUS OFFSET INSTRUMENT APPROACH	THE PUGET SOUND REGIONAL COUNCIL – PREPARING BUSY AIRPORTS FOR NEXTGEN TECHNOLOGY
AIRPORT SIZE	**MULTIPLE SMALL**	**SMALL**	**LARGE HUB**	**LARGE HUB**	**LARGE HUB**	**MULTIPLE MEDIUM AND SMALL**
Incorporation of NextGen into System, Master, or Environmental Planning Studies	X	X				X
PBN		X		X	X	X
Multiple Runway Operations	X			X	X	X
Reduced Airspace Interactions Through Decoupling of Adjacent Airports	X				X	X
Surface Operations Management			X			
TBFM			X		X	
Separation Management	X				X	X
Low-Visibility Approaches		X		X		X

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- One of the benefits of the project is in promoting the ability of an airport sponsor to enter into an agreement with the FAA to develop and integrate emerging NextGen technology into an airport system. This program allowed airport sponsors to have a role in implementing the improvement. It also allowed airport sponsors to factor in the increased capacity into their long-range planning. This program will allow airport sponsors to delay capital expenditures due to the increased capacity.
- The willingness of FAA to embrace a project where the need justifies a technologically feasible and economically beneficial solution was demonstrated.
- One of the more significant lessons learned from this program, and one of the primary reasons for its success, was a small, tenacious group of individuals that made it their mission to solve this problem. They never gave up, even when confronted with technical, political, and budget challenges. This group exhibited a unique blend of technical savvy, political astuteness, creativity, and force of will to make it all happen. It is doubtful airports, airlines, passengers, and businesses would be experiencing these benefits if it were not for this group.

Friedman Memorial Airport—RNP Approach Implementation

The Friedman Memorial Airport (SUN) is located in Hailey, Idaho, about 13 miles south of Ketchum, and about 4 miles north of the City of Bellevue. The airport is in the narrow valley of the Big Wood River and is surrounded on three sides—north, east, and west—by mountains. The weather, combined with the high terrain in the area, creates a challenging environment for aircraft operators. Instrument operations had been limited due to terrain and the high minimums associated with the conventional non-precision instrument approaches to the runways at the airport. As a result, many flights are delayed or cancelled each year, creating loss of revenue for area businesses and inconveniences for the traveling public.

The Friedman Memorial Airport Authority (FMAA) had attempted to improve the instrument approach capability at the airport through the implementation of available technology beginning in 1994, including a microwave landing system and a transponder landing system. In 2006, the FAA published an RNP AR SIAP in an attempt to improve all weather access to the airport. The procedure offered lower minimums and access to properly equipped aircraft. However, the missed approach segment of the procedure extended an extremely long distance of over 60 nm. The long missed approach segment exceeded the operating specifications of the airlines serving SUN, thereby inhibiting its use.

The FMAA has worked to solve the problems with the RNP approach since 2011. Recently, FAA issued a notice that due to criteria changes, the approach is in jeopardy of being cancelled. Best practices and lessons learned from this case study include:

- The promise of NextGen may delay long-term planning. The FMAA had been pursuing the relocation of SUN to a new site away from the valley, due to the terrain and resultant reliability issues. The process to build a new airport was compromised by the fact that opponents routinely made the argument that technology could solve the problem. In this case, it did not. From an airport sponsor perspective, it can be said that the FMAA worked very hard, with tenacity and diligence, to try and fix the problem.
- The promise of NextGen may not be fully realized at small airports with limited resources. The FMAA devoted significant resources (their own money) to trying to resolve the reliability issue through the implementation of NextGen technologies. It was a challenge to continue doing this, since there were other capital and operating expenses competing for limited budget resources.
- NextGen solutions do exist at smaller airports; however, the FAA does not have sufficient resources to address specialty projects at smaller airports. It is believed that the 60 mile missed approach issue, which is the reason commercial airlines cannot take advantage of the RNP approach, can be

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resolved. The FAA would have to change the rules in order to make that happen and they also face resource challenges, making it difficult to solve the problem.

- Changing instrument flight procedures criteria can adversely impact existing minimums at airports.

The New York John F. Kennedy International Airport Collaborative Decision Making

The Surface Management Program at The New York John F. Kennedy International Airport (JFK) aims to leverage the availability of comprehensive airport surface surveillance data and airline schedule information to better manage the taxi-out process, reduce taxi times, and improve efficiency. During periods when departure demand exceeds capacity, departing aircraft are held at the gate or another holding location, and released to the runway in time to join a short departure queue before taking off. As a result, aircraft absorb the delay with engines off and decrease their fuel burn, emissions, and engine maintenance costs.

A key element of the process was the use of predictive analytics to accurately forecast up to eight hours in advance the expected departure and arrival slot counts based on the weather forecast and past airport performance under similar predicted weather conditions. This in turn was used with the requested push-times sourced from the airlines to develop the initial allocation of flights to available taxi slot times over the forecast period.

Some of the best practices and lessons learned from this program include:

- A surface management program similar to the JFK program would only apply at airports that experience significant surface congestion or that have very limited taxiway and apron space for the staging and queuing of departures.
- One of the key enablers of such a surface management program is that it requires either available gates or ramp space in order to store aircraft that are in the virtual queue waiting for their clearance to taxi.
- Achieving the full benefits of such a surface management or departure metering program at some of the most congested airports could require substantial investment in additional gates or ramp space that may not be feasible or economically justifiable.

There was a general feeling that the program would have worked better if the local ATC facilities had been more involved. Recently, the FAA Surface Operations Office has been implementing changes at the Port Authority's airports without involving the Port Authority. This office views surface management as an ATC function; unlike the original surface management program that was developed at JFK, which was largely an airport and airlines function. Nevertheless, although the current JFK surface management system does not exactly follow the FAA Surface Operations process, the two programs have common goals and objectives and many common elements.

Newark Liberty International Airport Ground-Based Augmentation System

The goal of GBAS implementation is to provide an alternative to the ILS supporting the full range of Category I, II, and III approach and landing operations. The GLS is an ILS "look alike" system based on the use of the GPS. It is used only in the final approach phase of flight. Newark Liberty International Airport (EWR) was the first airport in the United States to have GBAS.

Some of the best practices and lessons learned from this program include:

- GBAS has more flexible siting criteria, allowing the GBAS to serve runways that ILS is unable to support.

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- A GBAS is sited to minimize critical areas, which places fewer restrictions on aircraft movement during ground taxi and air operations.
- The GBAS approach guidance is steadier than ILS approach guidance and does not fluctuate.
- GBAS installations are AIP eligible for funding using either AIP entitlement funds or PFC funds. However, these federal funds do not cover maintenance of the GBAS system.
- The Port Authority encountered soil problems at EWR that made it expensive to install the GBAS antenna.

The Port Authority now thinks it would have been more beneficial to implement GBAS at JFK first because it would have facilitated reducing the airspace conflicts between LaGuardia Airport (LGA) and JFK. In addition, Delta Air Lines, a predominant carrier at JFK, currently has 42 aircraft that are GBAS capable based out of JFK. In addition, one of the real benefits of GBAS would be that it would enable a straight-in approach to Runway 13R at JFK.

San Francisco International Airport—Simultaneous Offset Instrument Approach

During clear weather conditions SFO can accept up to 60 operations per hour to Runways 28L and 28R. During low cloud conditions, and conditions where the airport is not visible on approach due to broken cloud layers in the bay, simultaneous parallel runway operations (arrivals) are not achievable and the hourly acceptance rate drops to 30 operations. The drop in capacity is a result of IFR ATC separation requirements associated with closely spaced runways allowing only a single stream of traffic into the airport.

The FAA, RTCA, airlines, and SFO worked collaboratively in the late 1990s to develop concepts and demonstrate solutions using PBN and existing ground-based technology. A program was ultimately established named simultaneous offset instrument approach (SOIA) based on the implementation of PRM surveillance radar funded by the airport and the development of an offset ILS approach to Runway 28R.

At the time that the SOIA project was established, the NextGen initiative did not exist in FAA. The initial goal was to allow visual acceptance rates to the closely separated parallel runways at the airport during times when the cloud layers above 1,000 feet prevented visual sight of the runway in VFR conditions on the field. Multiple challenges needed to be addressed to enable the operation, including pilot procedures, ATC procedures, wake turbulence, adequate surveillance, and adequate navigation.

The solution to the problem involved implementation of a PRM and an offset LDA with glideslope equipment and approach procedure for Runway 28R. An offset localizer type directional aid (LDA) with glideslope is simply an ILS system that is offset more than 3 degrees from the runway centerline. The implementation of these technologies, combined with the conventional ILS serving Runway 28L, the establishment of controller and pilot standard procedures, and training enabled the operation.

Multiple lessons have been learned through the SOIA program, including:

- A local initiative in a collaborative working environment can lead to success.
- Airport funding of infrastructure may be an important factor in NextGen implementation.
- Unique uses of existing technology can create significant benefits in the NAS.
- Establishing and maintaining the SOIA program for the increase in capacity through technology has provided benefits over the course of the past 11 years.
- Interruption of service due to construction can be a driver to develop innovative technology applications and procedures to maintain operations.

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- Establishment of a successful technology-based program at an airport may lead to benefits at other airports. Additional capacity gained through NextGen may serve to delay the need for capital improvement projects.

Efforts continue today for the implementation of NextGen to improve operations and reduce cost.

The Puget Sound Regional Council—Preparing Busy Airports for NextGen Technology

The project represents the first NextGen study focused on a system of general aviation airports in the NAS and was funded via AIP grants. From the FAA's perspective, PSRC was an ideal organization to fulfill the objectives of this grant. It already served as a forum for communication with airports in the region. There were also clear economies of scale to be achieved by studying the impact of NextGen on many closely related airports at the same time. In addition, a number of NextGen technologies and operational improvements were being evaluated at the Seattle-Tacoma International Airport (SEA).

The project was divided into two phases. Phase I studied airports within the PSRC jurisdiction that could benefit from NextGen. The analysis provided an overview of the FAA NextGen program, details of NextGen technologies and how they may apply to airports in the Puget Sound system, an inventory of all airports within the Puget Sound area including economic analysis, establishment of the busiest airports in the system, a gap analysis for the busiest airports to enable NextGen, and a NextGen analysis template for busy airports. Phase II of PSRC's study focused on NextGen technologies to enhance airspace efficiency, reduce congestion and delay, and improve safety in the most congested airspace in the Puget Sound region.

The project has successfully brought together airlines, industry, airport management, regional planners, and FAA to plan for the NextGen implementation at smaller GA airports from a system perspective. Other outcomes include:

- Establishment of capital improvement programs for the Puget Sound area airports.
- Identification of survey requirements for airports in the Puget Sound area.
- Identification of obstacles to air navigation.
- Identification of airspace design and instrument approach alternatives for airports in the area to deconflict operations with operations at SEA.

The PSRC NextGen study has provided multiple lessons for application in all airport systems across the NAS including:

- AIP funds can be used for NextGen airport system planning.
- The approach taken by FAA and PSRC can be applied to other regional airport systems to enable GA airports to plan and be prepared for NextGen improvements.
- Development of capital improvement programs to enable NextGen technologies is a prudent step for small airports.
- Identification of aeronautical survey requirements is essential to enable the development of PBN instrument approach capability at small airports.
- Identification of obstacles to air NAVAIDS in obtaining timely airspace reviews and approvals of NextGen improvements.
- Development of plans for future PBN SIAPs will aid in the protection of airspace for the future implementation.

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C NextGen Elements and Applicability by Airport Sizes and Issues

The opportunities to enhance the NAS through the implementation of NextGen operational improvement and technologies can vary significantly among airports, regardless of an airport's size, role, and classification. There are numerous factors that will influence the direct and indirect benefits associated with each of the various NextGen initiatives for any given airport or system of airports. Therefore, the application of NextGen technologies and operational improvements, and the resulting benefits, will be unique for each airport. Nevertheless, the opportunities to realize the benefits of NextGen at small commercial service and GA airports can be differentiated from those for large and medium airports that have a significant composition of air carrier and cargo aircraft, which is what this appendix is intended to illustrate.

It is important to understand that the success of NextGen at these airports is as much a function of the equipage of the fleet operating at the airport as it is of the size or type of service the airport has. Any given small airport may be used by aircraft with Nav/Com capabilities ranging from minimal (no electrical system) to superior (full flat screen avionics suite). So, simply characterizing the impact of NextGen by airport size or type of service is difficult. Nevertheless, in keeping with this project's directive to address issues at small airports, the research team has tried to generalize the issues.

The following sections outline the general applicability of the various NextGen elements to airports based on airport size, although size alone is not the only criterion. While each airport's ability to leverage NextGen operational improvements is unique, the information contained herein is intended to distinguish the NextGen initiatives that are most likely to be applied at the various airport types and sizes, based on, but not limited to, the following criteria:

- **Benefit/Cost**—Prior to implementing any NextGen Initiative, the proponent needs to consider the benefit/cost relationships. Although achieving cost savings that outweigh the cost to implement and maintain equipment and facilities support a specific NextGen initiative is important, the evaluation of benefits should also consider other hard to quantify benefits, such as enhanced operational safety and reduction in emissions and noise exposure. Conducting a benefit/cost analysis, along with other economic and financial metrics (e.g., payback period, return on investment, etc.) can also assist in the evaluation of multiple solutions that are being considered to address a particular airfield or airspace need.
- **Aircraft Equipage Limitations**—Some of the avionics platforms necessary to enable certain NextGen initiatives are cost prohibitive to some aircraft owners and operators. While airlines and select cargo and GA aircraft operators can afford sophisticated avionics platforms, such as RNP, many smaller regional airlines, cargo operators, and other GA aircraft owners typically cannot justify the costs.
- **Aircraft Fleet Mix and Performance Characteristics**—In particular, the potential benefits associated with Wake RECAT, wake turbulence avoidance and mitigation initiatives offered

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by NextGen are primarily intended to benefit airports with significant operations by Heavy jet and B757 aircraft.

- **ATC and Flight Crew Training Requirements**—Certain NextGen initiatives, such as RNP, SOIA, and other closely spaced parallel runway operational enhancements require specialized ATC and/or flight crew training.
- **Airfield Configuration**—Many of the NextGen initiatives are intended to reduce the lateral separation requirements between parallel runways necessary to conduct independent or dependent operations. These initiatives may only benefit airfields with dual and triple parallel runway operations with sufficient lateral separations to meet the new requirements.
- **Airspace Constraints and Airport Dependencies—**While certain NextGen initiatives may not directly benefit a certain airport, indirect benefits may be realized when the NextGen initiative enhances the capacity of the local terminal airspace that it shares with other airports. For instance, certain airports may be unable to achieve optimal airspace capacity due to constraints in the local terminal airspace. Through the adoption of PBN at nearby airports, the arrival and/or departure corridors could be decoupled, thereby enabling other airports within the airspace system to achieve their optimal airfield capacity.

  Airport system planning also may offer the opportunity to leverage NextGen technologies that otherwise could not be implemented for the benefit of a single airport. For instance, the adoption of WAM by CDOT yielded capacity benefits to multiple airports that otherwise might not be able to justify the cost for the benefit of only a single airport.

  While NextGen may provide opportunities to enhance the efficiency of an airport or airspace system, the surrounding airspace system could also impair the ability to implement or achieve the benefit of certain NextGen Initiatives. For example, if an airport has significant airspace constraints that adversely affect departures (e.g., limited departure headings), it might be difficult to achieve the full benefits from Wake RECAT even if it has significant Heavy jet and B757 operations. In addition, airports that are in close proximity to other airports or nearby special use airspace, such as prohibited, restricted, military operations, and warning/alter areas, may inhibit or preclude certain NextGen technologies or procedures.
- **Surrounding Terrain and Obstacles**—As demonstrated by the case study for the Friedman Memorial Airport, the terrain surrounding an airport or system of airports can significantly impair the ability to either implement or utilize NextGen technologies. Airports that are located in or near areas designated as precipitous terrain or with a high concentration of obstacles could be precluded from realizing the benefits of certain NextGen initiatives.
- **Operational Needs**—Some NextGen initiatives are intended to benefit only certain types of airports or operators. CDM, ASDE-X radar installations, Intelligent Routing and Guidance/A-SMGCS/ALCMS systems, automated docking systems, and virtual ramp control towers are mostly limited to large constrained commercial service airports.
- **Airport Infrastructure Requirements/Funding Limitations**—Certain NextGen initiatives require infrastructure improvements at or near the airport. These infrastructure needs can be cost prohibitive at small commercial service and GA airports. Even large commercial airports with severe real estate constraints may not be able to provide the infrastructure improvements to enable the benefits of a particular NextGen capability. For example, certain large constrained airports that could benefit greatly from NextGen surface management and data sharing procedures do not have adequate space for the infrastructure (e.g., gates and hold pads to support such an operation).

To facilitate these discussions, the airport size classifications contained in the FAA's NPIAS serve the basis for classifying airports by airport size. Furthermore, the NextGen initiatives have been segregated

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to differentiate between the FAA's NextGen initiatives and other emerging technologies that are being developed by other organizations, but will either enhance or supplement the FAA's NextGen initiatives.

C.1 Airport Size Classifications for NextGen Applications

The NPIAS categorizes public use airports by type of activities, including commercial service, primary, cargo service, reliever, and GA airports. For the purpose of discussion within this guidebook, large and small airports generally include the following NPIAS Classifications:

Large Airports

Large Hub Primary Commercial Service (≥ 1.0 percent of total passenger boardings in the United States; 1.5 million boardings in 2014).

Medium Hub Primary Commercial Service (≥ 0.25 percent but less than 1.0 percent of total boardings in the United States; 380,000 and 1.5 million boardings, respectively in 2014).

Small Airports

- Small and Nonhub Primary Commercial Service (> 10,000 annual passenger boardings, but less than 0.25 percent of total passenger boardings in the United States; 380,000 boardings in 2014).
- Non-primary Commercial Service (no more than 10,000 annual passenger boardings).
- GA/reliever airports.

While private-use cargo service and military airports could benefit from NextGen technologies, their application is excluded from these discussions. Table C-1 presents the overall NPIAS airport classifications, while highlighting those that are considered as large and small airports. To better correlate the diverse range of GA airports with NextGen technologies, this guidebook further segregates GA airports in accordance with the four primary GA airport asset classifications defined in the FAA's 2012 Asset Report.

As of September, 2014, there were 3,283 airports contained in the NPIAS, and nearly 90 percent of these airports are classified as GA and/or reliever. GA airports either have no scheduled commercial service or less than 2,500 annual passenger boardings. Due to the unique physical constraints and demand characteristics, the operational complexities and aircraft fleet mix associated with GA airports can vary significantly. Many GA airports within urban areas have a high concentration of corporate jet activity, some with performance characteristics and avionics platforms similar to modern air carrier aircraft. Other municipal and rural airports may exclusively serve small piston and turboprop aircraft with limited avionics capabilities. Table C-2 correlates the subcategories of large and small airports with the typical aircraft fleet mix composition as it relates to NextGen opportunities.

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Table C-1. FAA airport classifications.

Categories of Airport Activities

Airport Classifications		Hub Type: Percentage of Annual Passenger Boardings	Common Name	
Commercial Service: Publicly owned airports that have *at least 2,500* passenger boardings each calendar year and receive scheduled passenger service **§47102(7)**	**Primary:** Have *more than 10,000* passenger boardings each year **§47102(16)**	**Large:** 1% or more	**Large Hub**	Medium and Large Airports[1/]
		Medium: At least 0.25%, but less than 1%	**Medium Hub**	
		Small: At least 0.05%, but less than 0.25%	**Small Hub**	Small Airports[1/]
		Nonhub: More than 10,000, but less than 0.05%	**Nonhub Primary**	
	Nonprimary	**Nonhub:** At least 2,500 and no more than 10,000	**Nonprimary Commercial Service**	
Nonprimary (Except Commercial Service)		Not Applicable	**Reliever §(47102(23))** **General Aviation (47102(8))**	

[1]Medium and Large Airports generally include the following NPIAS Classifications:

- Medium Hub Primary Commercial Service (at least 0.25% but less than 1% of annual passenger boardings).
- Large Hub Primary Commercial Service (1% or more of annual passenger boardings).

Small Airports generally include the following NPIAS Classifications:

- Nonhub Primary Commercial Service (>10,000 annual passenger boardings, but less than 0.25% of total passenger boardings in the United States).
- Non-primary Commercial Service (no more than 10,000 annual passenger boardings).
- General aviation/reliever airports.

Source: NPIAS, Federal Aviation Administration.

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Table C-2. Categorization of airports for NextGen opportunities.

CATEGORY/SUBCATEGORY	NPIAS CRITERIA	TYPICAL AIRCRAFT FLEET MIX COMPOSITION
Large Airports		
Large Hub Primary Commercial Service	≥ 1.0% of Total U.S. Passenger Boardings	Air Carrier, Cargo, and Corporate
Medium Hub Primary Commercial Service	< 1.0% of Total U.S. Passenger Boardings; ≥ 0.25% of Total U.S. Passenger Boardings	Air Carrier, Cargo, Corporate, and Limited Piston Aircraft
Cargo Service Airports	> 100 Million Pounds Landed Weight[1]	Air Carrier, Cargo, Corporate, and Limited Piston Aircraft
Small Airports		
Limited Commercial Service		
Small/Nonhub Primary Commercial Service	10,000 Annual Passenger Boardings - <0.25% of Total U.S. Passenger Boardings	Limited Air Carrier, Limited Cargo, Corporate and Piston Aircraft[1]
Non-primary Commercial Service	2,500 – 9,999 Annual Passenger Boardings	Limited Air Carrier, Corporate and Piston Aircraft
GA		
National	No Scheduled Service or < 2,500 Annual Passenger Boardings	Very High Levels of Activity with Many Jets and Multiengine Propeller Aircraft; Averaging About 200 Based Aircraft, Including 30 Jets.
Regional	No Scheduled Service or < 2,500 Annual Passenger Boardings	High Levels of Activity with Some Jets and Multiengine Propeller Aircraft; Averaging About 90 Based Aircraft, Including Three Jets.
Local	No Scheduled Service or < 2,500 Annual Passenger Boardings	Moderate Levels of Activity with Some Multiengine Propeller Aircraft; Averaging About 33 Based Propeller-Driven Aircraft and No Jets.
Basic	No Scheduled Service or < 2,500 Annual Passenger Boardings	Moderate to Low Levels of Activity; Averaging About 10 Propeller-Driven Aircraft and No Jets.

Note:

[1]"Landed weight" means the weight of aircraft transporting only cargo in intrastate, interstate, and foreign air transportation. An airport may be both a commercial service and a cargo service airport.

Source: NPIAS, Federal Aviation Administration; General Aviation Airports: A National Asset, Federal Aviation Administration.

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C.2 Typical NextGen Technologies and Initiatives Application by Airport Size

Tables C-3 and C-4 summarize the typical applicability of the various NextGen technologies and initiatives by airport size. These tables present the FAA's NextGen initiatives and other emerging technologies that are being developed by other organizations, respectively. Chapters 4 and 5 of this guidebook provide additional information pertaining to the applicability to large and small airports, respectively.

Table C-3. FAA NextGen operational benefits summary matrix (1 of 2).

	LARGE AIRPORTS		SMALL AIRPORTS		
NEXTGEN INITIATIVE BY FUNCTION	**LARGE HUB**	**MEDIUM HUB**	**SMALL COMMERCIAL SERVICE AIRPORTS[1]**	**NATIONAL/ REGIONAL GENERAL AVIATION**	**LOCAL/BASIC GENERAL AVIATION**
Improved Landing Systems					
RNP	X	X	Limited Benefits to Small Airports Within a Constrained Airspace System		
LPV	X	X	X	X	X
LP	X	X	X	X	X
LNAV	X	X	X	X	X
LNAV/VNAV	X	X	X	X	X
Ground-Based Augmentation Systems (GBAS)					
CAT I Approaches (multiple runway ends)[2]	X	X	Limited Application due to Aircraft Equipage Limitations		
CAT II/III Approaches (multiple runway ends)	X	X	Limited Application due to Aircraft Equipage Limitations		
Performance-Based Navigation (Terminal Area)					
RNAV	X	X	X	X	X
RNP	X	X	Limited Benefits to Small Airports Within a Constrained Airspace System		
Multilateration (Surveillance)					
Wide Area Multilateration (WAM) - En Route and Terminal Airspace	Not Applicable to Large Airports		Potential Benefits to Small Airports with Limited Radar Coverage		
Surface Movement (ASDE-X)	X	X	Not Applicable to Small Airports		

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Table C-3. Continued.

	LARGE AIRPORTS		SMALL AIRPORTS		
NEXTGEN INITIATIVE BY FUNCTION	**LARGE HUB**	**MEDIUM HUB**	**SMALL COMMERCIAL SERVICE AIRPORTS[1]**	**NATIONAL/ REGIONAL GENERAL AVIATION**	**LOCAL/BASIC GENERAL AVIATION**
Surface Operations & Data Sharing					
Collaborative Decision Making	X	X	Not Applicable to Small Airports		
SWIM	X	X	X	X	X
Independent Runways					
Wake Turbulence Recategorization					
Phase I (Aircraft Grouping Reclassification)	X	X	Limited to Non Hub Airports that Serve Heavy Aircraft Operations or Part of a System of Airports with a Significant Share of Heavy Aircraft Operations in the Terminal Airspace	Potential Benefit to General Aviation Airports That Are Part of a System of Airports with a Significant Share of Heavy Aircraft Operations in the Terminal Airspace	
Phase II (Aircraft Specific)	X	X	Potential Benefit to Small Airports That Are Congested or Part of a System of Airports with Congested Terminal Airspace		

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Table C-3. FAA NextGen operational benefits summary matrix (2 of 2).

NEXTGEN INITIATIVE BY FUNCTION	LARGE AIRPORTS		SMALL AIRPORTS		
	LARGE HUB	MEDIUM HUB	SMALL COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/ REGIONAL GENERAL AVIATION	LOCAL/BASIC GENERAL AVIATION
Dependent Runways					
Wake Turbulence Avoidance Procedures:					
Wake Turbulence for Parallel Runways (< 2,500' spacing - small/large leading)	X	X	Limited to Small Airports that May Serve Heavy Aircraft Operations		Not Applicable
Wake Turbulence Mitigation for Arrivals-Procedures (<2,500' spacing - B757/heavy leading)	X	X	Limited		
Wake Turbulence Mitigation for Departures (Upwind runways)	X	X	Limited to Small Airports that May Serve Heavy Aircraft Operations		Not Applicable
Closely Spaced Parallel Operations:					
Dual Independent Parallel Operations (>3,600' spacing)	X	X	Limited to Small Airports with Closely Spaced Parallel Runways Served by Precision Instrument Approach Procedures		Not Applicable
Dual Dependent Parallel Operations (2,500' - 3,600' spacing)	X	X	Limited to Small Airports with Closely Spaced Parallel Runways Served by Precision Instrument Approach Procedures		Not Applicable
Triple Dependent Parallel Operations (>3,900' spacing)	Limited to Airports Configured with Triple Parallel Runways		Not Applicable to Small Airports		
Dual Independent Parallel Operations with Offset (>3,000' spacing)	X	X	Not Applicable to Small Airports		
Dependent Parallel Operations (>4,300' spacing)	X	X	Not Applicable to Small Airports		
RNP Parallel Approaches with Transition (RPAT)	X	X	Not Applicable to Small Airports		
Separation Management					
Advanced Technologies and Oceanic Procedures (ATOP)	En Route Benefits Only - Not Directly Associated with Airport Operations				

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Table C-3. Continued.

NEXTGEN INITIATIVE BY FUNCTION	LARGE AIRPORTS		SMALL AIRPORTS		
	LARGE HUB	MEDIUM HUB	SMALL COMMERCIAL SERVICE AIRPORTS[1]	NATIONAL/ REGIONAL GENERAL AVIATION	LOCAL/BASIC GENERAL AVIATION
En Route Automation Modernization (ERAM)	En Route Benefits Only - Not Directly Associated with Airport Operations				
Terminal Automation Modernization and Replacement (TAMR)	X	X	Possibly Applicable to Small Airports Within a Congested Airspace System		
Established on RNP (EoR)	X	X	Possibly Applicable to Small Airports Within a Congested Airspace System		
CDTI Based Separation	X	X	Limited Use Due to Aircraft Equipage Limitations		
Equivalent Lateral Spacing Operations (ELSO) National Standard (reduced divergence angle)	X	X	Limited Use Due to Aircraft Equipage Limitations		
Unified Departure Operation Spacing (UDOS)	X	X	Not Applicable to Small Airports		
Established-on-Departure Operation (EDO)	X	X	Not Applicable to Small Airports		
ADS-B					
Traffic Information Services Broadcast (TIS-B)	X	X	Limited Use Due to Aircraft Equipage Limitations		
Automatic Dependent Surveillance-Rebroadcast (ADS-R)	X	X	Limited Use Due to Aircraft Equipage Limitations		
Flight Information Services Broadcast (FIS-B)	X	X	Limited Use Due to Aircraft Equipage Limitations		
Cockpit Display of Traffic Information	X	X	Limited Use Due to Aircraft Equipage Limitations		

Notes:

[1]Per the FAA's airport size classifications prescribed in the NPIAS, small commercial service airports includes small hub, nonhub primary and limited commercial service airports.

[2]Currently available for Category (CAT) I Instrument Landing Systems as a non-federal system. Research and development ongoing for CAT II/III operations.

Source: NPIAS, Federal Aviation Administration.

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Table C-4. Non-FAA NextGen operational benefits summary matrix.

	LARGE AIRPORTS		SMALL AIRPORTS		
OTHER TECHNOLOGY INITIATIVES BY FUNCTION	**LARGE HUB**	**MEDIUM HUB**	**SMALL COMMERCIAL SERVICE AIRPORTS[1]**	**NATIONAL/ REGIONAL GENERAL AVIATION**	**LOCAL/BASIC GENERAL AVIATION**
Improved Landing Systems					
LED PAPI	X	X	X	X	X
LED Approach Lighting	X	X	X	X	X
Performance-Based Navigation (Terminal Area)					
NA	Non-FAA Programs Currently Unavailable				
Multilateration (Surveillance)					
Virtual Air Traffic Control Towers	X	X	X	X	X
Virtual Ramp Control	X	X			
Noise Monitoring Systems	X	X	X	X	Not Applicable
Surface Operations & Data Sharing					
Collaborative Decision Making	X	X	Possibly Applicable to Small Airports Within a Congested Airspace System		
Intelligent Routing and Guidance/ A-SMGCS/ALCMS Systems	X	X	Not Applicable to Small Airports		
Automated Docking Systems	X	X	Not Applicable to Small Airports		
LED Lighting Technology	X	X	X	X	X
Ground Vehicle Tracking	X	X	X	X	X
Traffic Display and Analysis Systems (Non-ATC Automation)	X	X	X	X	X
Independent Runways					
NA	Non-FAA Programs Currently Unavailable				
Dependent Runways					
NA	Non-FAA Programs Currently Unavailable				

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Table C-4. Continued.

	LARGE AIRPORTS		SMALL AIRPORTS		
OTHER TECHNOLOGY INITIATIVES BY FUNCTION	**LARGE HUB**	**MEDIUM HUB**	**SMALL COMMERCIAL SERVICE AIRPORTS[1]**	**NATIONAL/ REGIONAL GENERAL AVIATION**	**LOCAL/BASIC GENERAL AVIATION**
Closely Spaced Parallel Operations					
NA	Non-FAA Programs Currently Unavailable				
Separation Management					
NA	Non-FAA Programs Currently Unavailable				
ADS-B					
NA	Non-FAA Programs Currently Unavailable				

Notes:

[1]Per the FAA's airport size classifications prescribed in the NPIAS, small commercial service airports includes small hub, nonhub primary, and limited commercial service airports.

Source: NPIAS, Federal Aviation Administration.

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D NextGen Resources, Organizations, and Contacts

There are multiple NextGen resources available to airport practitioners provided by FAA and other industry sources. This appendix highlights the organizational structure of the various sources and provides guidance to performance and web-based resources.

D.1 Federal Aviation Administration

The FAA core NextGen programs include:

- Automatic Dependent Surveillance-Broadcast (ADS-B)
- Data Communications (Data Comm)
- PBN
- SWIM
- NextGen Weather

NextGen plans and concepts are coordinated by the FAA Office of NextGen. Other divisions within the FAA are tasked with implementing the various NextGen programs. These other offices include ATO Mission Support Services, Flight Standards Service, and the ATO Program Management Office.

D.1.1 FAA Office of NextGen Organization

The Office of NextGen is structured under the ATO of the FAA. The mission of this office is to manage the implementation of NextGen technologies and capabilities into the NAS following the FAA Engineering Life Cycle Process depicted in Figure D-1.

The Assistant Administrator for NextGen and the Deputy Assistant Administrator for NextGen both work directly with the Chief of Staff, Office of Chief Scientist for NextGen, and Office of Chief Scientist for Software. There are six divisions responsible for the major activities associated with the core NextGen programs, including: (1) Management Services, (2) Engineering and Integrations, (3) Portfolio Management and Technology, (4) William J. Hughes Technical Center, (5) Performance and Outreach, and (6) Interagency Planning.

There are multiple offices below these six divisions supporting all activities associated with the FAA Engineering Life Cycle from program inception to in-service maintenance. For a list of positions and contact information, visit the FAA employee directory at **https://directory.faa.gov.**

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Figure D-1. *The FAA life cycle management process.*

D.1.2 FAA Airports

The FAA Office of Airports (ARP) ensures the national airport system is safe, efficient, environmentally responsible, and meets the needs of the traveling public. ARP is not directly responsible for the implementation of NextGen, but serves as a liaison in facilitating the implementation of the NextGen integration with airport planning initiatives and airport development activity.

D.2 Aviation Industry NextGen Resources

There are multiple industry-based groups that participate in addressing the challenges of implementing NextGen. Three of the major groups dedicated to representing the airline and airport industries include the RTCA, Airports Council International – North America (ACI-NA), and the American Association of Airport Executives (AAAE).

D.2.1 RTCA

Founded as the Radio Technical Commission for Aeronautics in 1935, RTCA is a private, not-for-profit corporation utilized as a federal advisory committee working in response to requests from the FAA to develop comprehensive, industry-vetted and endorsed recommendations for the government on issues ranging from technical performance standards to operational concepts for air transportation.

On September 23, 2010, the RTCA and FAA established the NAC, a 28-member federal advisory committee formed to provide advice on policy-level issues facing the aviation community in implementing NextGen (modernizing the aviation system). The NAC is tackling issues that are broader than air traffic management, including safety, airports, the environment, and global harmonization.

The goal of the NAC is to foster industry collaboration in an open and transparent manner. It includes a cross section of executives from the airlines, airports, general aviation, pilots, air traffic controllers,

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the Department of Defense, environmental interests, international interests, and providers of air traffic control technology—all committed to ensuring a successful transition to NextGen. This public-private partnership venue is addressing the critical policies and priorities for NextGen implementation, examining the business case for those who must invest in NextGen, and providing a venue for tracking progress and commitments.

In response to tasks from the FAA, the NAC forges consensus recommendations on implementation-related issues facing the community as it works to implement NextGen. The NextGen Advisory Committee SubCommittee (NACSC), which represents the spectrum of interested aviation industry parties, provides resources and support to the NAC to ensure that the perspectives of all in the aviation industry are considered in its recommendations to the NAC. In some cases, these recommendations are bundled into reports that are made available to the public. The RTCA office may be contacted for further information at:

1150 18th NW, Suite 910, Washington, DC 20036
Telephone: (202) 833-9339
Fax: (202) 833-9434
E-Mail: info@rtca.org

D.2.2 Airports Council International—North America

The mission of Airports Council International—North America (ACI-NA) is to advocate policies and provide services that strengthen the ability of airports to serve their passengers, customers, and communities. ACI-NA has established a NextGen Working Group to serve as the "voice of airports" regarding the ongoing efforts of the Federal Aviation Administration (FAA), RTCA, NAV CANADA, Transport Canada, and others to develop and implement NextGen technologies, procedures, and policies. The NextGen Working Group was formed to:

- Advise ACI-NA members and staff involved in NextGen advisory committees, industry working groups, and other stakeholder forums regarding airport NextGen priorities and issues.
- Identify and communicate opportunities for ACI-NA members to play a meaningful role in ongoing NextGen development and implementation activities.
- Monitor and disseminate information on NextGen development and implementation activities, their implications for airports, and document case studies where ACI-NA members have played a significant role.
- Serve as industry subject matter experts regarding NextGen and provide representation and speakers for various NextGen-related industry events and conferences.
- Assist ACI-NA staff with informing the industry and the public at large about airports' necessary role in NextGen development and implementation. Develop and maintain a list of research needs and opportunities for facilitating the development and implementation of NextGen technologies at airports and communicate those needs to ongoing research programs of the National Academies of Sciences, Engineering, and Medicine; NASA; FAA; and JPDO.

The ACI-NA office may be contacted for further information at:

Vice President, Safety and Regulatory Affairs
1615 L Street NW, Suite 300
Washington, DC 20036
Phone 202-293-8500
Fax 202-331-1362
www.aci-na.org

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D.2.3 AAAE

AAAE is the world's largest professional organization for airport executives, representing thousands of airport management personnel at public use commercial and general aviation airports. AAAE's members represent some 850 airports and hundreds of companies and organizations that support airports. AAAE serves its membership through results-oriented representation in Washington, D.C., and delivers a wide range of industry services and professional development opportunities including training, meetings and conferences, and a highly respected accreditation program.

AAAE hosts 13 member committees dedicated to identifying specific airport management and operation issues. Collectively these committees have addressed multiple NextGen issues related to surveillance, performance-based navigation, airspace, airport design, infrastructure needs, and environmental compliance. Each year AAAE sponsors conferences and workshop sessions, some in partnership with ACI-NA, FAA, and other organizations, devoted to increasing the understanding of airports on how NextGen could affect their facilities and operations.

The AAAE office may be contacted for further information at:

Staff Vice President, Regulatory and Legislative Affairs
The Barclay Building
601 Madison Street
Alexandria, VA 22314
703.824.0504
www.aaae.org

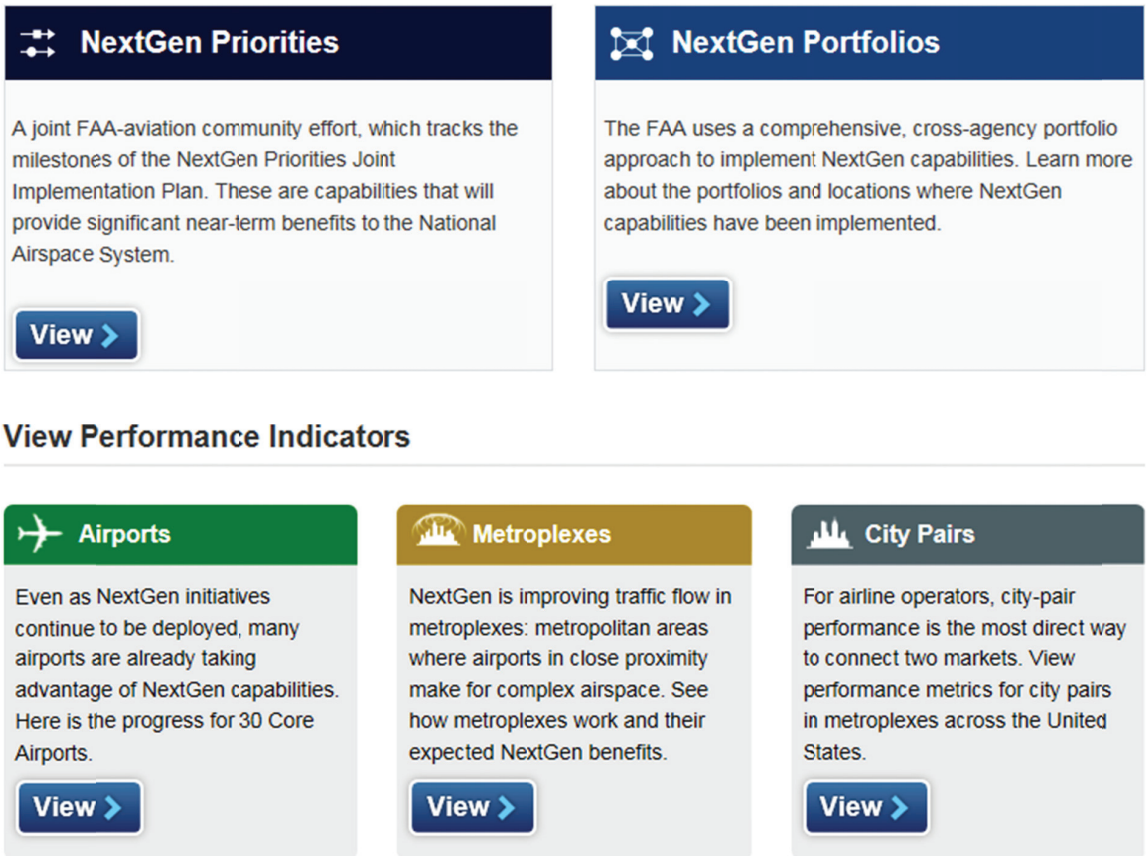
D.3 NextGen Implementation and Performance Resources

The FAA has engaged in the performance monitoring efforts tracking the milestones of NextGen implementation priorities and performance. This information can be found on the FAA website at http://www.faa.gov/nextgen/snapshots. Figure D-2 depicts a screenshot of this webpage, which allows a user to examine the NextGen priority milestone achievement, NextGen Portfolios, and performance indicators by Airport, Metroplex, and City Pairs.

NextGen Priorities information includes (1) Multiple Runway Operations (MRO), (2) Performance Based Navigation (PBN), (3) Surface Operations and Data Sharing, and (4) Data Communications (Data Comm). NextGen Portfolios provide a description of each program and the locations where each technology has been implemented.

Performance indicators or "Snap Shots" are included by Airports, Metroplex, and City Pairs. The Airports tab allows a user to view how performance indicators and NextGen capabilities change year by year for a particular airport. The Metroplexes tab allows a user to view the status of the current program as well as detailed performance metrics associated with the implementation of PBN. The City Pairs tab allows the user to compare how travel time trends change from year to year on a given route. Figure D-3 depicts a screenshot of the city pair data in the DC Metroplex.

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Source: FAA Webpage. http://www.faa.gov/nextgen/snapshots/

Figure D-2. *NextGen performance webpage.*

One can also see how the benefits of NextGen compare year by year on a national scale on the Scorecard when one clicks on the Across the National Airspace System tab (Figure D-4). These data are compiled using all the NextGen Data available from 104 NAC-recommended city pairs for select airlines and do include all flights between the city pairs.

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Average Airborne Time (FY) *Minutes*

Origin	Destination	2009	2010	2011	2012	2013	2014	2015
BOS	BWI	67.0	65.1	66.4	66.7	66.4	66.8	65.5
BOS	DCA	73.1	71.7	71.2	70.9	71.1	71.6	71.1
BOS	IAD	73.8	72.5	74.4	74.0	73.8	74.3	72.8
BWI	CLT	62.7	61.9	61.8	60.5	61.5	62.8	62.6
BWI	ORD	101.8	100.5	102.7	100.8	101.0	102.7	99.9
CLT	DCA	54.1	54.2	54.8	54.9	55.0	54.4	55.3
CLT	IAD	54.5	54.9	55.5	55.2	54.7	54.7	55.7
DCA	EWR	43.7	42.8	42.8	42.9	42.4	42.6	42.2
DCA	ORD	97.6	96.5	97.3	95.5	96.0	97.4	97.7
IAD	JFK	51.1	49.5	48.3	48.0	48.6	48.4	48.2
IAD	LGA	49.7	47.0	47.0	46.7	46.6	46.0	45.9

Source: FAA Webpage. http://www.faa.gov/nextgen/snapshots/airportPairs/?locationId=7

Figure D-3. *NextGen city pair data.*



Source: FAA Webpage. http://www.faa.gov/nextgen/snapshots/

Figure D-4. *Across the Nation Airspace System Data.*

D.4 FAA Instrument Flight Procedures Information Gateway

The FAA Instrument Flight Procedures Information Gateway is located at https://www.faa.gov/air_traffic/flight_info/aeronav/procedures/application/ and provides a centralized Instrument Flight Procedures (IFP) data portal, providing a single source for:

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- Charts — All Published Charts, Volume, and Type;
- IFP Production Plan — Current IFPs under Development or Amendments with Tentative Publication Date and Status;
- IFP Coordination — All coordinated developed/amended procedure forms forwarded to Flight Check or Charting for publication;
- IFP Documents - Navigation Database Review (NDBR) — Repository and Source Documents used for Data Validation of Coded IFPs.

The site provides a user the ability to look up existing and proposed IFPs by airport, state, region, and FAA service area. This site allows for coordination and exchange of information related the current production schedule of IFPs and the ability for an airport or aircraft operator to request the development of an IFP at an airport using an IFP request form.

The site also provides additional information including IFP Announcements, Optimization of Airspace Procedures in the Metroplex updates, PBN Implementation Plan, and IFP initiation information. Figure D-5 depicts a screen shot of the IFP Gateway.

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Source: FAA Webpage. https://www.faa.gov/air_traffic/flight_info/aeronav/procedures/application/

Figure D-5. *FAA Instrument Flight Procedure Gateway.*

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D.5 Conclusions

The foregoing organizations and web-based resources provide valuable information and insights into the objectives, status, benefits, and applications of NextGen to airport planning practitioners. The major challenge is that access to the information provided by these organizations and resources, and the contact information for individuals associated with these resources, is constantly changing. Therefore, it is suggested that airport practitioners using this guidebook consult with the representatives and websites of the foregoing organizations (i.e., FAA, RTCA, ACI-NA, and AAAE) to find the latest information on the applicability of NextGen capabilities and technologies to airport planning and development activities at their airports.

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E Relevant Airport Planning and NextGen References and Guidance Documents

This appendix summarizes the results of the research team's review and critique of the current knowledge, practice, and data relative to NextGen capabilities and airport planning. Included is an annotated bibliography of relevant existing literature for consideration by airport planners.

E.1 General Sources and References

This section includes a description of general sources and reference documents describing NextGen capabilities, priorities, and current status, as outlined below:

1. Federal Aviation Administration (FAA) NextGen Documents

The FAA Next Generation Air Transportation System (NextGen) website provides many resources that would be of interest to airport planning practitioners. Of particular interest is a separate page on "NextGen for Airports," which includes a list of frequently asked questions by operators of airports of all sizes with helpful answers. Below is a link to the FAA NextGen website: http://www.faa.gov/nextgen/.

In June 2016, the FAA published an updated PBN strategy document titled "PBN NAS Navigation Strategy 2016," which refocuses FAA priorities and milestones to transition to a PBN-centric NAS, that is, an NAS where PBN is used as the basis for daily operations. It describes a pathway to linking together many interdependent elements necessary to deliver PBN. This document is available at https://www.faa.gov/nextgen/media/PBN_NAS_NAV.pdf.

2. FAA Airport Planning Advisory Circulars and Air Traffic Control Orders

These documents are updated periodically, and drafts of the updates are made available to industry organizations for review and comment. In addition, individuals can sign-up to receive email notifications of drafts and updated ACs from the FAA. The following FAA websites provide further information regarding FAA Advisory Circulars and Orders:

http://www.faa.gov/regulations_policies/advisory_circulars/
http://www.faa.gov/airports/planning_capacity/
http://www.faa.gov/regulations_policies/orders_notices/

FAA rules and procedures for conducting air traffic control (ATC) and for operating in the National Airspace System (NAS), which usually are intended for air traffic controllers and pilots, are published as FAA Orders, Manuals, Bulletins, or Notifications. These documents are helpful because they contain the current standards for aircraft separations and parallel runway spacings, many of which are

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expected to change with the implementation of NextGen. The following FAA website provides further information regarding air traffic plans and publications: http://www.faa.gov/air_traffic/publications/.

The FAA will periodically publish guidelines for the methodologies, parameters, and assumptions to be used in NextGen-related airport planning and development studies. These publications tend to be issued irregularly when there are major changes in parameters, criteria, or methodologies. The various databases maintained by FAA are kept up to date and are available for use by the public with an FAA provided username and password. For example, the following FAA website includes various databases and information sources that are very helpful for conducting NextGen-related airport planning studies: https://aspm.faa.gov/.

3. Joint Planning and Development Office (JPDO), Interagency Planning Office (IPO), and the NextGen Institute Reports Analyses and Reports

NextGen was enacted in 2003 by President Bush and Congress under VISION 100—Century of Aviation Reauthorization Act (P.L. 108-176). Under this initiative, Congress created the Joint Planning and Development Office (JPDO) to manage the partnerships designed to bring NextGen online. These partnerships include private-sector organizations, academia, and the following government departments and agencies: Department of Transportation (DOT), Department of Commerce (DOC), Department of Defense (DOD), Department of Homeland Security (DHS), Federal Aviation Administration (FAA), National Aeronautics and Space Administration (NASA), White House Office of Science and Technology Policy (OSTP), and Office of the Director of National Intelligence (ODNI).

In January 2014 the Consolidated Appropriations Act, 2014 (P.L. 113-76), eliminated funding for the JPDO. To replace JPDO, the FAA has created the new Interagency Planning Office (IPO), which falls under the purview of the FAA office of NextGen. The NextGen Institute, which was formed in March 2005, uses its Web site and the FAA communications network to reach out to participants and inform them of FAA NextGen public meetings, study teams, and contractual work. Below is a link to the NextGen Institute website: http://nginstitute.org/library/documents/.

4. Airport Cooperative Research Program (ACRP) NextGen Reports

The Transportation Research Board of The National Academies of Sciences, Engineering, and Medicine manages ACRP, which is an industry-driven, applied research program that develops near-term, practical solutions to problems faced by airport operators. NextGen is one of many topics undertaken by ACRP, which also include topics related to NextGen such as airport planning, airfield capacity evaluation, and aircraft delays. Below are a few relevant projects (in addition to the ongoing ACRP NextGen Initiative Projects 01-27, 01-28, 03-33, 03-34, and 09-12):

- ACRP 03-13, “Understanding Airspace, Objects, and Their Effects on Airports”;
- ACRP 03-17, “Evaluating Airfield Capacity”;
- ACRP 03-20, “Defining and Measuring Aircraft Delay and Airport Capacity Thresholds”; and
- ACRP 10-19, ”Advancing Collaborative Decision Making (CDM) at Airports.”

The following TRB website provides further information regarding ACRP projects and publications: http://www.trb.org/acrp/acrp.aspx.

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5. NextGen Publications by Manufacturers/Vendors/Suppliers/MITRE/ Other Contractors/Magazines

Various contractors and vendors have prepared documents on NextGen related to airport planning and development either for FAA or the NextGen Institute, or as their own publications.

For example, TRB has published a paper on NextGen through its *TR News* magazine (http://trid.trb.org/view.aspx?id=1125059).

An example of a contractor source is MITRE Corporation, which does extensive research on NextGen for the FAA and publishes its results as technical papers or fact sheets. Please see link below: http://www.mitre.org/centers/center-for-advanced-aviation-system-development/where-we-focus/transforming-the-national.

6. NextGen Publications by Industry, Professional, and Academic Organizations

This category includes industry and academic journal articles, and conference presentations. This category often includes research that ultimately becomes part of one of the foregoing publications. Industry associations that are helpful resources for publications and presentations include the American Association of Airport Executives (AAAE), the American Institute of Aeronautics and Astronautics (AIAA), the Air Traffic Control Association (ATCA), the Airports Council International-North America (ACI-NA), the American Society of Civil Engineers (ASCE), and the Transportation Research Board (TRB). The following sample websites provide further information regarding the various industry and professional NextGen programs:

- Annual AAAE Conference and Exposition – http://www.aaae.org/meetings/meetings_calendar/mtgdetails.cfm?Meeting_ID=150501.
- AIAA Aviation Technology, Integration, and Operations (ATIO) Conference – http://www.aiaa.org/EventDetail.aspx?id=16562.
- ACI-NA Annual Conference and Exhibition – http://annual.aci-na.org/.
- ASCE Transportation & Development Institute (TDI) – http://www.asce.org/tdi/Content.aspx?id=25769809718.
- ATCA Annual Conference & Exposition – http://www.atca.org/59Annual.
- TRB Annual Meeting – http://www.trb.org/AnnualMeeting2015/AnnualMeeting2015.aspx.

7. National Aeronautics and Space Administration (NASA) NextGen Research Publications

NASA performs foundational research to support the modernization of the National Airspace System (NAS) through its Airspace Systems Program (ASP). ASP has a leadership role in NASA's partnership with other agencies supporting the Interagency Planning Office (IPO). Once they have been developed and established, the products of NASA's research are transferred to FAA. Recent examples of such technology transfer include the Precision Departure Release Capability (PDRC) and the Terminal Sequencing and Spacing (TSS) tool. The following NASA website provides further information regarding the various NASA NextGen research programs: http://www.aeronautics.nasa.gov/programs_asp.htm.

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8. Radio Technical Commission for Aeronautics (RTCA)/NextGen Advisory Committee (NAC) Reports

The NAC is a 28-member federal advisory committee formed to provide advice on policy-level issues in implementing NextGen. It includes a cross section of executives from the airlines, airports, general aviation, pilots, air traffic controllers, the Department of Defense, environmental interests, international interests, and providers of air traffic control technology. In response to tasks from the FAA, the NextGen Advisory Committee issues reports that are made available to the public. Further information on available documents can be found at the following RTCA websites:

http://www.rtca.org/store_list.asp.
http://www.rtca.org/content.asp?pl=33&sl=61&contentid=61.

9. Government Accountability Office (GAO) Reports

The GAO is an independent, nonpartisan agency that conducts investigative audits at the request of Congress. GAO conducts interviews with officials from FAA and industry and reviews agency documents to provide an objective assessment of the key issues and challenges of implementing NextGen and provide recommendations focused on accountability and performance of FAA's processes. Based on the work GAO has conducted, they have published several reports summarizing their findings. These reports can be found at: http://www.gao.gov/browse/topic/Transportation/Aviation/.

Annotated Bibliography of Relevant Airport Planning Documents

This section includes a description of the key relevant documents, particularly those focusing on impacts of NextGen on airport planning and development. The documents are organized by the potential areas of airport planning that will be impacted, as outlined below:

1. General NextGen Guidance with Airport Planning Elements

These documents provide a high-level description of selected capabilities that are relevant to airports, as well as the airport site locations where the capabilities are already in operation or planned to be operational in the near-term. The GAO documents outline the key challenges in rolling out some of these implementations.

Federal Aviation Administration (May 2015). ***NextGen Implementation Plan 2015*****.** https://www.faa.gov/nextgen/media/NextGen_Implementation_Plan-2015.pdf.

This document is an update of the NextGen Implementation Plan, which has been published annually since 2008. It is divided into two sections: key programs and milestones (ADS-B, Data Comm, ERAM, TAMR, NVS, SWIM), and implementation portfolios. Each portfolio includes a description of the target users, target areas, and anticipated benefits. Those related to airport planning are: Improved Surface Operations, Improved Approaches and Low-Visibility Operations, Improved Multiple Runway Operations, Performance Based Navigation (PBN), Time Based Flow Management (TBFM), Collaborative Air Traffic Management (CATM), and Separation Management.

Federal Aviation Administration (October 2014). ***NextGen Priorities Joint Implementation Plan.*** http://www.faa.gov/nextgen/media/ng_priorities.pdf.

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The priorities referred to in this plan originated with the RTCA NextGen Advisory Committee (NAC), its Subcommittee (NACSC), and various working groups. Based on the work of meetings among FAA subject matter experts and aviation industry representatives, this document summarizes the high-level commitments to which the FAA and the aviation community have agreed, and provides a timeline of capability milestones and airport site locations where they collectively committed to accomplishing these milestones within the next three years. The plan is organized into four focus areas: Multiple Runway Operations, Performance-Based Navigation, Surface Operations and Data Sharing, Data Communications.

Federal Aviation Administration (2014). ***NextGen for Airports Brochure.*** https://www.faa.gov/nextgen/media/nextgenForAirports.pdf.

This document is divided into two parts: General Aviation (GA) airports and commercial airports. It outlines the current implementation status of key, relevant NextGen technologies. For example, both GA and commercial service airports can benefit from WAAS/LPV Options for Low Visibility Approaches, and information provided freely via TIS-B and FIS-B. For commercial airports with ASDE-X/Multilateration installations, they can benefit from sharing of real-time surface surveillance data, allowing them to keep track of ground vehicles and enabling surface collaborate decision making; PBN procedures and OAPM; and closely spaced parallel runway operations.

RTCA NextGen Advisory Committee (NAC) (October 2014). ***NextGen Integration Working Group Final Report.*** http://www.rtca.org/content. asp?pl=33&sl=61&contentid=61.

It summarizes the work by the NextGen Integration Working Group in developing top priority capabilities and industry recommendations for the four focus areas: Closely Spaced Parallel Runways/MRO, Data Comm, PBN, and Surface and Data Sharing. Recommendations for each focus area include detailed implementation plan, expected benefits, identified airport locations and timing, as well as relevant metrics to be used in evaluation.

Air Traffic Control Association (Fall 2014). ***Key NextGen Achievements in the First 10 Years, Journal of Air Traffic Control.*** http://www.atca.org/Journal-of-Air-Traffic-Control.

This article summarizes operational improvements that are about to be completed. These include ADS-B, En Route Automation Modernization (ERAM), the FAA telecommunication infrastructure, performance-based navigation (PBN), transoceanic initiatives, safety, and collaboration.

Government of Accountability Office (2012). ***Next Generation Air Transportation System: FAA Faces Implementation Challenges.*** http://www.gao.gov/assets/650/648122.pdf.

One of the key challenges highlighted in the report was FAA's current process for implementing or amending flight procedures, resulting in the implementation of low or no-benefit flight procedures that have to be reworked or amended. Moreover, the report suggested that NextGen alone is not likely to sufficiently expand the capacity to meet ongoing and planned demand. The airports have to continuously look for other ways to expand capacities, for example: (1) certifying and approving standards for using closely spaced parallel runways; (2) developing policies to address situations where demand exceeds capacity (e.g., pricing, administrative rules, service priorities); (3) planning infrastructure projects to increase capacity as they can be a lengthy process and will require substantial advance planning and cost analyses; and (4) understanding that improved runway and airspace capacity from NextGen may exacerbate capacity constraints in other areas, such as taxiways, terminal gates, or parking areas.

Government of Accountability Office (2013). ***FAA Has Made Some Progress in Midterm Implementation, but Ongoing Challenges Limit Expected Benefits.*** http://www.gao.gov/assets/660/653626.pdf.

This report recommended that FAA should better develop processes for selecting new PBN procedures that will provide the maximum benefits (e.g., relieve congestion and improve efficiency) and not rely

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on requests from airlines and other stakeholders. The report also discusses the lengthy environmental reviews which have been identified as an obstacle to timely implementation of PBN, and the current challenges with the National Environmental Policy Act (NEPA) for the development of new flight procedures.

The MITRE Corporation (October 2014). ***NextGen Independent Assessment and Recommendations.*** http://www.faa.gov/nextgen/media/MITRE_NextGen_Independent_Assessment_and_Recommendations.pdf.

The report provides an independent assessment of the NextGen progress to date, and identifies key fundamental challenges and remaining gaps for moving NextGen forward. The largest gap is that more effective transition planning is needed for maturing NextGen capabilities (e.g., including the development of procedures and best practices). It is also expected that the aviation community will not meet the 2020 mandate for ADS-B at the current pace of equipage. As a result of lack of user adoption of required avionics, Trajectory-Based Operations (TBO) implementation should be deferred. The report provides recommendations for moving NextGen forward for 2020 and beyond, which include (but are not limited to): maximizing operational use of available aircraft and ground capabilities, as well as procedurally permit initial operations of new entrants (e.g., UAS) leveraging existing policy and introduce new rules to enable their evolving operations. The Appendix includes more detailed description of the NextGen capabilities implemented to date (e.g., closely spaced parallel operations, wake turbulence separation reductions, oceanic airspace operations, airspace redesign, PBN procedures, better leveraging of existing automation).

2. NextGen Guidance Documents Relevant to Potential Airport Planning Areas

2.1 Master Planning

Federal Aviation Administration: Office of Airport Planning & Programming, Planning & Environmental Division (2015). ***Advisory Circular 150/5070-6B – Airport Master Plans.*** http://www.faa.gov/documentLibrary/media/Advisory_Circular/150-5070-6B-Change-2-Consolidated.pdf.

This document provides guidance to prepare master plans for airports of a range of sizes and functions. In particular, the section on airfield and airspace requirements contains best practices for capacity analysis and runway requirements as part of a master plan. Guidance is given on the threshold for annual average delay as a measure for determining the practical capacity of an airfield.

2.2 Regional Airport System Planning

Federal Aviation Administration (2014). ***Optimization of Airspace & Procedures in the Metroplex (OAPM).*** https://www.faa.gov/air_traffic/flight_info/aeronav/procedures/oapm/.

Rather than focusing on a single airport or set of procedures, the FAA has identified 21 sites that include several commercial and general aviation airports in close proximity serving large metropolitan areas, and putting integrated NextGen capabilities in place to improve air traffic flow for an entire region, or Metroplex. The website includes the scheduled plan regarding selected OAPM sites, as well as detailed information for the current and completed Metroplexes.

2.3 Airfield and Airspace Planning

Federal Aviation Administration (1995). ***Advisory Circular 150/5060-5 – Airport Capacity and Delay.*** http://www.faa.gov/documentLibrary/media/advisory_circular/150-5060-5/150_5060_5.pdf.

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This document contains methods to compute the hourly airport capacity and annual aircraft delay for airport planning and design purposes. This circular is primarily intended for airport planners and is the primary resource from FAA on the subject of capacity and delay. The circular includes lookup tables in Chapter 2 and charts and nomographs in Chapter 3 for calculating hourly runway throughput based on runway layout and fleet mix.

Federal Aviation Administration (2014). ***Advisory Circular 150/5300-13A – Airport Design.*** http://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5300-13.

This document contains the FAA's standards and recommendations for airport design. Most relevant to capacity and delay are the sections on runway design, parallel runway separation requirements, and taxiway design.

Federal Aviation Administration (2007). ***Advisory Circular 150/5300-16A – General Guidance and Specifications for Aeronautical Surveys: Establishment of Geodetic Control and Submission to the National Geodetic Survey.*** http://www.faa.gov/documentlibrary/media/advisory_circular/150-5300-16a/150_5300_16a.pdf.

This document has been established for engineering and airport sponsors and explains the specifications for establishing geodetic control on or near an airport. It also describes how to submit the information to the National Geodetic Survey (NGS) for approval and inclusion in the National Spatial Reference System (NSRS) in support of aeronautical information surveys.

Federal Aviation Administration (2009). ***Advisory Circular 150/5300-17-C – Standards for Using Remote Sensing Technologies in Airport Surveys.*** http://www.faa.gov/documentLibrary/media/Advisory_Circular/150_5300_17c.pdf.

This document applies to airport proponents contracting airport surveying services utilizing remote sensing technologies, such as aerial or satellite imagery or Light Detection and Ranging (LIDAR), which are used as part of aeronautical surveys associated with Performance Based Navigation (PBN) instrument flight procedures. The AC outlines guidance regarding the use of remote sensing technologies in the collection of data describing the physical infrastructure of an airport. This AC describes the acceptable uses and standards for use of different remote sensing technologies in the data collection process. This AC also provides data providers the standards and recommended practices for using remote sensing technologies in the collection of airport data.

Federal Aviation Administration (2009). ***Advisory Circular 150/5300-18B – General Guidance and Specifications for Submission of Aeronautical Surveys to NGS: Field Data Collection and Geographic Information Systems (GIS) Standards.*** http://www.faa.gov/documentLibrary/media/Advisory_Circular/150_5300_18b.pdf.

This Advisory Circular (AC) provides the specifications for the collection of airport data through field and office methodologies in support of the Federal Aviation Administration (FAA). It also explains how to submit data to the FAA, who will forward the safety-critical data to the National Geodetic Survey (NGS) for independent verification and validation. The primary purpose of these general guidelines and specifications is to list the requirements for data collection conducted at airports in support of the FAA Airport Surveying – Geographic Information System (GIS) Program.

The standards covered in the document provide critical information for the operation and safety of the National Airspace System (NAS) and are classified as critical by the International Civil Aviation Organization (ICAO). Most of this information is source data, acquired by field survey and/or remote sensing methods.

Federal Aviation Administration (2014). ***Order JO 7110.65W, Air Traffic Control.*** http://www.faa.gov/documentLibrary/media/Order/ATC.pdf.

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This order prescribes air traffic control procedures and phraseology for use by persons providing air traffic control services. It covers procedures for various types of flights, flight plan information (its needs and type of information), flight strips, communications, signals, and reporting information. Of particular interest is Chapter 3, Airport Traffic Control—Terminal, which covers departure and arrival procedures and separation.

Federal Aviation Administration (2015). ***Order JO 7110.308A, 1.5-Nautical Mile Dependent Approaches to Parallel Runways Spaced Less Than 2,500 Feet Apart.*** http://www.faa.gov/documentLibrary/media/Order/Order_7110.308A.pdf.

This order describes the use of 1.5-nautical mile dependent staggered approaches for closely spaced runways less than 2,500 feet apart at selected airports. This order is for aiding air traffic management and air traffic control personnel.

Federal Aviation Administration (2015). ***Order JO 7110.659B, Wake Turbulence Recategorization.*** https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/1026926.

This order supplements 7110.65 by defining separations for different classes of heavy aircraft. The order defines six classes of aircraft, based on maximum takeoff weight and wingspan, and the updated requisite spacing behind each based on recent studies of wake turbulence characteristics.

Federal Aviation Administration (2013). ***Order N JO 7110.625, Simultaneous Independent Close Parallel Approaches – High-Update Radar Not Required.*** http://www.faa.gov/documentLibrary/media/Notice/N_JO_7110.625.pdf.

This order supplements 7110.65 by reducing the separation needed between parallel runways to support simultaneous independent approaches. The order allows runways with centerline spacing of 3,600 feet to be authorized for simultaneous independent approaches without the use of a high-update radar, but a high-resolution color monitor with alert algorithms, such as the Final Monitor Aid (FMA), must be used to monitor the close parallel approaches.

Federal Aviation Administration (2014). ***Order N JO 7110.652, Converging Runway Operations.*** http://www.faa.gov/documentLibrary/media/Notice/N_JO_7110.652_Converging_Runway_Operations.pdf.

This order has been cancelled as of January 14, 2015. Please refer to FAA Order JO 7110.65W, Change 1, Paragraph 3-9-9, NONINTERSECTING CONVERGING RUNWAY OPERATIONS, MAY 26, 2016.

Federal Aviation Administration (2012). ***Order 8260.49A, Simultaneous Offset Instrument Approach (SOIA).*** http://www.faa.gov/documentLibrary/media/Order/8260.49A%20CHG-1%20Consolidated.pdf.

This order provides criteria and guidance for constructing and operating simultaneous offset instrument approaches to parallel runways spaced at least 750 feet apart, and less than 3,000 feet apart at airports identified by the FAA for SOIA. This order establishes criteria for conducting closely spaced parallel operations, and identifies procedural requirements to ensure safety. Implementation of SOIA procedures requires additional analysis and study at most locations.

Federal Aviation Administration (2015). ***Aeronautical Information Manual: Official Guide to Basic Flight Information and ATC Procedures.*** https://www.faa.gov/air_traffic/publications/media/aim.pdf.

The Aeronautical Information Manual provides basic flight information and explanation of ATC procedures for use in the United States. Of particular interest here are sections 3-1 (General) which highlights VFR minimums, 3-2 (Controlled Airspace) which highlights the operating rules for various airspaces, 4-4 (ATC Clearances and Aircraft Separation) which identifies role of pilot for implementing separation, 5-2 (Departure Procedures), 5-3 (En Route Procedures), 5-4 (Arrival Procedures), and 5-5 (Pilot/Controller Roles and Responsibilities).

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Federal Aviation Administration (2016). ***Order 8260.58A, United States Standards for Performance Based Navigation (PBN) Instrument Procedure Design.*** https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/1029267.

This document provides a consolidated United States Performance Based Navigation (PBN) procedure design criteria. Procedures design criteria is provided for all segments of Standard Instrument Arrivals (STARs), Standard Instrument Approach Procedures (SIAPs), and Standard Instrument Departures (SIDs). Procedure design criteria is based on multiple Area Navigation (RNAV) navigational systems, including Lateral Navigation (LNAV), Lateral/Vertical Navigation (LNAV/VNAV), Required Navigational Performance (RNP), RNP Authorization Required, Localizer Performance (LP), and Localizer Performance with Vertical (LPV).

Federal Aviation Administration (2010). ***Order 6884.1, Siting Criteria for Ground Based Augmentation System (GBAS).*** http://www.faa.gov/documentLibrary/media/Order/6884_1.pdf.

This order establishes procedures for determining, evaluating, and approving the siting requirements for the installation of a Ground Based Augmentation System (GBAS) Ground Facility supporting Category I, II, and III precision instrument approach. GBAS is not fixed by function (e.g., proximity to runway centerline) increasing the siting location options and potentially reduces the need for an extensive site preparation effort typically associated with ground-based precision approach and landing systems. This criteria enables the selection of the optimum siting of the GBAS Ground Facility as part of the planning and development of the system. The document covers installation requirements, facility requirements for the installation, and requirements for maintaining the areas around the antennas associated with the GBAS Ground Facility.

Federal Aviation Administration (2014). ***Order 7100.41, Performance-Based Navigation Implementation Process.*** http://www.faa.gov/documentLibrary/media/Order/FAA_JO_7100.41_Performance_Based_Navigation_Implementation_Process.pdf.

This order provides a standardized five-phase implementation process related to Performance-Based Navigation (PBN) routes and procedures, referred to as the "Performance Based Navigation Implementation Process," which has been deemed compliant by the Office of Safety and meets the requirements set forth by the Federal Aviation Administration (FAA) Air Traffic Organization's (ATO) Safety Management System (SMS).

This order applies to the development and implementation of PBN procedures and routes; specifically, Area Navigation (RNAV)/Required Navigation Performance (RNP) Standard Instrument Departures (SID), RNAV/RNP Standard Terminal Arrivals (STAR), and RNP Authorization Required (AR) Standard Instrument Approach Procedures (SIAP), Q, Tango or "T," and TK (helicopter) Routes, and RNAV/RNP transitions to SIAPs.

Federal Aviation Administration (2014). ***Airport Capacity Profiles.*** http://www.faa.gov/airports/planning_capacity/profiles/http://www.faa.gov/airports/planning_capacity/profiles/media/Airport-Capacity-Profiles-2014.pdf.

The capacity profiles provide a high-level assessment of airport runway capacity now and in the future (with assumptions of NextGen improvements), replacing the previous Airport Capacity Benchmark Report (2004). The airport capacity profiles also provide context for strategic infrastructure discussions by providing a succinct estimate of the current and future state of capacity at the nation's major airports. The capacity profiles serve as a basis for the Future Airport Capacity Task 3 (FACT3), an in-depth evaluation of airport capacity needs.

RTCA NextGen Advisory Committee (NAC) (October 2014). ***Blueprint for Success to Implementing PBN.*** http://www.rtca.org/content.asp?pl=33&sl=61&contentid=61.

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As a follow-on response to FAA's request to analyze lessons learned from prior PBN implementations, this blueprint covers the entire lifecycle, from planning to execution, and both technical and non-technical components, and serves as a checklist for future success in PBN implementations. It is emphasized that collaboration with airport authorities, including representatives from airport planning and development, airport operations, airport noise abatement office, community affairs/public relations, and the airport owner (e.g., city, county, state, or authority), is critical during all project phases.

2.4 Airport Environmental Planning

Federal Aviation Administration (2013). ***Noise Control and Compatibility Planning for Airports.*** http://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5020-1.

This document provides general guidance for noise control and compatibility planning for airports as well as specific guidance for preparation of airport noise exposure maps and airport noise compatibility programs by airport operators for submission under Title 14, Code of Federal Regulations, Part 150, and the Aviation Safety and Noise Abatement Act of 1979.

Federal Aviation Administration (2006). ***Federal Aviation Administration Order 1050.1E, Environmental Impacts: Policies and Procedures, CHG 1, March 20, 2006.*** http://www.faa.gov/documentLibrary/media/Order/1050.1E.pdf.

This order describes the FAA agency-wide policies and procedures for compliance with the National Environmental Policy Act (NEPA) and implementing regulations issued by the Council on Environmental Quality (40 CFR parts 1500-1508), as applied to federal actions undertaken by the FAA (2006).

Airport Cooperative Research Program (2009). ***ACRP Synthesis 17: Approaches to Integrating Airport Development and Federal Environmental Review Processes.*** http://onlinepubs.trb.org/onlinepubs/acrp/acrp_syn_017.pdf.

This document examines why the disconnect occurs between the NEPA process and steps that precede it, and provides guidance for airport sponsors to integrate FAA's environmental review processes into airport planning efforts. The study presents case studies for 17 projects. Each case study includes a description of the proposed project, an identification of the key planning and environmental issues, and strategies and other factors that were used to integrate the planning and environmental processes.

Airport Cooperative Research Program (2009). ACRP Report 15: Aircraft Noise: A Toolkit for Managing Community Expectations. http://onlinepubs.trb.org/onlinepubs/acrp/acrp_rpt_015.pdf.

This project was focused on developing a toolkit that airport decision makers can use to manage expectations related to aircraft noise within the community, based on reviews of relevant literature, as well as surveys from airports ranging in different sizes and categories. Chapter 8 summarizes flight management techniques to abate the noise of aircraft in flight, which includes evaluation of Continuous Descent Approach (CDA).

Airports Council International – North America (2013). ***Integrating Planning and NEPA Processes for Airport Development Projects.*** http://aci-na.org/sites/default/files/integrating_nepa_and_planning_document_5.13.13.pdf.

This report, an outcome of the ACI-NA Task Force efforts, provides airport operators and their consultants with practical recommendations regarding how to improve the integration of airport planning and NEPA review processes.

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Airports Council International – North America (2013). ***Airports' Role in the Development and Implementation of Performance-Based Navigation Procedures.*** http://www.aci-na.org/sites/default/files/aci-na_pbn-paper-version_1.1_7mar13.pdf.

This paper, developed by the ACI-NA's NextGen Working Group, emphasizes the need for airports to be actively involved in the design and implementation of PBN procedures. Airports should position themselves as a critical resource for the FAA ATO and other industry stakeholders to ensure local expectations are understood and respected.

2.5 Airport Financial and Strategic Planning

Federal Aviation Administration (2013). ***National Plan of Integrated Airport Systems (NPIAS) 2013–2017.*** http://www.faa.gov/airports/planning_capacity/npias/reports/media/2013/npias2013Narrative.pdf.

This is an annual report submitted to Congress summarizing capital development needs of the public-use airports in the United States. Of particular interest is the section on capacity, which addresses FAA guidance on triggers at which construction of capacity projects should begin and alternative capacity enhancement methods aside from capital development.

Federal Aviation Administration (1999). ***FAA Airport Benefit Cost Analysis Guidance.*** http://www.faa.gov/regulations_policies/policy_guidance/benefit_cost/media/1999_FAA_Airport_Benefit_Cost_Analysis_Guidance.pdf.

This guidance provides the methodology for airport sponsors to prepare a project-level benefit-cost analysis for capacity-related projects. The quantification of capacity benefits and monetization of airfield delay savings are discussed.

Federal Aviation Administration (2013*). NextGen, The Business Case.* http://www.faa.gov/nextgen/media/Business_Case_for_NextGen_2013.pdf.

This report presents the FAA's business case for NextGen, which focuses on direct benefits to aircraft operators, passengers, and taxpayers from the rollout of NextGen improvements. This includes monetary benefits from avoided delay, reduced flight time, fewer flight cancellations, and reduced emissions. It is worth noting that this business case is conducted at the national level, and not focused on benefits or costs specific to airport operators.

Federal Aviation Administration (2014). ***NextGen Investments for Operators and Airports.*** http://www.faa.gov/nextgen/media/investments.pdf.

This pamphlet outlines the investments for operators and airports, and provides an overview of existing and planned capabilities. Of particular note, airport enhancements are described.

3. Specialized Airport Planning Guidance Documents with Impacts on:

3.1 General Aviation and Small Airports

Federal Aviation Administration (2014). ***NextGen for General Aviation.*** https://www.faa.gov/nextgen/media/generalAviation.pdf.

This document highlights the benefits of NextGen for General Aviation (GA). It describes NextGen accomplishments to date that have directly benefitted GA and small airports, e.g., ADS-B-enabled TIS-B and FIS-B, WAAS-enabled approach procedures.

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3.2 Unmanned Aircraft Systems (UAS)

UAS are aircraft and associated equipment that do not carry a pilot aboard, but instead operate on pre-programmed routes or are manually controlled by pilot-operated ground stations. Although current non-military, domestic uses of UAS are limited to activities such as law enforcement, forensic photography, border security, and scientific data collection, UAS have a wide range of other potential commercial uses and the market for UAS use is expected to grow.

The Federal Aviation Administration (FAA) maintains a website to keep the industry updated of the progress and regulations regarding UAS. The website can be accessed at https://www.faa.gov/uas/. Of particular interest:

Federal Aviation Administration (2013). ***UAS Comprehensive Plan.*** https://www.faa.gov/about/office_org/headquarters_offices/agi/reports/media/UAS_Comprehensive_Plan.pdf.

This document outlines the nation's UAS goals and objectives. Of particular interest, Section 332 requires recommendations for airspace designations and establishment of a process to develop flight standards and air traffic requirements for unmanned aircraft systems to be made before September 30, 2015.

Federal Aviation Administration (2013). ***UAS Roadmap.*** https://www.faa.gov/uas/legislative_programs/uas_roadmap/.

This document outlines the tasks and considerations necessary to achieve UAS integration into the NAS. First published in 2013, subsequent publications will incorporate lessons learned and related findings, which will further define goals, metrics, and target dates for the five-year Roadmap.

Federal Aviation Administration (2013). ***FAA Modernization and Reform Act of 2013, Section 333.*** https://www.faa.gov/uas/legislative_programs/section_333/.

Section 333, "Special Rules for Certain Unmanned Aircraft Systems," provides FAA the authority to establish an interim policy that bridges the gap between the current state and NAS operations before the UAS rules are finalized. In September 2014, the FAA has granted exemptions to six aerial movie production companies for commercial UAS operation in the NAS, under Section 333.

Department of Transportation (June 2014). ***Office of Inspector General Audit Report: FAA Faces Significant Barriers to Safely Integrate Unmanned Aircraft Systems into the National Airspace System.*** https://www.oig.dot.gov/sites/default/files/FAA%20Oversight%20of%20Unmanned%20Aircraft%20Systems%5E6-26-14.pdf.

The report describes the background and current status of integrating UAS into NAS. One of the key challenges is that FAA has not established a regulatory framework for integrating UAS into the NAS, such as when and how UAS are authorized to operate in U.S. airspace. Instead, FAA currently allows UAS operations on a case-by-case basis, and only separates UAS into two classes (under and over 55 pounds). In March 2013, a UAS rulemaking committee recommended that FAA apply weight distinctions that are currently established for manned aircraft; however, no consensus was reached for the need for additional classifications. UAS Integration Office will need to interface with personnel in ATO who must develop airspace policy that considers the operational needs of both manned and unmanned aircraft, as well as reaching out to FAA NextGen office and other organizations.

4. NextGen Airport Planning and Environmental Studies

Puget Sound Regional Council (May 2013). ***Preparing Busy General Aviation Airports for Next Generation Technologies.*** http://www.psrc.org/assets/7340/NextGen.pdf.

Preparing the Region's Busy General Aviation Airports for NextGen Technologies is a first of a kind effort funded by FAA Airport Improvement Program. It presents a regional/system planning approach

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to identifying the General Aviation benefits in the Puget Sound metropolitan area that can be realized through the deployment and implementation of the Federal Aviation Administration's NextGen program.

The premise of the PSRC NextGen study is to help the region's airports get ready for emerging aviation technology. Taking advantage of new technology will require airports to meet FAA design standards. The report includes a system planning analysis for five tasks:

1. Identification of busy general aviation airports where NextGen technology could be beneficial.
2. Inventory preparation of these airports to determine their preparedness to implement NextGen.
3. Identification of airport design and operational deficiencies ("gap" analysis) showing the difference between existing airport conditions and FAA design requirements to meet NextGen criteria.
4. Development of a capital improvement plan (CIP) for each airport listing improvements to implement NextGen.
5. Identification of individual airport NextGen strategies and PSRC system applications.

The report ultimately serves as a regional system plan identifying opportunities and priorities for NextGen implementation supporting General Aviation.

Federal Aviation Administration (2012). ***Finding of No Significant Impact (FONSI) and the Record of Decision (ROD) for the Implementation of RNAV/RNP Procedures at Seattle-Tacoma International Airport (Greener Skies over Seattle).*** http://www.greenerskiesea.com.

This document serves as the Federal Aviation Administration's (FAA) Finding of No Significant Impact and Record of Decision (FONSI/ROD) and provides final agency determinations and approvals for the proposed action, namely utilization of Performance-Based Navigation (PBN) by implementing new Area Navigation (RNAV) procedures, including Required Navigation Performance (RNP) and Optimized Profile Descent (OPD) at Seattle-Tacoma International Airport (SEA). The proposed routes and procedures are designed to improve the safety and efficiency of the SEA airspace, which includes the Terminal Radar Approach Control (TRACON) as well as high-altitude Air Route Traffic Control Center (ARTCC) airspace. This FONSI/ROD is based on the information and analysis contained in the Final Environmental Assessment (Final EA) dated October 2012.

Federal Aviation Administration Environmental Reviews. http://www.faa.gov/air_traffic/environmental_issues/ared_documentation/.

Environmental reviews are a critical element in FAA's efforts to manage airspace capacity. FAA provides environmental reviews for all of FAA's airspace redesign and RNAV/RNP procedure development projects concerning air space use and air traffic. Environmental reviews of air traffic projects are conducted under the guidelines and regulations of the National Environmental Policy Act (NEPA) and related statutory and regulatory environmental laws such as the Clean Air Act and National Historic Preservation Act, as well as internal FAA environmental regulations:

- FAA Order 1050.1E (Environmental Impacts: Policies and Procedures).
- FAA Order 7400.2K (Chapter 32) (Procedures for Handling Airspace Matters).

Airspace Redesign Environmental Reviews

- Las Vegas Area Optimization of Airspace and Procedures.
- New York/New Jersey/Philadelphia Airspace Redesign.
- Houston Area Air Traffic System (HAATS).

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Metroplex Environmental Reviews

- Final Environmental Assessment – FONSI/ROD for Atlanta, Georgia; Optimization of the Airspace and Procedures in the Metroplex (signed July 31, 2014).
- Final Environmental Assessment – FONSI/ROD for Northern California; Optimization of the Airspace and Procedures in the Metroplex (signed July 31, 2014).
- Final Environmental Assessment – FONSI/ROD for North Texas Optimization of the Airspace and Procedures in the Metroplex (North Texas OAPM) (June 2013).
- Final Environmental Assessment – FONSI/ROD for the DC OAPM Project (signed December 30, 2013).
- Final Environmental Assessment – FONSI/ROD for the Houston Optimization of Airspace and Procedures in the Metroplex (OAPM) (signed June 13, 2013).

Performance-Based Navigation (PBN) Environmental Reviews

- Boston Runway 33L, PBN.
- **Final Environmental Assessment** – FONSI/ROD for the implementation of a new air traffic control Area Navigation (RNAV) standard instrument departure (SID) procedure on Runway 33 Left (33L) at Boston-Logan International Airport (signed June 4, 2013).
- Chicago-Midway PBN.
- Denver PBN.
- Greener Skies Over Seattle.

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F NextGen—Airport Planning and Development List of Acronyms

Below is a table of commonly used NextGen-related acronyms found in this guidebook and in the NextGen literature cited in Appendix E:

4D	Four Dimensional
ADS	Automated Docking Systems
ADS-B	Automatic Dependent Surveillance-Broadcast
ADS-R	Automatic Dependent Surveillance-Rebroadcast
AEFS	Advanced Electronic Flight Strips
ANOMS	Airport Noise and Operations Monitoring System
ANSP	Air Navigation Service Provider
AR	Authorization Required
ARTCC	Air Route Traffic Control Center
ARTS	Automated Radar Terminal System
ASDE-X	Airport Surface Detection Equipment-Model X
ASSC	Airport Surface Surveillance Capability
ATC	Air Traffic Control
ATCT	Airport Traffic Control Tower
ATM	Air Traffic Management
ATO	Air Traffic Organization
ATOP	Advanced Technologies and Oceanic Procedures
CAT	Category
CDM	Collaborative Decision Making
CDTI	Cockpit Display of Traffic Information
CNS	Communication, Navigation, and Surveillance
CPDLC	Controller Pilot Data Link Communications
CSPO	Closely Spaced Parallel Operations
Data Comm	Data Communications

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EDO	Established-on-Departure
EVS	Enhanced Flight Vision System
ELSO	Enhanced Lateral Spacing Operations
EOBT	Earliest Off Block Time
EOR	Established on RNP
ERAM	En Route Automation Modernization
FIS-B	Flight Information Services Broadcast
FMA	Final Monitor Aid
FMS	Flight Management System
GA	General Aviation
GBAS	Ground-Based Augmentation System
GPS	Global Positioning System
ICAO	International Civil Aviation Organization
ILS	Instrument Landing System
LED	Light Emitting Diodes
LNAV	Lateral Navigation
LP	Localizer Performance
LPV	Localizer Performance with Vertical Guidance
MLAT	Multilateration
MMR	Multi-Mode Receiver
MOPS	Minimum Operational Performance Standards
MRO	Multiple Runway Operations
NAC	NextGen Advisory Committee
NAS	National Airspace System
NextGen	Next Generation Air Transportation System
NPIAS	National Plan of Integrated Airport Systems
OCS	Obstacle Clearance Surface
OPD	Optimized Profile Descent

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PAPI	Precision Approach Path Indicator
PBN	Performance-Based Navigation
PRM	Precision Runway Monitor
RECAT	Recategorization
RNAV	Area Navigation
RNP	Required Navigation Performance
RPAT	RNP Parallel Approaches with Transition
SBAS	Satellite-Based Augmentation System
SID	Standard Instrument Departure
SMS	Safety Management System
SOIA	Simultaneous Offset Instrument Approach
STAR	Standard Terminal Arrival
STARS	Standard Terminal Automation Replacement System
SVS	Synthetic Vision System
SWAP	Severe Weather Avoidance Plan
SWIM	System Wide Information Management
TAMR	Terminal Automation Modernization and Replacement
TBFM	Time-Based Flow Management
TBO	Trajectory-Based Operations
TDLS	Tower Data Link Services
TDOA	time difference of arrival
TFDM	Terminal Flight Data Management
TFM	Traffic Flow Management
TIS-B	Traffic Information Services Broadcast
TMI	Traffic Management Initiative
TRACON	Terminal Radar Approach Control
UAS	Unmanned Aircraft Systems
UDOS	Uniform Departure Operation Spacing
VNAV	Vertical Navigation

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WAAS	Wide Area Augmentation System
Wake RECAT	Wake Turbulence Recategorization
WAM	Wide Area Multilateration
WTMD	Wake Turbulence Mitigation for Departures

Source: Selected and augmented from several more comprehensive lists of acronyms found on the FAA Website and MITRE Corporation Reports (e.g., http://www.faa.gov/airports/resources/acronyms/; http://www.faa.gov/nextgen/media/MITRE_NextGen_Independent_Assessment_and_Recommendations.pdf \; and http://www.faa.gov/nextgen/media/ng_priorities.pdf).

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S.DOT	United States Department of Transportation

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